Towards Agentic RAG with Deep Reasoning: A Survey of RAG-Reasoning Systems in LLMs

Yangning Li[1*], Weizhi Zhang[2*], Yuyao Yang[2], Wei-Chieh Huang[2], Yaozu Wu[3]
Junyu Luo[4], Yuanchen Bei[5], Henry Peng Zou[2], Xiao Luo[6], Yusheng Zhao[4]
Chunkit Chan[7], Yankai Chen[2], Zhongfen Deng[2], Yinghui Li[1], Hai-Tao Zheng[1],
Dongyuan Li[3], Renhe Jiang[3], Ming Zhang[4], Yangqiu Song[7], Philip S. Yu[1]
[1]Tsinghua University [2]University of Illinois Chicago [3]The University of Tokyo
[4]Peking University [5]University of Illinois Urbana-Champaign
[6]University of California, Los Angeles [7]HKUST
`ynli23@mails.tsinghua.edu.cn`, `wzhan42@uic.edu`

Abstract

Retrieval-Augmented Generation (RAG) lifts the factuality of Large Language Models (LLMs) by injecting external knowledge, yet it falls short on problems that demand multi-step inference; conversely, purely reasoning-oriented approaches often hallucinate or mis-ground facts. This survey synthesizes both strands under a unified reasoning-retrieval perspective. We first map how advanced reasoning optimizes each stage of RAG (**Reasoning-Enhanced RAG**). Then, we show how retrieved knowledge of different type supply missing premises and expand context for complex inference (**RAG-Enhanced Reasoning**). Finally, we spotlight emerging **Synergized RAG-Reasoning** frameworks, where (agentic) LLMs iteratively interleave search and reasoning to achieve state-of-the-art performance across knowledge-intensive benchmarks. We categorize methods, datasets, and open challenges, and outline research avenues toward deeper RAG-Reasoning systems that are more effective, multimodally-adaptive, trustworthy, and human-centric. The collection is available at `https://github.com/DavidZWZ/Awesome-RAG-Reasoning`.

1 Introduction

The remarkable progress in Large Language Models (LLMs) has transformed a wide array of fields, showcasing unprecedented capabilities across diverse tasks (Zhao et al., 2023). Despite these advancements, the effectiveness of LLMs remains hindered by two fundamental limitations: knowledge hallucinations, due to the static and parametric manner of their knowledge storage (Huang et al., 2025b); and struggles with complex reasoning, especially when tackling real-world problems (Chang et al., 2024). These limitations have driven the development of two major directions: Retrieval-Augmented Generation (RAG) (Fan et al., 2024a), which provides LLMs with external knowledge; and various methods aimed at enhancing their inherent reasoning abilities (Chen et al., 2025c).

The two limitations are inherently intertwined: missing knowledge can impede reasoning, and flawed reasoning hinders knowledge utilization (Tonmoy et al., 2024). Naturally, researchers have increasingly explored combining retrieval with reasoning, though early work followed *two separate, one-way enhancements*. The first, **Reasoning-enhanced RAG** (Gao et al., 2023b) (Reasoning → RAG), leverages reasoning to improve specific stages of the RAG pipeline. The second path, **RAG-enhanced Reasoning** (Fan et al., 2024a) (RAG → Reasoning), supplies external factual grounding or contextual cues to bolster LLM reasoning.

While beneficial, the above methods remain bound to a static Retrieval-Then-Reasoning (RTR) framework, offering only localized improvements to individual components. Several inherent limitations persist: *(1) Retrieval Adequacy and Accuracy* cannot be guaranteed; Pre-retrieved knowledge may fail to align with the actual knowledge needs that emerge during reasoning, especially in complex tasks (Zheng et al., 2025; Li et al., 2025d). *(2) Reasoning Depth* remains constrained. When retrieved knowledge contains errors or conflicts, it can adversely interfere with the model's inherent reasoning capabilities (Li et al., 2025b; Chen et al., 2025a). *(3) System Adaptability* proves insufficient. The RTR framework lacks mechanisms for iterative feedback or dynamic retrieval during reasoning. This rigidity limits its effectiveness in scenarios that require adaptive reasoning, such as open-domain QA or scientific discovery (Xiong et al., 2025; Alzubi et al., 2025).

As shown in Figure 1, these shortcomings have catalyzed a paradigm shift toward **Synergized Re-**

[*] Equal Contribution.

面向具能动性的深度推理RAG：LLM中 RAG-推理系统的综述

杨宁李[1*], 张伟志[2*], 杨雨瑶[2],黄伟哲[2],吴瑶祖[3]罗军宇[4], 贝元辰[5], 彭亨利[2], 罗晓[6],赵宇升[4]陈坤铁[7], 陈言凯[2], 邓中芬[2],李英会[1],郑海涛[1],李东元[3], 蒋仁和[3],张明[4],宋阳秋[7],余培思[1][1]清华大学 [2]伊利诺伊大学芝加哥分校 [3]东京大学[4]北京大学 [5]伊利诺伊大学厄巴纳-香槟分校[6]加州大学洛杉矶分校 [7]香港科技大学
ynli23@mails.tsinghua.edu.cn, wzhan42@uic.edu

arXiv:2507.09477v2 [cs.CL] 16 Jul 2025

摘要

检索增强生成（RAG）通过注入外部知识提升了大型语言模型（LLM）的事实性，但在需要多步推理的问题上仍显不足；相反，纯粹以推理为导向的方法则容易产生幻觉或错误地理解事实。本综述在统一的推理-检索视角下整合了这两种思路。我们首先梳理了高级推理如何优化RAG的每个阶段（推理增强RAG）。接着，展示了不同类型的检索知识如何提供缺失的前提并扩展上下文以支持复杂推理（RAG增强推理）。最后，我们重点介绍了新兴的协同RAG-推理框架，其中（具能动性的）LLM通过迭代交替搜索和推理，在知识密集型基准测试中实现了最先进性能。我们对方法、数据集和开放挑战进行了分类，并勾勒了更深层次RAG-推理系统的研究方向，使其更有效、多模态适应、值得信赖且以人为本。相关资料可在https://github.com/DavidZWZ/Awesome-RAG-Reasoning获取。

1 引言

大型语言模型（LLM）的显著进展已改变众多领域，展现出其在不同任务上的前所未有的能力（赵等，2023）。尽管如此，LLM的有效性仍受两大基本限制阻碍：知识幻觉，源于其静态和参数化的知识存储方式（黄等，2025b）；以及处理复杂推理时的困难，尤其是在应对现实世界问题时（常等，2024）。这些限制推动了两大主要方向的发展：检索增强生成（RAG）（范等，2024a），为LLM提供外部知识；以及旨在增强其内在推理能力的各种方法（陈等，2025c）。

这两个局限性本质上相互交织：缺失的知识会阻碍推理，而推理缺陷又会阻碍知识的利用（Tonmoy 等人，2024）。自然地，研究人员越来越多地探索将检索与推理相结合，尽管早期工作遵循了两个独立的单向增强路径。第一个是推理增强的 RAG（Gao 等人，2023b）（推理 →RAG），它利用推理来改进 RAG 管道的特定阶段。第二条路径是 RAG 增强的推理（Fan 等人，2024a）（RAG →推理），它提供外部事实基础或上下文线索来增强 LLM 推理。

虽然有益，但上述方法仍然局限于静态的检索后推理（RTR）框架，仅对单个组件提供局部改进。仍然存在一些固有的局限性：(1) 检索的充分性和准确性无法保证；预先检索的知识可能与推理过程中实际出现的知识需求不一致，尤其是在复杂任务中（Zheng 等人，2025；Li 等人，2025d）。(2) 推理深度仍然受限。当检索到的知识包含错误或冲突时，它可能会对模型的内在推理能力产生不利干扰（Li 等人，2025b；Chen 等人，2025a）。(3) 系统适应性不足。RTR 框架缺乏推理过程中的迭代反馈或动态检索机制。这种僵化限制了它在需要自适应推理的场景中的有效性，例如开放域问答或科学发现（Xiong 等人，2025；Alzubi 等人，2025）。

如图1所示，这些不足促使了向协同式检索与推理（RAG {v1}推理）的范式转变。

* 平等贡献。

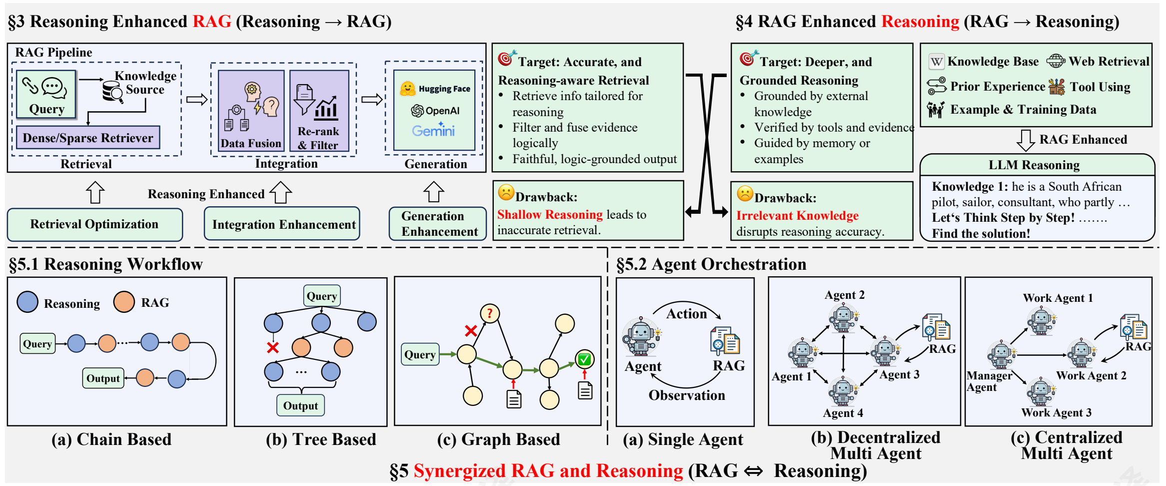

Figure 1: Overview of the RAG-Reasoning System. The *Reasoning-Enhanced RAG* methods and *RAG-Enhanced Reasoning* methods represent **one-way** enhancements. In contrast, the *Synergized RAG-Reasoning System* performs reasoning and retrieval **iteratively**, enabling mutual enhancements.

trieval and Reasoning within LLMs (RAG ⇔ Reasoning). These methods support a dynamic, iterative interplay where reasoning actively guides retrieval, and newly retrieved knowledge, in turn, continuously refines the reasoning process. This trend is further exemplified by recent "Deep Research" products from OpenAI[1], Gemini[2], Perplexity[3], and others, which emphasize tightly coupled retrieval and reasoning (Zhang et al., 2025f). These systems employ agentic capabilities to orchestrate multi-step web search and leverage reasoning to comprehensively interpret retrieved content, solving problems demanding in-depth investigation.

This survey charts the shift from isolated enhancements to cutting-edge synergized frameworks where retrieval and reasoning are deeply interwoven and co-evolve. While surveys on RAG (Fan et al., 2024a; Gao et al., 2023b) and LLM Reasoning (Chen et al., 2025c; Li et al., 2025e) exist, a dedicated synthesis focusing on their integration remains lacking. Our goal is to provide a comprehensive overview of how the symbiosis between retrieval and reasoning is advancing LLM capabilities, with particular emphasis on the move towards a synergized RAG and Reasoning framework.

The survey is structured as follows: Section 2 introduces the background; Section 3 and 4 review two one-way enhancements, respectively. Section 5 unifies both lines into synergized RAG–Reasoning frameworks. Section 6 lists benchmarks, and Section 7 outlines open challenges.

2 Background and Preliminary

RAG mitigates knowledge cut-off of LLMs through three sequential stages: *(i) Retrieval*, fetching task-relevant content from external knowledge stores; *(ii) Integration*, deduplicating, resolving conflicts, and re-ranking the retrieved content; and *(iii) Generation*, reasoning over the curated context to produce the final answer. Concurrently, Chain-of-Thought technique has significantly enhanced the reasoning capabilities of modern LLMs by encouraging them to "*think step by step*" before answering. The synergy between the structured RAG pipeline and these multi-step reasoning capacities grounds the emerging RAG-Reasoning paradigm explored in this survey.

3 Reasoning-Enhanced RAG

Traditional RAG methods first retrieve relevant documents, then concatenate the retrieved knowledge with the original query to generate the final answer. These methods often fail to capture the deeper context or intricate relationships necessary for complex reasoning tasks. By integrating reasoning capabilities across **Retrieval**, **Integration**, and **Generation** stages of the RAG pipeline, the system can identify and fetch the most relevant information, reducing hallucinations and improving response accuracy.[4]

[1] https://openai.com/index/introducing-deep-research/

[2] https://gemini.google/overview/deep-research/

[3] https://www.perplexity.ai/hub/blog/introducing-perplexity-deep-research

[4] If reasoning only serves to better leverage **fixed** retrieved knowledge in a unidirectional manner, it is considered within

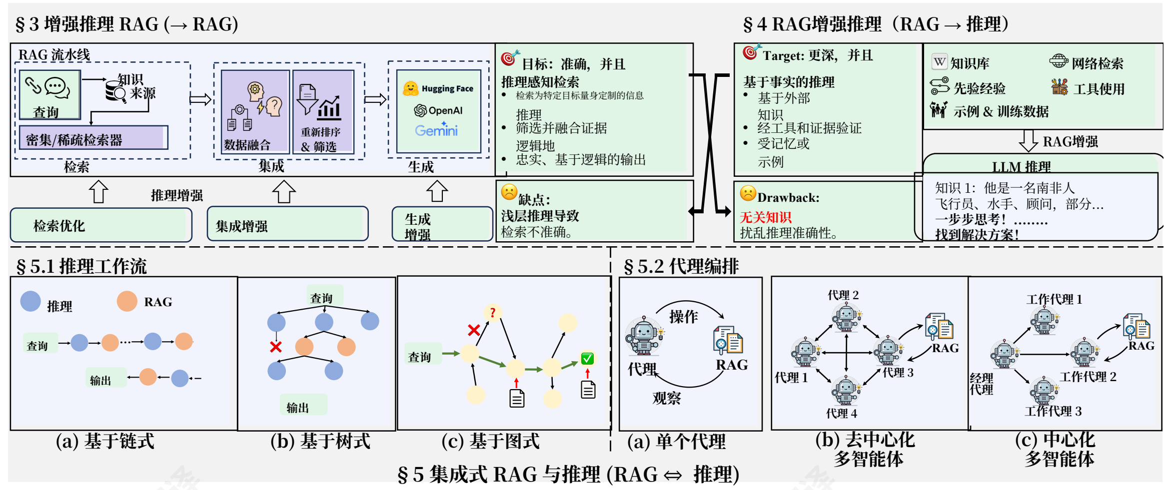

图 1：RAG-推理系统概述。推理增强型 RAG 方法和 RAG 增强型推理方法代表单向增强。相比之下，协同型 RAG-推理系统通过推理和检索迭代执行，实现相互增强。

这些方法支持动态、迭代的交互过程，其中推理主动指导检索，而新检索到的知识反过来又持续优化推理过程。这一趋势在OpenAI[1], Gemini[2], Perplex- ity[3], 等公司最近的"深度研究"产品中得到了进一步体现，这些产品强调紧密耦合的检索与推理（Zhang et al., 2025f）。这些系统运用代理能力来协调多步网络搜索，并利用推理全面解读检索到的内容，解决需要深入探究的问题。

这项调查追踪了从孤立增强到尖端协同框架的转变，其中检索和推理深度融合、协同进化。虽然已有关于RAG（Fan等人，2024a；Gao等人，2023b）和LLM推理（Chen等人，2025c；Li等人，2025e）的调查，但专注于其整合的专门综述仍然缺失。我们的目标是全面概述检索与推理之间的共生关系如何提升LLM能力，尤其强调向协同RAG和推理框架的迈进。

该调查的结构如下：第2节介绍背景；第3节和第4节分别回顾两种单向增强。第5节将两条线统一为协同的RAG–推理框架。第6节列出基准测试，第7节概述开放性挑战。

2 背景与初步

RAG通过三个连续阶段缓解LLM的知识断点：（i）检索，从外部知识库获取与任务相关的内容；（ii）整合，去重、解决冲突并重新排序检索到的内容；（iii）生成，对精选的上下文进行推理以生成最终答案。与此同时，思维链技术通过鼓励LLM在回答前"逐步思考"，显著增强了现代LLM的推理能力。结构化的RAG流程与这些多步推理能力的协同作用，奠定了本调查中探索的RAG-推理新范式的基础。

3 基于推理的增强式 RAG

传统的 RAG 方法首先检索相关文档，然后将检索到的知识与原始查询连接起来生成最终答案。这些方法往往无法捕捉到复杂推理任务所需的深层上下文或复杂关系。通过在 RAG 管道的检索、整合和生成阶段集成推理能力，系统可以识别并获取最相关的信息，减少幻觉并提高响应准确性。[4]

[1] https://openai.com/index/ 介绍深度研究/[2] https://gemini.google/overview/ 深度研究/[3] https://www.perplexity.ai/hub/blog/ 介绍困惑度深度研究

[4] 如果推理仅用于以单向方式更好地利用固定的检索知识，那么它被认为属于

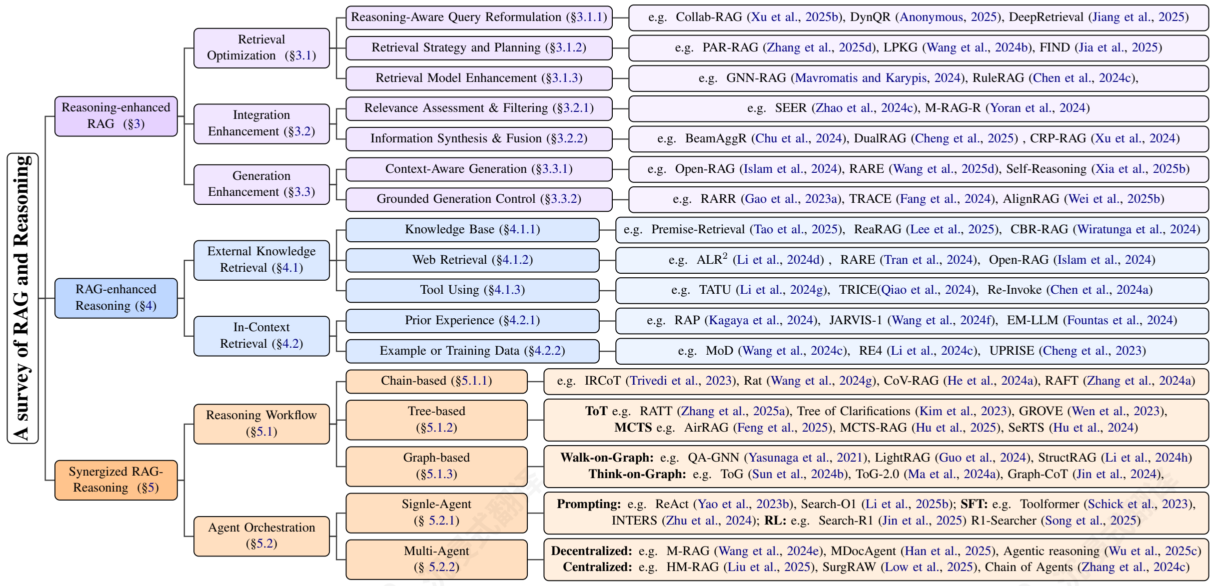

Figure 2: Taxonomy of Recent Advances in RAG-Reasoning System.

3.1 Retrieval Optimization

Retrieval optimization leverages reasoning to improve result relevance and quality. Existing methods are broadly categorized (1) Reasoning-Aware Query Reformulation, (2) Retrieval Strategy and Planning, and (3) Retrieval Model Enhancement.

3.1.1 Reasoning-Aware Query Reformulation

It reformulates the original query to better retrieve reasoning-relevant context. First, query decomposition breaks down complex queries into simpler sub-queries (Xu et al., 2025b). Second, query reformulation recasts ambiguous queries into more clear ones. To align with reasoning needs of generator, certain works train rewrites with RL signals (Anonymous, 2025; Wang et al., 2025c). Third, query expansion enrich the semantic richness of the query via CoT reasoning (Dhuliawala et al., 2024; Li et al., 2024e; Lee et al., 2024).

3.1.2 Retrieval Strategy and Planning

This section covers global retrieval guidance. Advance planning uses a reasoning model to generate a complete retrieval blueprint prior to execution. PAR-RAG (Zhang et al., 2025d) applies CoT for multi-step planning, mitigating local optima. LPKG (Wang et al., 2024b) fine-tunes LLMs on knowledge graphs to encode relational structure. In contrast, adaptive retrieval decision methods make a one-step prediction on whether and how to retrieve. FIND (Jia et al., 2025) and adaptive RAG

§3.3. In contrast, if reasoning **dynamically triggers new retrieval**, it is discussed in §5.

(Jeong et al., 2024) use classifiers to assess query complexity and select retrieval strategies, reducing unnecessary calls. Marina et al. (2025) further adds features like entity popularity and question type.

3.1.3 Retrieval Model Enhancement

A line of work enhances retrievers with reasoning via two strategies. The first one leverages structured knowledge: GNN-RAG (Mavromatis and Karypis, 2024) encodes knowledge graphs with GNNs for implicit multi-hop reasoning, while RuleRAG (Chen et al., 2024c) appends symbolic rules to guide retrieval toward logical consistency. Another strategy integrates explicit reasoning: Ji et al. (2024) combines CoT with the query to improve intermediate knowledge recall in multi-hop QA.

3.2 Integration Enhancement

Integration enhancement uses reasoning to assess relevance and merge heterogeneous evidence, preventing irrelevant content from disrupting generation. Methods fall into two categories: (1) relevance assessment and (2) information synthesis.

3.2.1 Relevance Assessment & Filtering

These methods assess the relevance of each retrieved fragment to the user query through deeper reasoning. SEER (Zhao et al., 2024c) employs assessor experts to select faithful, helpful, and concise evidence while discarding irrelevant content. Yoran et al. (2024) improves robustness by filtering non-entailing passages using an NLI model, then

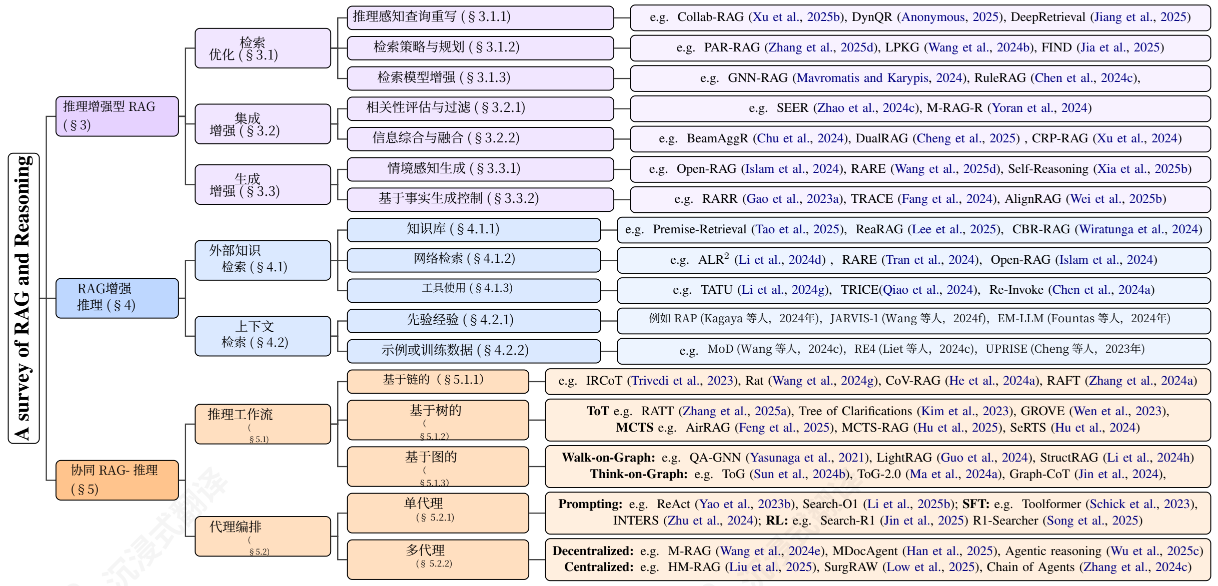

图2：RAG推理系统近期进展的分类学

3.1 检索优化

检索优化利用推理来提高结果的相关性和质量。现有方法大致分为三类：(1)推理感知查询重写，(2)检索策略与规划，(3)检索模型增强。

3.1.1 基于推理的查询重写

它将原始查询重写为更好地检索与推理相关的上下文。首先，查询分解将复杂查询分解为更简单的子查询（Xu 等人，2025b）。其次，查询重写将模糊查询改写为更清晰的查询。为了与生成器的推理需求保持一致，某些工作使用 RL 信号训练重写（Anonymous，2025；Wang 等人，2025c）。第三，查询扩展通过思维链推理（Dhuliawala 等人，2024；Li 等人，2024e；Lee 等人，2024）丰富查询的语义丰富性。

3.1.2 检索策略与规划

本节涵盖全局检索指南。高级规划使用推理模型在执行前生成完整的检索蓝图。PAR-RAG（Zhang 等人，2025d）应用思维链（CoT）进行多步规划，缓解局部最优问题。LPKG（Wang 等人，2024b）在知识图谱上微调大型语言模型（LLMs）以编码关系结构。相比之下，自适应检索决策方法通过一步预测来决定是否以及如何检索。FIND（Jia 等人，2025）和自适应 RAG

§3.3. 相比之下，如果推理动态触发新的检索，则在 §5 中讨论。

(Jeong 等人，2024)使用分类器评估查询复杂度并选择检索策略，减少不必要的调用。Marina 等人（2025）进一步添加了诸如实体流行度和问题类型等特征。

3.1.3 检索模型增强

一条工作线通过两种策略增强检索器的推理能力。第一种策略利用结构化知识：GNN-RAG（Mavromatis 和 Karypis，2024）使用 GNN 对知识图谱进行编码，实现隐式多跳推理，而 RuleRAG（Chen 等人，2024c）则附加符号规则，引导检索以实现逻辑一致性。另一种策略整合显式推理：Ji 等人（2024）将 CoT 与查询结合，以在多跳问答中提升中间知识召回率。

3.2 集成增强

集成增强使用推理来评估相关性并合并异构证据，防止无关内容干扰生成。方法分为两类：（1）相关性评估和（2）信息合成。

3.2.1 相关性评估与过滤

这些方法通过更深层次的推理来评估每个检索到的片段与用户查询的相关性。SEER（赵等，2024c）采用评估专家来选择忠实、有帮助且简洁的证据，同时丢弃不相关的内容。Yor等（2024）通过使用NLI模型过滤非蕴含性段落来提高鲁棒性，然后

fine-tuning the LLM on mixed relevant/irrelevant contexts to help it ignore residual noise.

3.2.2 Information Synthesis & Fusion

Once relevant snippets are identified, the challenge is to fuse them into a coherent evidence set. Beam-AggR (Chu et al., 2024) enumerates sub-question answer combinations and aggregates them via probabilistic reasoning. DualRAG (Cheng et al., 2025) combines reasoning-augmented querying with progressive knowledge aggregation to filter and organize retrieved information into an evolving outline. CRP-RAG (Xu et al., 2024) builds a reasoning graph to retrieve, evaluate, and aggregate knowledge at each node, dynamically selecting knowledge-sufficiency paths before generation.

3.3 Generation Enhancement

Even with retrieved context, traditional RAG may still generate unfaithful content without reasoning. Reasoning during generation addresses this issue through two main approaches: (1) context-aware synthesis and (2) grounded generation control.

3.3.1 Context-Aware Synthesis Strategies

Context-aware generation ensures outputs remain relevance while reducing noise. Selective–context utilization prunes or re-weights content based on task relevance. Open-RAG (Islam et al., 2024) uses a sparse expert mixture to dynamically select knowledge modules, while RARE (Wang et al., 2025d) adds domain knowledge to prompts to promote reliance on external context over memorization. Reasoning path generation builds explicit logical chains to enhance transparency, e.g., Ranaldi et al. (2024) generate contrasting explanations by comparing paragraph relevance step-by-step, guiding the model toward accurate conclusions. Self-Reasoning (Xia et al., 2025b) constructs structured reasoning chains through sequential evidence selection and verification.

3.3.2 Grounded Generation Control

Grounded generation control introduces verification mechanisms to ensure outputs remain anchored to retrieved evidence through reasoning. Fact verification methods use reasoning to assess factual consistency between generated content and retrieved evidence, e.g., Self-RAG (Asai et al., 2023) introduces reflection markers during decoding to trigger critical review and correction. Citation generation links generated content to source materials to enhance traceability and credibility, as in RARR (Gao et al., 2023a), which inserts citations while preserving stylistic coherence. Faithful reasoning ensures that each reasoning step adheres to retrieved evidence without introducing unverified content. TRACE (Fang et al., 2024) builds knowledge graphs to form coherent evidence chains, while AlignRAG (Wei et al., 2025b) applies criticism alignment to refine reasoning paths.

4 RAG-Enhanced Reasoning

Integrating external knowledge or in-context knowledge during reasoning can help LLMs reduce hallucinations and bridge logical gaps. External retrieval leverages structured sources like databases or web content, providing factual grounding, like IAG (Zhang et al., 2023). In-context retrieval utilizes internal contexts like prior interactions or training examples, enhancing contextual coherence, like RA-DT (Schmied et al., 2024). Both strategies collectively improve factual accuracy, interpretability, and logical consistency of reasoning processes.

4.1 External Knowledge Retrieval

External knowledge retrieval incorporates web content, database information, or external tools into reasoning, effectively filling knowledge gaps. Targeted retrieval improves factual accuracy, enabling language models to reliably address complex queries by grounding reasoning steps in verified external evidence.

4.1.1 Knowledge Base

Knowledge base (KB) typically stores arithmetic, commonsense, or logical knowledge in databases, books, or documents, with retrieval approaches varying by task. For question answering (QA) reasoning, AlignRAG (Wei et al., 2025b), MultiHop-RAG (Tang and Yang, 2024), and CRP-RAG (Xu et al., 2025a) retrieve interconnected factual entries from general KBs to enhance sequential reasoning. In specialized reasoning tasks, mathematical approaches like Premise-Retrieval (Tao et al., 2025) and ReaRAG (Lee et al., 2025) utilize formal lemmas from theorem libraries for structured deduction; legal approaches like CASEGPT (Yang, 2024) and CBR-RAG (Wiratunga et al., 2024) extract judicial precedents for analogical reasoning. For code generation tasks, CodeRAG (Li et al., 2025a) and Koziolek et al. (2024) access code snippets from repositories, ensuring syntactic correctness.

在混合相关/不相关上下文中微调LLM，以帮助其忽略残留噪声。

3.2.2 信息合成与融合

一旦识别出相关片段，挑战在于将它们融合成一个连贯的证据集。Beam-AggR（Chu等人，2024）枚举子问题答案组合，并通过概率推理进行聚合。DualRAG（Cheng等人，2025）结合推理增强查询与渐进式知识聚合，以过滤和组织检索到的信息到一个不断发展的提纲中。CRP-RAG（Xu等人，2024）构建推理图，在每个节点检索、评估和聚合知识，在生成前动态选择知识充足路径。

3.3 生成增强

即使有检索到的上下文，传统RAG仍可能在无推理的情况下生成不忠实的内容。生成过程中的推理通过两种主要方法解决这个问题：(1) 上下文感知合成和(2) 基于事实的生成控制。

3.3.1 上下文感知合成策略

上下文感知生成确保输出保持相关性，同时减少噪声。选择性上下文利用根据任务相关性对内容进行剪枝或重新加权。Open-RAG（Islam 等人，2024）使用稀疏专家混合体动态选择知识模块，而RARE（Wang 等人，2025d）向提示中添加领域知识，以促进对外部上下文的依赖而非记忆。推理路径生成构建显式的逻辑链以增强透明度，例如Ranaldi 等人（2024）通过逐步比较段落相关性生成对比性解释，引导模型得出准确结论。自推理（Xia 等人，2025b）通过顺序证据选择和验证构建结构化推理链。

3.3.2 基于证据生成控制

基于检索的生成控制引入验证机制，以确保输出通过推理与检索到的证据保持一致。事实验证方法使用推理来评估生成内容与检索到证据之间的事实一致性，例如，Self-RAG（Asai 等人，2023）在解码过程中引入反思标记以触发批判性审查和修正。引用生成将生成内容与源材料链接起来，以增强可追溯性和可信度，如在 RARR（Gao 等人，2023a）中，它在保持风格连贯性的同时插入引用。忠实的推理确保每一步推理都遵循检索到的证据，不会引入未经验证的内容。TRACE（Fang 等人，2024）构建知识图谱以形成连贯的证据链，而 AlignRAG（Wei 等人，2025b）应用批评对齐来优化推理路径。

4 RAG增强推理

在推理过程中整合外部知识或情境知识可以帮助LLM减少幻觉并弥合逻辑鸿沟。外部检索利用结构化来源如数据库或网络内容，提供事实基础，例如IAG（Zhang 等人，2023）。情境检索利用内部上下文如先前的交互或训练示例，增强上下文连贯性，例如RA-DT（Schmied 等人，2024）。这两种策略共同提高了推理过程的事实准确性、可解释性和逻辑一致性。

4.1 外部知识检索

外部知识检索将网络内容、数据库信息或外部工具融入推理，有效填补知识空白。定向检索可提高事实准确性，使语言模型能够通过基于已验证的外部证据来可靠地处理复杂查询，从而为推理步骤提供基础。

4.1.1 知识库

知识库（KB）通常在数据库、书籍或文档中存储算术、常识或逻辑知识，检索方法因任务而异。对于问答（QA）推理，AlignRAG（Wei 等人，2025b）、MultiHop-RAG（Tang 和 Yang，2024）和 CRP-RAG（Xu 等人，2025a）从通用知识库中检索相互关联的事实条目，以增强序列推理。在专业推理任务中，像前提检索（Tao 等人，2025）和 ReaRAG（Lee 等人，2025）这样的数学方法利用定理库中的正式公理进行结构化演绎；像 CASEGPT（Yang, 2024）和 CBR-RAG（Wiratunga 等人，2024）这样的法律方法提取司法先例进行类比推理。对于代码生成任务，CodeRAG（Li 等人，2025a）和 Koziolek 等人（2024）从代码库中访问代码片段，确保语法正确性。

4.1.2 Web Retrieval

Web retrieval accesses dynamic online content like web pages, news or social media. Specifically, in fact-checking tasks, approaches such as Ver-aCT Scan (Niu et al., 2024), Ragar (Khaliq et al., 2024), PACAR (Zhao et al., 2024b), and STEEL (Li et al., 2024b) verify claims step-by-step using evidence from news or social media, enhancing logical reasoning. Meanwhile, QA-based reasoning like RARE (Tran et al., 2024), RAG-Star (Jiang et al., 2024), MindSearch (Chen et al., 2024b), and OPEN-RAG (Islam et al., 2024) iteratively refine reasoning with broad web content, aligning with current trends in agentic search, which involve synthesizing complex online materials to enhance context-aware and robust reasoning. Conversely, in specialized areas like medical reasoning, approaches such as FRVA (Fan et al., 2024b) and ALR^2 (Li et al., 2024d), retrieve literature for accurate diagnostics.

4.1.3 Tool Using

Tool-using approaches leverage external resources like calculators, libraries, or APIs to enhance reasoning interactively. In QA-based reasoning, Re-Invoke (Chen et al., 2024a), AVATAR (Wu et al., 2024), ToolkenGPT (Hao et al., 2023), and Tool-LLM (Qin et al., 2023) invoke calculators or APIs (e.g., Yahoo Finance, Wikidata), improving numerical accuracy and factual precision. Within the context of scientific modeling, SCIAGENT (Ma et al., 2024b) and TRICE (Qiao et al., 2024) integrate symbolic computation tools (e.g., WolframAlpha), strengthening computational robustness. Similarly, in mathematical computation, llm-tool-use (Luo et al., 2025b) autonomously employs calculators for accurate numerical reasoning. Distinctively in code generation tasks, RAR (Dutta et al., 2024) retrieves code documentation via OSCAT libraries, ensuring syntactic accuracy and executable logic.

4.2 In-context Retrieval

In-context retrieval leverages a model's internal experiences or retrieved examples from demonstrations and training data to guide reasoning. This retrieval provides relevant exemplars, guiding models to emulate reasoning patterns and enhancing accuracy and logical coherence in novel questions.

4.2.1 Prior Experience

Prior experience refers to past interactions or successful strategies stored in a model's internal memory, with retrieval varying by task. In tasks involving planning and decision-making tasks such as robot path finding, RAHL (Sun et al., 2024a) and RA-DT (Schmied et al., 2024) leverage past decisions and reinforcement signals for sequential reasoning. For interactive reasoning tasks, JARVIS-1 (Wang et al., 2024f), RAP (Kagaya et al., 2024), and EM-LLM (Fountas et al., 2024) dynamically recall multimodal interactions and conversational histories, facilitating adaptive reasoning for personalized interactions. In the domain for **logical reasoning**, CoPS (Yang et al., 2024a) retrieves structured prior cases like medical and legal cases for robust logical reasoning in medical and legal scenarios.

4.2.2 Example or Training Data

Unlike approaches relying on prior experiences, example-based reasoning retrieves external examples from demonstrations or training data. For example, In complex text-understanding, RE4 (Li et al., 2024c) and Fei et al. (2024) utilize annotated sentence pairs to enhance relation recognition. Addressing QA-based reasoning, OpenRAG (Zhou and Chen, 2025), UPRISE (Cheng et al., 2023), MoD (Wang et al., 2024c), and Dr.ICL (Luo et al., 2023) select demonstrations closely matching queries, improving generalization. Additionally, in code generation tasks, PERC (Yoo et al., 2025) retrieves pseudocode by semantic or structural similarity from datasets like HumanEval, ensuring alignment with target code.

5 Synergized RAG-Reasoning

Many real-world problems, such as open-domain question answering (Yang et al., 2015; Chen and Yih, 2020) and scientific discovery (Lu et al., 2024; Wang et al., 2023; Baek et al., 2024; Schmidgall et al., 2025), require an iterative approach where new evidence continuously informs better reasoning and vice versa. A single retrieval step may not provide sufficient information, and a single round of reasoning may overlook key insights (Trivedi et al., 2023). By tightly integrating retrieval and reasoning in a multi-step, interactive manner, these systems can progressively refine both the search relevance of retrieved information and the reasoning-based understanding of the original query. We focus on two complementary perspectives within existing approaches: reasoning workflows, which emphasize structured, often pre-defined inference formats for multi-step reasoning; and agent orches-

4.1.2 网络检索

网络检索访问动态在线内容，如网页、新闻或社交媒体。具体来说，在事实核查任务中，像 Ver- aCT Scan（Niu 等人，2024）、Ragar（Khaliq 等人，2024）、PACAR（Zhao 等人，2024b）和 STEEL（Li 等人，2024b）这样的方法使用来自新闻或社交媒体的证据逐步验证声明，增强逻辑推理。同时，基于问答的推理，如 RARE （Tran 等人，2024）、RAG-Star（Jiang 等人，2024）、MindSearch（Chen 等人，2024b）和 OPEN-RAG（Islam 等人，2024）通过迭代使用广泛的网络内容来改进推理，这与代理搜索的当前趋势一致，代理搜索涉及综合复杂的在线材料以增强上下文感知和鲁棒推理。相反，在像医疗推理这样的专业领域，像 FRVA（Fan 等人，2024b）和 ALR^2（Li 等人，2024d）这样的方法检索文献以进行准确诊断。

4.1.3 工具使用

工具使用方法借助外部资源（如计算器、库或 API）来增强交互式推理。在基于问答的推理中，Re-Invoke（Chen 等人，2024a）、AVATAR（Wu 等人，2024）、Toolken GPT（Hao 等人，2023）和 Tool-LLM（Qin 等人，2023）调用计算器或 API（例如 Yahoo Finance、Wikidata），提高了数值准确性和事实精确性。在科学建模的背景下，SCIAGENT（Ma 等人，2024b）和 TRICE（Qiao 等人，2024）集成了符号计算工具（例如 WolframAlpha），增强了计算鲁棒性。类似地，在数学计算中，llm-tool-use（Luo 等人，2025b）自主使用计算器进行精确的数值推理。在代码生成任务中，RAR（Dutta 等人，2024）通过 OSCAT 库检索代码文档，确保了句法和可执行逻辑的准确性。

4.2 上下文检索

上下文检索利用模型的内部经验或从示例和训练数据中检索到的示例来指导推理。这种检索提供相关的示例，指导模型模仿推理模式，并在新问题中提高准确性和逻辑连贯性。

4.2.1 先验经验

先前经验指的是存储在模型内部过去的交互或成功策略

先前经验指的是模型内部存储的过去交互或成功策略，其检索方式因任务而异。在涉及规划和决策的任务中，例如机器人路径规划，RAHL（Sun 等人，2024a）和 RA-DT（Schmied 等人，2024）利用过去的决策和强化信号进行序列推理。对于交互式推理任务，JARVIS-1（Wang 等人，2024f）、RAP（Kagaya 等人，2024）和 EM-LLM（Fountas 等人，2024）动态回忆多模态交互和对话历史，促进个性化交互的自适应推理。在逻辑推理领域，CoPS（Yang 等人，2024a）检索结构化的先前案例，例如医疗和法律案例，以在医疗和法律场景中进行稳健的逻辑推理。

4.2.2 示例或训练数据

与依赖先验经验的方法不同，基于示例的推理从示例或训练数据中检索外部示例。例如，在复杂的文本理解中，RE4（Li 等人，2024c）和 Fei 等人（2024）利用标注的句子对来增强关系识别。针对基于问答的推理，OpenRAG（Zhou 和 Chen，2025）、UPRISE（Cheng 等人，2023）、MoD（Wang 等人，2024c）和 Dr.ICL（Luo 等人，2023）选择与查询紧密匹配的示例，以提高泛化能力。此外，在代码生成任务中，PERC（Yoo 等人，2025）通过语义或结构相似性从 HumanEval 等数据集中检索伪代码，确保与目标代码的一致性。

5 融合式 RAG 推理

许多现实世界问题，例如开放域问答（Yang 等人，2015年；Chen和Yih，2020年）和科学发现（Lu等人，2024年；Wang等人，2023年；Baek等人，2024年；Sch midgall等人，2025年），需要迭代方法，其中新证据不断为更好的推理提供信息，反之亦然。单次检索步骤可能无法提供足够的信息，单轮推理可能忽略关键见解（Trivedi等人，2023年）。通过紧密集成检索和推理，以多步交互方式，这些系统可以逐步完善检索信息的搜索相关性以及基于推理的原始查询理解。我们关注现有方法中的两个互补视角：推理工作流，它强调结构化、通常预定义的推理格式用于多步推理；以及代理协调，它关注代理如何与环境互动以及如何相互协调。

tration, which focus on how agents interact with environment and coordinate with each others.

5.1 Reasoning Workflow

Broadly, the reasoning workflows can be categorized as chain-based, tree-based, or graph-based, reflecting an evolution from linear reasoning chains to branching and expressive reasoning structures.

5.1.1 Chain-based

Chain-of-Thought (CoT) (Wei et al., 2022) structures the reasoning process as a linear sequence of intermediate steps. However, relying solely on the parametric knowledge of LLMs can lead to error propagation. To solve this, IRCoT (Trivedi et al., 2023) and Rat (Wang et al., 2024g) interleave retrieval operations between reasoning steps. Several recent methods further improve the robustness and rigor of this chain-based paradigm via verification and filtering. CoV-RAG (He et al., 2024a) introduces a chain-of-verification that checks and corrects each reasoning step against retrieved references. To combat noisy or irrelevant context, approaches like RAFT (Zhang et al., 2024a) fine-tune LLMs to ignore distractor documents, while Chain-of-Note (Yu et al., 2024) prompts the model to take sequential "reading notes" on retrieved documents to filter out unhelpful information.

5.1.2 Tree-based

Tree-based reasoning methods typically adopt either Tree-of-Thought (ToT) (Yao et al., 2023a) or Monte Carlo Tree Search (MCTS) (Browne et al., 2012) approaches. **ToT** extends the CoT to explicitly construct a deterministic reasoning tree and branch multiple logical pathways. Examples include RATT (Zhang et al., 2025a), which construct retrieval-augmented thought trees to simultaneously evaluate multiple reasoning trajectories. Such ToT principles avoid LLM being trapped by an early mistaken assumption and have been applied to address ambiguous questions (Kim et al., 2023), to cover different diagnostic possibilities (Yang and Huang, 2025), and to create complex stories (Wen et al., 2023). Conversely, **MCTS**-based approaches like AirRAG (Feng et al., 2025), MCTS-RAG (Hu et al., 2025), and SeRTS (Hu et al., 2024) employ probabilistic tree search, dynamically prioritizing exploration based on heuristic probabilities. To ensure retrieval and reasoning quality, AirRAG (Feng et al., 2025) incorporates self-consistency checks, and MCTS-RAG (Hu et al., 2025) integrates adaptive MCTS retrieval to refine evidence and reduce hallucinations.

5.1.3 Graph-based

Walk-on-Graph methods mainly rely on graph learning techniques for the retrieval and reasoning. For example, PullNet (Sun et al., 2019), QA-GNN (Yasunaga et al., 2021), and GreaseLM (Zhang et al., 2022b) directly integrate graph neural networks (GNNs) to iteratively aggregate information from neighbor nodes, excelling at modeling the intricate relationships inherent in graph-structured data. Methods such as SR (Zhang et al., 2022a), LightRAG (Guo et al., 2024), and StructRAG (Li et al., 2024h) employ lightweight graph techniques such as vector indexing and PageRank to efficiently retrieve and reason in multi-hop context, providing the LLM with high-quality, structured content tailored for the queries. In contrast, **Think-on-Graph** methods integrate graph structures directly into the LLM reasoning loop, enabling dynamic and iterative retrieval and reasoning processes guided by the LLMs themselves. In the Think-on-Graph (ToG) framework (Sun et al., 2024b; Ma et al., 2024a), the LLM uses the KG as a "reasoning playground": at each step, it decides which connected entity or relation to explore next, gradually building a path that leads to the answer. While Graph-CoT (Jin et al., 2024) introduces a three-stage iterative loop (reasoning, graph interaction, and execution), KGP (Wang et al., 2024d) prioritize first constructing a document-level KG, both enabling LLM-driven graph traversal agent to navigate passages in each step with globally coherent context. GraphReader (Li et al., 2024f) further refines this paradigm by coupling LLM reasoning with explicit subgraph retrieval and evidence anchoring at each step

5.2 Agent Orchestration

According to agent architectures (Luo et al., 2025a), we organize existing work into single-agent and multi-agent. Particularly, we have attached recent advances in agentic deep research and implementations in Appendix B.

5.2.1 Single-Agent

Single agentic system interweaves knowledge retrieval (search) into an LLM's reasoning loop, enabling dynamic information lookup at each step of problem solving and incentivizing it to actively seek out relevant evidence when needed.

代理协调，它关注代理如何与环境互动以及如何相互协调。

5.1 推理工作流

总体而言，推理工作流可以分为基于链、基于树或基于图，反映了从线性推理链到分支化和表达性推理结构的演变。

5.1.1 基于链

思维链 (CoT) (Wei 等人, 2022) 将推理过程构建为一系列中间步骤的线性序列。然而，仅依赖大型语言模型的参数化知识会导致错误传播。为解决这个问题，IRCoT (Trivedi 等人, 2023) 和 Rat (Wang 等人, 2024g) 在推理步骤之间插入检索操作。一些最近的方法通过验证和过滤进一步增强了基于链范式的鲁棒性和严谨性。CoV-RAG (He 等人, 2024a) 引入了一个验证链，用于检查和纠正每个推理步骤相对于检索到的参考。为对抗噪声或不相关的上下文，像 RAFT (Zhang 等人, 2024a) 这样的方法微调大型语言模型以忽略干扰文档，而思维笔记 (Yu 等人, 2024) 则提示模型对检索到的文档进行顺序“阅读笔记”，以过滤掉无用的信息。

5.1.2 基于树

基于树的推理方法通常采用要么树形思维链（ToT）（Yao 等人，2023a）要么蒙特卡洛树搜索（MCTS）（Browne 等人，2012）的方法。ToT 扩展了 CoT，以显式构建确定性推理树并分支多个逻辑路径。例如 RATT（Zhang 等人，2025a），它构建检索增强的思维树以同时评估多个推理轨迹。这种 ToT 原则避免了 LLM 被早期的错误假设困住，并已被应用于解决模糊问题（Kim 等人，2023）、涵盖不同的诊断可能性（Yang 和 Huang，2025）以及创建复杂故事（Wen 等人，2023）。相反，AirRAG（Feng 等人，2025）、MCTS-RAG（Hu 等人，2025）和 SeRTS（Hu 等人，2024）等基于 MCTS 的方法采用概率树搜索，根据启发式概率动态优先考虑探索。为确保检索和推理质量，AirRAG（Feng 等人，2025）包含了自我一致性检查，和MCTS-RAG(Hu et al.,2025) 集成了自适应MCTS检索来优化证据并减少幻觉。

5.1.3 基于图

Walk-on-Graph 方法主要依赖图学习技术进行检索和推理。例如，PullNet（Sun 等人，2019 年）、QA-GNN（Yasunaga 等人，2021 年）和 GreaseLM（Zhang 等人，2022b）直接将图神经网络（GNN）集成起来，以迭代地聚合来自相邻节点的信息，擅长对图结构数据中固有的复杂关系进行建模。SR（Zhang 等人，2022a）、LightRAG（Guo 等人，2024 年）和 StructRAG（Li 等人，2024h）等方法采用轻量级的图技术，如向量索引和 PageRank，以高效地在多跳上下文中进行检索和推理，为 LLM 提供高质量、结构化的内容，以适应查询。相比之下，Think-on-Graph 方法将图结构直接集成到 LLM 推理循环中，使 LLM 能够动态和迭代地进行检索和推理过程。在 Think-on-Graph（ToG）框架（Sun 等人，2024b；Ma 等人，2024a）中，LLM 将 KG 作为“推理游乐场”：在每一步，它决定要探索下一个连接的实体或关系，逐渐构建一条通往答案的路径。虽然 Graph-CoT（Jin 等人，2024 年）引入了一个三阶段的迭代循环（推理、图交互和执行），但 KGP（Wang 等人，2024d）优先考虑首先构建文档级 KG，两者都使 LLM 驱动的图遍历代理能够在每一步中导航具有全局一致性的上下文的路径。GraphReader（Li 等人，2024f）通过将 LLM 推理与每一步的显式子图检索和证据锚定相结合，进一步完善了这一范式

5.2 代理编排

根据代理架构（Luo 等人，2025a），我们将现有工作组织为单代理和多代理。特别是，我们在附录 B 中附上了代理深度研究的最新进展和实现。

5.2.1 单代理

单一智能体系统将知识检索（搜索）交织到LLM的推理循环中，使其能够在问题解决的每一步进行动态信息查询，并激励其在需要时主动寻找相关证据。

The ReAct (Yao et al., 2023b) paradigm and its derivatives (Li et al., 2025b; Alzubi et al., 2025) have pioneered this **prompting** strategy by guiding LLMs to explicitly alternate between reasoning steps and external tool interactions, such as database searches. Different from ReAct that separates reasoning and action, with explicit commands like "search" triggering external retrieval, methods such as Self-Ask (Press et al., 2023) and IR-CoT (Trivedi et al., 2023) prompt the model to recursively formulate and answer sub-questions, enabling interleaved retrieval within the Chain-of-Thought (step-by-step retrieval and reasoning). Involving self-reflection strategies, DeepRAG (Guan et al., 2025) and Self-RAG (Asai et al., 2024) empower LLMs to introspectively assess their knowledge limitations and retrieve only when necessary.

Rather than relying solely on prompting or static retrievers, Toolformer (Schick et al., 2023) and INTERS (Zhu et al., 2024) represent a complementary approach via **supervised fine-tuning** (SFT) LLMs on instruction-based or synthetic datasets that interleave search and reasoning. Synthetic data generation (Schick et al., 2023; Mao et al., 2024; Zhang et al., 2024a) aims to create large-scale, diverse, and task-specific datasets for search without the need for extensive human annotation. In contrast, instruction-based data reformulation (Zhu et al., 2024; Wang et al., 2024a; Lin et al., 2023; Nguyen et al., 2024) repurposes existing datasets into instructional formats to fine-tune models for improved generalization and alignment with human-like reasoning. INTERS (Zhu et al., 2024) exemplifies this approach by introducing a SFT dataset encompassing 20 tasks, derived from 43 distinct datasets with manually written templates.

Reinforcement learning (RL)-incentivized approaches provides a mechanism to optimize answer quality via reward signals on incentivizing agents' behaviors – what to search, how to integrate retrieved evidence, and when to stop, aiming at complex knowledge-intensive tasks (or "deep research" questions). Notable efforts like WebGPT (Nakano et al., 2021) and RAG-RL (Huang et al., 2025a) focus on improving reasoning fidelity by rewarding outputs based on factual correctness or human preference. More recent contributions operate directly in dynamic environments (e.g., live web search, local search tools), training agents to explore, reflect, and self-correct in noisy real-world conditions. For example, Search-R1 (Jin et al., 2025) learns to generate <search> token during reasoning and concurrently R1-Searcher (Song et al., 2025) builds on RL-driven search demonstrating strong generalization across domains. Deep-Researche (Zheng et al., 2025) make step further by introducing the first end-to-end RL-trained research agent that interacts with the open web. These settings showcase emergent capabilities, like decomposition, iterative verification, and retrieval planning, that supervised methods often hard to instill. Moreover, ReSearch (Chen et al., 2025b) and ReARTeR (Sun et al., 2025c) tackle a deeper challenge: not just producing correct answers, but aligning reasoning steps with both factuality and interpretability.

5.2.2 Multi-Agent

The exploration of multi-agent collaboration within RAG and reasoning has led to diverse orchestrations: centralized architectures (harness collective intelligence from workers-manager paradigm) and decentralized architectures (leverage complementary capabilities from role-specialized agents).

Decentralized architectures deploy multiple agents to collaboratively perform retrieval, reasoning, and knowledge integration, aiming to broaden coverage of relevant information and fully exploit the heterogeneous strengths of specialized agents. Wang et al. (2024e) and Salve et al. (2024) introduce multi-agent systems where each agent retrieves from a partitioned database or a specific data source (relational databases, NoSQL document stores, etc.). Beyond retrieval, Collab-RAG (Xu et al., 2025b) and RAG-KG-IL (Yu and McQuade, 2025) integrate different model capacities and assign them different roles in reasoning and knowledge integration. This philosophy extends to multi-modal settings as in MDocAgent (Han et al., 2025), which employs a team of text and image agents to process and reason the document-based QA. A general formulation is seen in Agentic reasoning (Wu et al., 2025c), which unites tool-using agents for search, computation, and structured reasoning, orchestrated to solve complex analytical tasks.

Centralized architectures structure agents in hierarchical centralized patterns, supporting efficient task decomposition and progressive refinement. HM-RAG (Liu et al., 2025) and SurgRAW (Low et al., 2025) both employ decomposer-retriever-decider architectures, where different agent roles isolate subproblems such as multimodal processing or surgical decision-making. Wu et al. (2025a) and Iannelli et al. (2024) emphasize dynamic routing and system reconfiguration, respectively—enabling

ReAct（Yao等人，2023b）范式及其衍生方法（Li等人，2025b；Alzubi等人，2025）通过指导LLM在推理步骤和外部工具交互（如数据库搜索）之间明确切换，开创了这种提示策略。与将推理和行动分离的ReAct不同，ReAct使用“搜索”等显式命令触发外部检索，Self-Ask（Press等人，2023）和IR-CoT（Trivedi等人，2023）等方法则提示模型递归地构建和回答子问题，在思维链（逐步检索和推理）中实现交错检索。涉及自我反思策略的DeepRAG（Guan等人，2025）和Self-RAG（Asai等人，2024）则赋予LLM直觉评估其知识局限性的能力，仅在必要时进行检索。

与其完全依赖提示或静态检索器，Toolformer（Schick等人，2023）和inters（Zhu等人，2024）则通过在基于指令或合成数据集上进行监督微调（SFT）的大型语言模型（LLM）代表了一种补充方法，这些数据集交错包含搜索和推理。合成数据生成（Schick等人，2023；Mao等人，2024；Zhang等人，2024a）旨在创建大规模、多样化且特定于任务的数据集，用于搜索，而无需大量人工标注。相比之下，基于指令的数据重述（Zhu等人，2024；Wang等人，2024a；Lin等人，2023；Nguyen等人，2024）则将现有数据集改造成指令格式，以微调模型，从而提高泛化能力并与人类推理保持一致。inters（Zhu等人，2024）通过引入一个包含20项任务的SFT数据集，该数据集源自43个不同的数据集，并使用手动编写的模板，展示了这种方法。

强化学习（RL）激励型方法提供了一种机制，通过奖励信号优化答案质量，激励代理的行为——要搜索什么、如何整合检索到的证据、何时停止——旨在解决复杂知识密集型任务（或“深度研究”问题）。像WebGPT（Nakano等人，2021年）和RAG-RL（Huang等人，2025a）这样的显著努力专注于通过基于事实正确性或人类偏好的输出奖励来提高推理保真度。更近期的贡献直接在动态环境中（例如，实时网络搜索、本地搜索工具）运行，训练代理在嘈杂的真实世界条件下进行探索、反思和自我纠正。例如，Search-R1（Jin等人，2025年）学习在推理过程中生成 <搜索>标记，目前R1-Searcher（Song等人，2025年）基于RL驱动的搜索，展示了跨领域的强大泛化能力。Deep-Researche（Zheng等人，2025年）更进一步，引入了第一个端到端RL训练的研究代理，该代理能与开放网络交互。这些设置展示了涌现能力，如分解、迭代验证和检索规划，这些能力受控方法通常难以植入。此外，ReSearch（Chen等人，2025b）和ReARTeR（Sun等人，2025c）应对了一个更深层次的挑战：不仅生成正确答案，还要使推理步骤与事实性和可解释性保持一致。

5.2.2 多智能体

对 RAG 和推理中多智能体协作的探索已催生出多样化的编排方式：集中式架构（从工人-管理者范式中 harness 集体智能）和分布式架构（利用角色专业化的智能体提供的互补能力）。

分布式架构部署多个智能体协同执行检索、推理和知识融合，旨在拓宽相关信息的覆盖范围并充分利用专业智能体的异构优势。Wang 等人 (2024e) 和 Salve 等人 (2024) 介绍了多智能体系统，其中每个智能体从分区数据库或特定数据源（关系数据库、NoSQL 文档存储等）中检索信息。除了检索之外，Collab-RAG (Xu 等人，2025b) 和 RAG-KG-IL (Yu 和 McQuade，2025) 集成不同的模型能力，并在推理和知识融合中分配不同的角色。这种理念扩展到 MDocAgent (Han 等人，2025) 的多模态设置中，后者采用一组文本和图像智能体来处理和推理基于文档的问答。Agentic reasoning (Wu 等人，2025c) 中可见的通用公式将使用工具的智能体联合起来进行搜索、计算和结构化推理，编排以解决复杂的分析任务。

集中式架构将智能体结构化为分层集中模式，支持高效的任务分解和渐进式细化。HM-RAG（刘等人，2025）和SurgRAW（Low等人，2025）均采用分解器-检索器-决策器架构，其中不同的智能体角色隔离了多模态处理或手术决策等子问题。Wu等人（2025a）和Iannelli等人（2024）分别强调动态路由和系统重构——实现基于任务相关性或资源约束的智能体选择。

intelligent agent selection based on task relevance or resource constraints. Chain of Agents (Zhang et al., 2024c) and the cooperative multi-agent control framework for on-ramp merging (Zhang et al., 2025c) illustrate hierarchical agent designs where layered processing enables long-context summarization or policy refinement. Collectively, these works demonstrate how centralized control and hierarchical pipelining foster efficiency and adaptability in multi-agent RAG-reasoning systems.

6 Benchmarks and Datasets

Benchmarks and datasets for simultaneously evaluating knowledge (RAG) and reasoning capability cover a wide range of complexities, from basic fact retrieval to intricate multi-step reasoning in general or specific domains. We categorize notable benchmarks in several tasks and list them in Table 1 and highlight their details and properties. These representative tasks include Web browsing, such as BrowseComp (Wei et al., 2025a), single-hop QA, such as TriviaQA (Joshi et al., 2017), multi-hop QA, such as HotpotQA (Yang et al., 2018), multiple-choice QA, such as MMLU-Pro (Wang et al., 2025b), mathematics, such as MATH (Hendrycks et al., 2021), and code-centric evaluations from LiveCodeBench (Jain et al., 2024). More tasks can refer to Appendix A and Table 2.

7 Future Work

Future research directions for Synergized RAG-Reasoning systems center around enhancing both reasoning and retrieval capabilities to meet real-world demands for accuracy, efficiency, trust, and user alignment. We outline several key challenges and opportunities below.

- **Reasoning Efficiency.** Despite their advantages in complex reasoning, Synergized RAG-Reasoning systems can suffer significant latency due to iterative retrieval and multi-step reasoning loops (Sui et al., 2025). For instance, executing a single deep research query can take over 10 minutes in practical settings. This issue is especially pronounced in chain-based workflows discussed in Section 5. Future research should explore reasoning efficiency through latent reasoning approaches and strategic control over reasoning depth via thought distillation and length-penalty (Xia et al., 2025a; Zhang et al., 2025b). Beyond reasoning itself, emerging directions in models compression like quantization, pruning, and knowledge distillation is worth to explore for efficient small RAG-reasoning systems.

- **Retrieval Efficiency.** On the retrieval side, efficiency demands budget-aware query planning and memory-aware mechanisms that cache prior evidence or belief states to reduce redundant access (Zhao et al., 2024a). Additionally, adaptive retrieval control, learning when and how much to retrieve based on uncertainty signals can reduce wasteful operations. These technical paths push the system beyond static RAG, toward dynamic self-regulation of efficient retreival behaviors under real-world constraints.

- **Human-Agent Collaboration.** Many applications of RAG-Reasoning, such as literature reviews or interactive programming, are inherently personalized and cannot assume users know precisely what to ask or how to process retrieved results (Sun et al., 2025b). Corresponding to Section 5.2, humans can act as advanced agents, providing nuanced feedback to steer reasoning processes. Future systems should develop methods for modeling user intent under uncertainty (Zhang et al., 2025e; Yang et al., 2025), building interactive interfaces for iterative clarification, and designing agents that adapt reasoning strategies based on user expertise and preferences (Zhang et al., 2025g). This human-in-the-loop approach (Zou et al., 2025) is essential for creating robust and user-aligned RAG-Reasoning systems in open-ended domains.

- **Agentic Structures and Capabilities.** A key feature of Synergized RAG-Reasoning is its agentic architecture, where the system autonomously decides the roles of different agents and which tools or retrieval strategies to invoke during inference stages (Luo et al., 2025a; Bei et al., 2025). To fully exploit this potential, future research should focus on developing agent frameworks capable of dynamic tool selection, retrieval planning, and adaptive orchestration across reasoning workflows. Such capabilities enable flexible, context-aware problem solving and are critical for handling diverse, complex tasks (Schneider, 2025).

- **Multimodal Retrieval.** As also shown in our benchmark analysis, most existing Synergized RAG-Reasoning systems remain confined to text-only tasks. However, real-world applications increasingly require the ability to retrieve and integrate multimodal content (Liang et al., 2024).

智能体链（Zhang等人，2024c）和用于匝道合并的协同多智能体控制框架（Zhang等人，2025c）展示了分层智能体设计，其中分层处理支持长上下文摘要或策略细化。总体而言，这些工作表明集中控制和分层流水线如何提升多智能体RAG推理系统的效率和适应性。

6 基准测试和数据集

用于同时评估知识（RAG）和推理能力的基准测试和数据集涵盖了从基本事实检索到通用或特定领域中的复杂多步推理的广泛范围。我们将著名的基准测试按任务分类，并在表1中列出它们，并突出其详细信息和属性。这些代表性任务包括网络浏览，如BrowseComp（Wei等人，2025a）、单跳问答，如TriviaQA（Joshi等人，2017）、多跳问答，如HotpotQA（Yang等人，2018）、选择题问答，如MMLU-Pro（Wang等人，2025b）、数学，如MATH（Hendrycks等人，2021），以及来自LiveCodeBench的以代码为中心的评估（Jain等人，2024）。更多任务可参考附录A和表2。

7 未来工作

Synergized RAG-推理系统的未来研究方向主要集中在提升推理和检索能力，以满足实际对准确性、效率、信任和用户一致性的需求。我们概述了以下几个关键挑战和机遇。

- 推理效率。尽管在复杂推理方面具有优势，Synergized RAG-推理系统由于迭代检索和多步骤推理循环（Sui 等人，2025），可能会出现显著的延迟。例如，在实际环境中执行单个深度研究查询可能需要超过10分钟。这个问题在第五章讨论的链式工作流程中尤为突出。未来研究应通过潜在推理方法探索推理效率，并通过思维蒸馏和长度惩罚（Xia 等人，2025a；Zhang 等人，2025b）对推理深度进行策略性控制。除了推理本身，模型压缩领域的新兴方向，如量化，剪枝和知识蒸馏对于高效的 smallRAG 推理系统来说是值得探索的。

- 检索效率。在检索方面，效率要求预算感知查询规划和内存感知机制，这些机制缓存先前的证据或信念状态以减少冗余访问（Zhao 等人，2024a）。此外，自适应检索控制，根据不确定性信号学习何时以及检索多少，可以减少浪费的操作。这些技术路径将系统推向动态自我调节的效率检索行为，使其在现实世界约束下运作。

- 人机协作。许多 RAG-推理的应用，如文献综述或交互式编程，本质上都是个性化的，无法假设用户确切知道该问什么或如何处理检索到的结果（Sun 等人，2025b）。对应于第 5.2 节，人类可以作为高级代理，提供细致的反馈来引导推理过程。未来系统应开发在不确定性下模拟用户意图的方法（Zhang 等人，2025e；Yang 等人，2025），构建用于迭代澄清的交互界面，并设计能够根据用户经验和偏好调整推理策略的代理（Zhang 等人，2025g）。这种人在回路的方法（Zou 等人，2025）对于在开放式领域中创建稳健且与用户一致的 RAG-推理系统至关重要。

- 代理结构和能力。Synergized RAG-推理的一个关键特征是其代理架构，系统自主决定不同代理的角色以及在推理阶段调用哪些工具或检索策略（Luo 等人，2025a；Bei 等人，2025）。为了充分挖掘这一潜力，未来研究应专注于开发能够动态选择工具、检索规划以及在推理工作流中进行自适应协调的代理框架。这些能力支持灵活、上下文感知的问题解决，对于处理多样化、复杂的任务至关重要（Schneider，2025）。

- 多模态检索。正如我们的基准分析也所示，大多数现有的协同 RAG 推理系统仍然局限于仅文本的任务。然而，现实世界的应用越来越多地要求具备检索和整合多模态内容的能力（Liang 等人，2024）。

Task	*Dataset*	Domain	Knowledge Source	Knowledge Type	Reasoning	Size	Input	Output
Web Browsing	*BrowseComp (Wei et al., 2025a)*	General	Human, Internet	Commonsense, Logical	Deductive	1,266	Question/Text	Natural Language
	GAIA (Mialon et al., 2023)	General	Internet, TooL	Commonsense, Logical	Deductive	466	Question/Text, Image/File/Code	Natural Language
	WebWalkerQA (Wu et al., 2025b)	General	Human, LLM	Commonsense, Logical	Deductive	680	Question/Text	Natural Language
Single-hop QA	*TriviaQA (Joshi et al., 2017)*	General	Internet	Commonsense, Logical	Deductive	650,000+	Question/Text	Natural Language
	NQ (Kwiatkowski et al., 2019)	General	Internet	Commonsense, Logical	Deductive	307,373	Question/Text	Natural Language
Multi-hop QA	*2WikiMultiHopQA (Ho et al., 2020)*	General	Internet	Commonsense, Logical	Deductive	192,606	Question/Text	Natural Language
	HotpotQA (Yang et al., 2018)	General	Internet	Commonsense	Deductive	113,000	Question/Text	Natural Language
	MuSiQue (Trivedi et al., 2022)	General	Previous Resource, Internet	Commonsense, Logical	Deductive	25,000	Question/Text	Natural Language
Multi-choice QA	*QuALITY (Pang et al., 2022)*	Narrative	Books	Commonsense, Logical	Deductive, Abductive	6,737	Question/Text, Options	Options
	MMLU-Pro (Wang et al., 2025b)	Science	Previous Resource, Internet	Arithmetic, Commonsense, Logical	Deductive, Inductive	12,032	Question/Text, Options	Natural Langue, Number, Options
Math	*MATH (Hendrycks et al., 2021)*	Math	Exam	Arithmetic, Logic	Deductive	12,500	Question/Text, Figure, Equation	Natural Langue, Number
	AQuA (Ling et al., 2017)	Math	Exam, Internet, Previous Resource	Arithmetic, Logic	Deductive	100,000	Question/Text, Options, Equation	Natural Langue, Options
Code	*Refactoring Oracle (Tsantalis et al., 2020)*	Software	Internet, Human	Logical	Deductive	7,226	Code, Instruction	Code
	LiveCodeBench (Jain et al., 2024)	Contest	Internet	Logical	Deductive, Abductive	500+	Question/Text, Code, Instruction	Code, Test Output

Table 1: Overview of representative knowledge and reasoning intensive benchmarks by task category.

Future research should move beyond the traditional vision-text paradigm to achieve genuine multimodality. This advancement necessitates strengthening foundational abilities of MLLMs, including grounding and cross-modal reasoning (Liang et al., 2024). Additionally, enhancing the agentic capabilities of these models through hybrid-modal chain-of-thought reasoning is crucial, enabling interaction with the real world via multimodal search tools (Wang et al., 2025a). Concurrently, developing unified multimodal retrievers that can jointly embed images, tables, text, and heterogeneous documents is essential.

• **Retrieval Trustworthiness.** Synergized RAG-Reasoning systems remain vulnerable to adversarial attacks through poisoned or misleading external knowledge sources. Ensuring the trustworthiness of retrieved content is therefore crucial for maintaining fully reliable downstream reasoning (Huang et al., 2024). Techniques like watermarking and digital fingerprinting have been employed to enhance system traceability. However, there's a pressing need to develop more dynamic and adaptive methods that can keep pace with the evolving landscape of LLMs, emerging attack techniques, and shifting model contexts (Liu et al., 2024). Existing studies have also individually explored uncertainty quantification and robust generation to bolster system reliability (Shorinwa et al., 2025). Future research should aim to integrate these approaches, as their combination can mutually reinforce system robustness and trustworthiness. Moreover, future efforts should also focus on extending current benchmarks to encompass multi-dimensional trust metrics beyond mere accuracy.

8 Conclusion

This survey charts the rapid convergence of retrieval and reasoning in LLMs. We reviewed three evolutionary stages: (1) Reasoning-Enhanced RAG, which uses multi-step reasoning to refine each stage of RAG; (2) RAG-Enhanced Reasoning, which leverages retrieved knowledge to bridge factual gaps during long CoT; and (3) Synergized RAG-Reasoning systems, where single- or multi-agents iteratively refine both search and reasoning, exemplified by recent "Deep Research" platforms. Collectively, these lines demonstrate that tight retrieval–reasoning coupling improves factual grounding, logical coherence, and adaptability beyond one-way enhancement. Looking forward, we identify research avenues toward synergized RAG-Reasoning systems that are more effective, multimodally-adaptive, trustworthy, and human-centric.

Limitations

While this survey synthesizes over 200 research papers across RAG and reasoning with large language models, its scope favors breadth over depth. In striving to provide a unified and comprehensive taxonomy, we may not delve deeply into the technical nuances or implementation details of individual methods-especially within specialized subfields of either RAG (e.g., sparse vs. dense retrieval, memory-augmented retrievers) or reasoning (e.g., formal logic solvers, symbolic methods, or long-context reasoning). Moreover, our cate-

Task	数据集	领域	知识来源	知识类型	推理	Size	输入	输出
网络浏览	浏览组件 *(Wei 等人, 2025a)*	通用	人类, 互联网	常识, 逻辑	演绎式	1,266	问题/文本	自然语言
	GAIA (Mialon 等人，2023)	通用	互联网，工具	常识，逻辑	演绎	466	问题/文本，图像/文件/代码	自然语言
	WebWalkerQA (吴等人，2025b)	通用	人类，大语言模型	常识，逻辑	演绎	680	问题/文本	自然语言
单跳问答	*TriviaQA (Joshi 等人，2017)*	通用	互联网	常识、逻辑	演绎式	650,000+	问题/文本	自然语言
	NQ（Kwiatkowski 等人，2019年）	通用	互联网	常识，逻辑	演绎	307,373	问题/文本	自然语言
多跳问答	*2WikiMultiHopQA (Ho 等人，2020)*	通用	互联网	常识、逻辑	演绎	192,606	问题/文本	自然语言
	HotpotQA (Yang 等人，2018)	通用	互联网	常识	演绎	113,000	问题/文本	自然语言
	MuSiQue (Trivedi 等人，2022)	通用	上一个资源，互联网	常识，逻辑	演绎	25,000	问题/文本	自然语言
多选题问答	*QuALITY（Pang等人，2022年）*	叙事	书籍	常识，逻辑	演绎，溯因	6,737	问题/文本，选项	选项
	MMLU-Pro (王等人，2025b)	科学	先前资源，互联网	算术，常识，逻辑	演绎，归纳	12,032	问题/文本，选项	自然语言，数量，选项
Math	数学 *(Hendrycks 等人，2021)*	Math	Exam	算术，逻辑	演绎	12,500	问题/文本，图表，公式	自然语言，数字
	AQuA（Ling 等人，2017年）	Math	考试，互联网，上一个资源	算术、逻辑	演绎	100,000	问题/文本，选项、方程	自然语言，选项
Code	重构神谕（*Tsantalis* 等人，2020）	软件	互联网，人类	逻辑	演绎	7,226	代码，指令	Code
	LiveCodeBench (Jain 等人，2024)	竞赛	互联网	逻辑	演绎，溯因	500+	问题/文本，代码，指令	代码，测试输出

表1：按任务类别划分的代表性知识和推理密集型基准概述。

未来的研究应该超越传统的视觉-文本范式，以实现真正的多模态。这一进步需要加强 MLLM 的基础能力，包括接地和跨模态推理（Liang 等人，2024 年）。此外，通过混合模态思维链推理来增强这些模型的能动性至关重要，能够通过多模态搜索工具与现实世界交互（Wang 等人，2025a 年）。同时，开发能够联合嵌入图像、表格、文本和异构文档的统一多模态检索器也是必不可少的。

• 检索可信度。协同式 RAG- 推理系统仍容易受到通过被污染或误导的外部知识源发起的对抗性攻击。因此，确保检索内容可信度对于维持完全可靠的下游推理至关重要（Huang 等人，2024）。水印和数字指纹等技术已被用于增强系统可追溯性。然而，迫切需要开发更动态和自适应的方法，以跟上 LLM 发展、新兴攻击技术和模型环境变化的步伐（Liu 等人，2024）。现有研究也分别探索了不确定性量化和鲁棒生成，以增强系统可靠性（Shorinwa 等人，2025）。未来研究应致力于整合这些方法，因为它们的结合可以相互增强系统鲁棒性和可信度。此外，未来工作还应关注扩展当前基准，以涵盖超越单纯准确性的多维可信度指标。

8 结论

本调查描绘了大型语言模型中检索和推理的快速融合。我们回顾了三个进化阶段：(1) 推理增强型RAG，它使用多步推理来优化RAG的每个阶段；(2) RAG增强型推理，它利用检索到的知识在长CoT过程中弥补事实差距；(3) 协同型RAG-推理系统，其中单代理或多代理迭代优化搜索和推理，以最近的"深度研究"平台为例。总体而言，这些路线表明紧密的检索-推理耦合在单向增强之外，提高了事实基础性、逻辑一致性和适应性。展望未来，我们确定了协同型RAG-推理系统的研究方向，这些系统更有效、多模态适应性强、值得信赖且以人为本。

局限性

尽管这项调查综合了超过200篇关于RAG和大型语言模型推理的研究论文，但其范围更侧重广度而非深度。在努力提供一个统一且全面的分类法时，我们可能不会深入探讨各个方法的技术细节或实现细节——尤其是在RAG（例如，稀疏与密集检索、记忆增强检索器）或推理（例如，形式逻辑求解器、符号方法或长上下文推理）的专门子领域内。此外，我们的分类框架（推理增强RAG、RAG增强推理以及协同RAG和推理）跨越了多种不同的方法论。虽然这有助于从高层次理解设计模式，但它可能会掩盖每种方法类别独特的细微权衡、假设和局限性。

gorization framework (reasoning-enhanced RAG, RAG-enhanced reasoning, and synergized RAG and reasoning) abstracts across diverse methodologies. While this facilitates a high-level understanding of design patterns, it may obscure the finer-grained trade-offs, assumptions, and limitations unique to each class of approach.

References

Vaibhav Adlakha, Shehzaad Dhuliawala, Kaheer Suleman, Harm de Vries, and Siva Reddy. 2022. Topiocqa: Open-domain conversational question answering with topic switching. *Transactions of the Association for Computational Linguistics*, 10:468–483.

Firoj Alam, Ferda Ofli, and Muhammad Imran. 2018. Crisismmd: Multimodal twitter datasets from natural disasters. In *Proceedings of the international AAAI conference on web and social media*, volume 12.

Salaheddin Alzubi, Creston Brooks, Purva Chiniya, Edoardo Contente, Chiara von Gerlach, Lucas Irwin, Yihan Jiang, Arda Kaz, Windsor Nguyen, Sewoong Oh, et al. 2025. Open deep search: Democratizing search with open-source reasoning agents. *arXiv preprint arXiv:2503.20201*.

Anonymous. 2025. DynQR: Dynamic uncertainty-guided query rewriting for effective retrieval-augmented generation. In *Submitted to ACL Rolling Review - December 2024*. Under review.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Self-reflective retrieval augmented generation. In *NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following*.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *The Twelfth International Conference on Learning Representations*.

Jinheon Baek, Sujay Kumar Jauhar, Silviu Cucerzan, and Sung Ju Hwang. 2024. Researchagent: Iterative research idea generation over scientific literature with large language models. *arXiv preprint arXiv:2404.07738*.

Yuanchen Bei, Weizhi Zhang, Siwen Wang, Weizhi Chen, Sheng Zhou, Hao Chen, Yong Li, Jiajun Bu, Shirui Pan, Yizhou Yu, et al. 2025. Graphs meet ai agents: Taxonomy, progress, and future opportunities. *arXiv preprint arXiv:2506.18019*.

Cameron B Browne, Edward Powley, Daniel Whitehouse, Simon M Lucas, Peter I Cowling, Philipp Rohlfshagen, Stephen Tavener, Diego Perez, Spyridon Samothrakis, and Simon Colton. 2012. A survey of monte carlo tree search methods. *IEEE Transactions on Computational Intelligence and AI in games*, 4(1):1–43.

Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, et al. 2024. A survey on evaluation of large language models. *ACM transactions on intelligent systems and technology*, 15(3):1–45.

Danqi Chen and Wen-tau Yih. 2020. Open-domain question answering. In *Proceedings of the 58th annual meeting of the association for computational linguistics: tutorial abstracts*, pages 34–37.

Mingyang Chen, Tianpeng Li, Haoze Sun, Yijie Zhou, Chenzheng Zhu, Haofen Wang, Jeff Z Pan, Wen Zhang, Huajun Chen, Fan Yang, et al. 2025a. Research: Learning to reason with search for llms via reinforcement learning. *arXiv preprint arXiv:2503.19470*.

Mingyang Chen, Tianpeng Li, Haoze Sun, Yijie Zhou, Chenzheng Zhu, Fan Yang, Zenan Zhou, Weipeng Chen, Haofen Wang, Jeff Z Pan, et al. 2025b. Learning to reason with search for llms via reinforcement learning. *arXiv preprint arXiv:2503.19470*.

Qiguang Chen, Libo Qin, Jinhao Liu, Dengyun Peng, Jiannan Guan, Peng Wang, Mengkang Hu, Yuhang Zhou, Te Gao, and Wanxiang Che. 2025c. Towards reasoning era: A survey of long chain-of-thought for reasoning large language models. *arXiv preprint arXiv:2503.09567*.

Yanfei Chen, Jinsung Yoon, Devendra Sachan, Qingze Wang, Vincent Cohen-Addad, Mohammadhossein Bateni, Chen-Yu Lee, and Tomas Pfister. 2024a. Re-invoke: Tool invocation rewriting for zero-shot tool retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4705–4726.

Zehui Chen, Kuikun Liu, Qiuchen Wang, Jiangning Liu, Wenwei Zhang, Kai Chen, and Feng Zhao. 2024b. Mindsearch: Mimicking human minds elicits deep ai searcher. *arXiv preprint arXiv:2407.20183*.

Zhongwu Chen, Chengjin Xu, Dingmin Wang, Zhen Huang, Yong Dou, Xuhui Jiang, and Jian Guo. 2024c. Rulerag: Rule-guided retrieval-augmented generation with language models for question answering. *arXiv preprint arXiv:2410.22353*.

Daixuan Cheng, Shaohan Huang, Junyu Bi, Yuefeng Zhan, Jianfeng Liu, Yujing Wang, Hao Sun, Furu Wei, Weiwei Deng, and Qi Zhang. 2023. Uprise: Universal prompt retrieval for improving zero-shot evaluation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12318–12337.

Rong Cheng, Jinyi Liu, Yan Zheng, Fei Ni, Jiazhen Du, Hangyu Mao, Fuzheng Zhang, Bo Wang, and Jianye Hao. 2025. Dualrag: A dual-process approach to integrate reasoning and retrieval for multi-hop question answering. *arXiv preprint arXiv:2504.18243*.

Zheng Chu, Jingchang Chen, Qianglong Chen, Haotian Wang, Kun Zhu, Xiyuan Du, Weijiang Yu, Ming Liu,

分类框架（推理增强RAG、RAG增强推理以及协同RAG和推理）跨越了多种不同的方法论。虽然这有助于从高层次理解设计模式，但它可能会掩盖每种方法类别独特的细微权衡、假设和局限性。

参考文献

Vaibhav Adlakha, Shehzaad Dhuliawala, Kaheer Suleman, Harm de Vries, and Siva Reddy. 2022. Topiocqa: Open-domain conversational question answering with topic switching. *Transactions of the Association for Computational Linguistics*, 10:468–483.

Firoj Alam, Ferda Ofli, and Muhammad Imran. 2018. Crisismmd: Multimodal twitter datasets from natural disasters. In *Proceedings of the international AAAI conference on web and social media*, volume 12.

Salaheddin Alzubi, Creston Brooks, Purva Chiniya, Edoardo Contente, Chiara von Gerlach, Lucas Irwin, Yihan Jiang, Arda Kaz, Windsor Nguyen, Sewoong Oh, et al. 2025. Open deep search: Democratizing search with open-source reasoning agents. *arXiv preprint arXiv:2503.20201*.

Anonymous. 2025. DynQR: Dynamic uncertainty-guided query rewriting for effective retrieval-augmented generation. In *Submitted to ACL Rolling Review - December 2024*. Under review.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Self-reflective retrieval augmented generation. In *NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following*.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *The Twelfth International Conference on Learning Representations*.

Jinheon Baek, Sujay Kumar Jauhar, Silviu Cucerzan, and Sung Ju Hwang. 2024. Researchagent: Iterative research idea generation over scientific literature with large language models. *arXiv preprint arXiv:2404.07738*.

Yuanchen Bei, Weizhi Zhang, Siwen Wang, Weizhi Chen, Sheng Zhou, Hao Chen, Yong Li, Jiajun Bu, Shirui Pan, Yizhou Yu, et al. 2025. Graphs meet ai agents: Taxonomy, progress, and future opportunities. *arXiv preprint arXiv:2506.18019*.

Cameron B Browne, Edward Powley, Daniel Whitehouse, Simon M Lucas, Peter I Cowling, Philipp Rohlfshagen, Stephen Tavener, Diego Perez, Spyridon Samothrakis, and Simon Colton. 2012. A survey of monte carlo tree search methods. *IEEE Transactions on Computational Intelligence and AI in games*, 4(1):1–43.

Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, et al. 2024. A survey on evaluation of large language models. *ACM transactions on intelligent systems and technology*, 15(3):1–45.

Danqi Chen and Wen-tau Yih. 2020. Open-domain question answering. In *Proceedings of the 58th annual meeting of the association for computational linguistics: tutorial abstracts*, pages 34–37.

Mingyang Chen, Tianpeng Li, Haoze Sun, Yijie Zhou, Chenzheng Zhu, Haofen Wang, Jeff Z Pan, Wen Zhang, Huajun Chen, Fan Yang, et al. 2025a. Research: Learning to reason with search for llms via reinforcement learning. *arXiv preprint arXiv:2503.19470*.

Mingyang Chen, Tianpeng Li, Haoze Sun, Yijie Zhou, Chenzheng Zhu, Fan Yang, Zenan Zhou, Weipeng Chen, Haofen Wang, Jeff Z Pan, et al. 2025b. Learning to reason with search for llms via reinforcement learning. *arXiv preprint arXiv:2503.19470*.

Qiguang Chen, Libo Qin, Jinhao Liu, Dengyun Peng, Jiannan Guan, Peng Wang, Mengkang Hu, Yuhang Zhou, Te Gao, and Wanxiang Che. 2025c. Towards reasoning era: A survey of long chain-of-thought for reasoning large language models. *arXiv preprint arXiv:2503.09567*.

Yanfei Chen, Jinsung Yoon, Devendra Sachan, Qingze Wang, Vincent Cohen-Addad, Mohammadhossein Bateni, Chen-Yu Lee, and Tomas Pfister. 2024a. Re-invoke: Tool invocation rewriting for zero-shot tool retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4705–4726.

Zehui Chen, Kuikun Liu, Qiuchen Wang, Jiangning Liu, Wenwei Zhang, Kai Chen, and Feng Zhao. 2024b. Mindsearch: Mimicking human minds elicits deep ai searcher. *arXiv preprint arXiv:2407.20183*.

Zhongwu Chen, Chengjin Xu, Dingmin Wang, Zhen Huang, Yong Dou, Xuhui Jiang, and Jian Guo. 2024c. Rulerag: Rule-guided retrieval-augmented generation with language models for question answering. *arXiv preprint arXiv:2410.22353*.

Daixuan Cheng, Shaohan Huang, Junyu Bi, Yuefeng Zhan, Jianfeng Liu, Yujing Wang, Hao Sun, Furu Wei, Weiwei Deng, and Qi Zhang. 2023. Uprise: Universal prompt retrieval for improving zero-shot evaluation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12318–12337.

Rong Cheng, Jinyi Liu, Yan Zheng, Fei Ni, Jiazhen Du, Hangyu Mao, Fuzheng Zhang, Bo Wang, and Jianye Hao. 2025. Dualrag: A dual-process approach to integrate reasoning and retrieval for multi-hop question answering. *arXiv preprint arXiv:2504.18243*.

Zheng Chu, Jingchang Chen, Qianglong Chen, Haotian Wang, Kun Zhu, Xiyuan Du, Weijiang Yu, Ming Liu,

10

and Bing Qin. 2024. Beamaggr: Beam aggregation reasoning over multi-source knowledge for multi-hop question answering. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1229–1248.

Debrup Das, Debopriyo Banerjee, Somak Aditya, and Ashish Kulkarni. 2024. Mathsensei: A tool-augmented large language model for mathematical reasoning. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 942–966.

Chao Deng, Jiale Yuan, Pi Bu, Peijie Wang, Zhong-Zhi Li, Jian Xu, Xiao-Hui Li, Yuan Gao, Jun Song, Bo Zheng, et al. 2024. Longdocurl: a comprehensive multimodal long document benchmark integrating understanding, reasoning, and locating. *arXiv preprint arXiv:2412.18424*.

Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2024. Chain-of-verification reduces hallucination in large language models. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 3563–3578.

Avik Dutta, Mukul Singh, Gust Verbruggen, Sumit Gulwani, and Vu Le. 2024. Rar: Retrieval-augmented retrieval for code generation in low resource languages. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21506–21515.

Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. 2024a. A survey on rag meeting llms: Towards retrieval-augmented large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 6491–6501.

Yue Fan, Hu Zhang, Ru Li, Yujie Wang, Hongye Tan, and Jiye Liang. 2024b. Frva: Fact-retrieval and verification augmented entailment tree generation for explainable question answering. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 9111–9128.

Jinyuan Fang, Zaiqiao Meng, and Craig Macdonald. 2024. Trace the evidence: Constructing knowledge-grounded reasoning chains for retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 8472–8494.

Weizhi Fei, Xueyan Niu, Guoqing Xie, Yanhua Zhang, Bo Bai, Lei Deng, and Wei Han. 2024. Retrieval meets reasoning: Dynamic in-context editing for long-text understanding. *arXiv preprint arXiv:2406.12331*.

Wenfeng Feng, Chuzhan Hao, Yuewei Zhang, Jingyi Song, and Hao Wang. 2025. Airrag: Activating intrinsic reasoning for retrieval augmented generation via tree-based search. *arXiv preprint arXiv:2501.10053*.

James Ferguson, Matt Gardner, Hannaneh Hajishirzi, Tushar Khot, and Pradeep Dasigi. 2020. Iirc: A dataset of incomplete information reading comprehension questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1137–1147.

Zafeirios Fountas, Martin A Benfeghoul, Adnan Oomerjee, Fenia Christopoulou, Gerasimos Lampouras, Haitham Bou-Ammar, and Jun Wang. 2024. Human-like episodic memory for infinite context llms. *arXiv preprint arXiv:2407.09450*.

Luyu Gao, Zhuyun Dai, Panupong Pasupat, Anthony Chen, Arun Tejasvi Chaganty, Yicheng Fan, Vincent Zhao, Ni Lao, Hongrae Lee, Da-Cheng Juan, and Kelvin Guu. 2023a. RARR: Researching and revising what language models say, using language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16477–16508, Toronto, Canada. Association for Computational Linguistics.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yixin Dai, Jiawei Sun, Haofen Wang, and Haofen Wang. 2023b. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2:1.

Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. 2021. Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361.

Xinyan Guan, Jiali Zeng, Fandong Meng, Chunlei Xin, Yaojie Lu, Hongyu Lin, Xianpei Han, Le Sun, and Jie Zhou. 2025. Deeprag: Thinking to retrieval step by step for large language models. *arXiv preprint arXiv:2502.01142*.

Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. 2024. Lightrag: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779*.

Siwei Han, Peng Xia, Ruiyi Zhang, Tong Sun, Yun Li, Hongtu Zhu, and Huaxiu Yao. 2025. Mdocagent: A multi-modal multi-agent framework for document understanding. *arXiv preprint arXiv:2503.13964*.

Shibo Hao, Tianyang Liu, Zhen Wang, and Zhiting Hu. 2023. Toolkengpt: Augmenting frozen language models with massive tools via tool embeddings. In *Advances in Neural Information Processing Systems*, volume 36, pages 45870–45894.

Bolei He, Nuo Chen, Xinran He, Lingyong Yan, Zhenkai Wei, Jinchang Luo, and Zhen-Hua Ling.

and Bing Qin. 2024. Beamaggr: Beam aggregation reasoning over multi-source knowledge for multi-hop question answering. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1229–1248.

Debrup Das, Debopriyo Banerjee, Somak Aditya, and Ashish Kulkarni. 2024. Mathsensei: A tool-augmented large language model for mathematical reasoning. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 942–966.

Chao Deng, Jiale Yuan, Pi Bu, Peijie Wang, Zhong-Zhi Li, Jian Xu, Xiao-Hui Li, Yuan Gao, Jun Song, Bo Zheng, et al. 2024. Longdocurl: a comprehensive multimodal long document benchmark integrating understanding, reasoning, and locating. *arXiv preprint arXiv:2412.18424*.

Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2024. Chain-of-verification reduces hallucination in large language models. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 3563–3578.

Avik Dutta, Mukul Singh, Gust Verbruggen, Sumit Gulwani, and Vu Le. 2024. Rar: Retrieval-augmented retrieval for code generation in low resource languages. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21506–21515.

Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. 2024a. A survey on rag meeting llms: Towards retrieval-augmented large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 6491–6501.

Yue Fan, Hu Zhang, Ru Li, Yujie Wang, Hongye Tan, and Jiye Liang. 2024b. Frva: Fact-retrieval and verification augmented entailment tree generation for explainable question answering. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 9111–9128.

Jinyuan Fang, Zaiqiao Meng, and Craig Macdonald. 2024. Trace the evidence: Constructing knowledge-grounded reasoning chains for retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 8472–8494.

Weizhi Fei, Xueyan Niu, Guoqing Xie, Yanhua Zhang, Bo Bai, Lei Deng, and Wei Han. 2024. Retrieval meets reasoning: Dynamic in-context editing for long-text understanding. *arXiv preprint arXiv:2406.12331*.

Wenfeng Feng, Chuzhan Hao, Yuewei Zhang, Jingyi Song, and Hao Wang. 2025. Airrag: Activating intrinsic reasoning for retrieval augmented generation via tree-based search. *arXiv preprint arXiv:2501.10053*.

James Ferguson, Matt Gardner, Hannaneh Hajishirzi, Tushar Khot, and Pradeep Dasigi. 2020. Iirc: A dataset of incomplete information reading comprehension questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1137–1147.

Zafeirios Fountas, Martin A Benfeghoul, Adnan Oomerjee, Fenia Christopoulou, Gerasimos Lampouras, Haitham Bou-Ammar, and Jun Wang. 2024. Human-like episodic memory for infinite context llms. *arXiv preprint arXiv:2407.09450*.

Luyu Gao, Zhuyun Dai, Panupong Pasupat, Anthony Chen, Arun Tejasvi Chaganty, Yicheng Fan, Vincent Zhao, Ni Lao, Hongrae Lee, Da-Cheng Juan, and Kelvin Guu. 2023a. RARR: Researching and revising what language models say, using language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16477–16508, Toronto, Canada. Association for Computational Linguistics.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yixin Dai, Jiawei Sun, Haofen Wang, and Haofen Wang. 2023b. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2:1.

Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. 2021. Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361.

Xinyan Guan, Jiali Zeng, Fandong Meng, Chunlei Xin, Yaojie Lu, Hongyu Lin, Xianpei Han, Le Sun, and Jie Zhou. 2025. Deeprag: Thinking to retrieval step by step for large language models. *arXiv preprint arXiv:2502.01142*.

Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. 2024. Lightrag: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779*.

Siwei Han, Peng Xia, Ruiyi Zhang, Tong Sun, Yun Li, Hongtu Zhu, and Huaxiu Yao. 2025. Mdocagent: A multi-modal multi-agent framework for document understanding. *arXiv preprint arXiv:2503.13964*.

Shibo Hao, Tianyang Liu, Zhen Wang, and Zhiting Hu. 2023. Toolkengpt: Augmenting frozen language models with massive tools via tool embeddings. In *Advances in Neural Information Processing Systems*, volume 36, pages 45870–45894.

Bolei He, Nuo Chen, Xinran He, Lingyong Yan, Zhenkai Wei, Jinchang Luo, and Zhen-Hua Ling.

11

2024a. Retrieving, rethinking and revising: The chain-of-verification can improve retrieval augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 10371–10393.

Jie He, Nan Hu, Wanqiu Long, Jiaoyan Chen, and Jeff Z Pan. 2024b. Mintqa: A multi-hop question answering benchmark for evaluating llms on new and tail knowledge. *arXiv preprint arXiv:2412.17032*.

Xiaoxin He, Yijun Tian, Yifei Sun, Nitesh Chawla, Thomas Laurent, Yann LeCun, Xavier Bresson, and Bryan Hooi. 2024c. G-retriever: Retrieval-augmented generation for textual graph understanding and question answering. *Advances in Neural Information Processing Systems*, 37:132876–132907.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625.

Minda Hu, Licheng Zong, Hongru Wang, Jingyan Zhou, Jingjing Li, Yichen Gao, Kam-Fai Wong, Yu Li, and Irwin King. 2024. Serts: Self-rewarding tree search for biomedical retrieval-augmented generation. *arXiv preprint arXiv:2406.11258*.

Yunhai Hu, Yilun Zhao, Chen Zhao, and Arman Cohan. 2025. Mcts-rag: Enhancing retrieval-augmented generation with monte carlo tree search. *arXiv preprint arXiv:2503.20757*.

Jerry Huang, Siddarth Madala, Risham Sidhu, Cheng Niu, Julia Hockenmaier, and Tong Zhang. 2025a. Rag-rl: Advancing retrieval-augmented generation via rl and curriculum learning. *arXiv preprint arXiv:2503.12759*.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. 2025b. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55.

Xiaowei Huang, Wenjie Ruan, Wei Huang, Gaojie Jin, Yi Dong, Changshun Wu, Saddek Bensalem, Ronghui Mu, Yi Qi, Xingyu Zhao, et al. 2024. A survey of safety and trustworthiness of large language models through the lens of verification and validation. *Artificial Intelligence Review*, 57(7):175.

Yulong Hui, Yao Lu, and Huanchen Zhang. 2024. Uda: A benchmark suite for retrieval augmented generation in real-world document analysis. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Michael Iannelli, Sneha Kuchipudi, and Vera Dvorak. 2024. Sla management in reconfigurable multi-agent rag: A systems approach to question answering. *arXiv preprint arXiv:2412.06832*.

Shayekh Islam, Md Asib Rahman, KSM Tozammel Hossain, Enamul Hoque, Shafiq Joty, and Md Rizwan Parvez. 2024. Open-rag: Enhanced retrieval augmented reasoning with open-source large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 14231–14244.

Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. 2024. Livecodebench: Holistic and contamination free evaluation of large language models for code. *arXiv preprint arXiv:2403.07974*.

Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C Park. 2024. Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7029–7043.

Yixin Ji, Kaixin Wu, Juntao Li, Wei Chen, Mingjie Zhong, Xu Jia, and Min Zhang. 2024. Retrieval and reasoning on kgs: Integrate knowledge graphs into large language models for complex question answering. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 7598–7610.

Mingyi Jia, Junwen Duan, Yan Song, and Jianxin Wang. 2025. Find: Fine-grained information density guided adaptive retrieval-augmented generation for disease diagnosis. *arXiv preprint arXiv:2502.14614*.

Jinhao Jiang, Jiayi Chen, Junyi Li, Ruiyang Ren, Shijie Wang, Wayne Xin Zhao, Yang Song, and Tao Zhang. 2024. Rag-star: Enhancing deliberative reasoning with retrieval augmented verification and refinement. *arXiv preprint arXiv:2412.12881*.

Pengcheng Jiang, Jiacheng Lin, Lang Cao, Runchu Tian, SeongKu Kang, Zifeng Wang, Jimeng Sun, and Jiawei Han. 2025. Deepretrieval: Hacking real search engines and retrievers with large language models via reinforcement learning. *arXiv preprint arXiv:2503.00223*.

Bowen Jin, Chulin Xie, Jiawei Zhang, Kashob Kumar Roy, Yu Zhang, Zheng Li, Ruirui Li, Xianfeng Tang, Suhang Wang, Yu Meng, et al. 2024. Graph chain-of-thought: Augmenting large language models by reasoning on graphs. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 163–184.

2024a. Retrieving, rethinking and revising: The chain-of-verification can improve retrieval augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 10371–10393.

Jie He, Nan Hu, Wanqiu Long, Jiaoyan Chen, and Jeff Z Pan. 2024b. Mintqa: A multi-hop question answering benchmark for evaluating llms on new and tail knowledge. *arXiv preprint arXiv:2412.17032*.

Xiaoxin He, Yijun Tian, Yifei Sun, Nitesh Chawla, Thomas Laurent, Yann LeCun, Xavier Bresson, and Bryan Hooi. 2024c. G-retriever: Retrieval-augmented generation for textual graph understanding and question answering. *Advances in Neural Information Processing Systems*, 37:132876–132907.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625.

Minda Hu, Licheng Zong, Hongru Wang, Jingyan Zhou, Jingjing Li, Yichen Gao, Kam-Fai Wong, Yu Li, and Irwin King. 2024. Serts: Self-rewarding tree search for biomedical retrieval-augmented generation. *arXiv preprint arXiv:2406.11258*.

Yunhai Hu, Yilun Zhao, Chen Zhao, and Arman Cohan. 2025. Mcts-rag: Enhancing retrieval-augmented generation with monte carlo tree search. *arXiv preprint arXiv:2503.20757*.

Jerry Huang, Siddarth Madala, Risham Sidhu, Cheng Niu, Julia Hockenmaier, and Tong Zhang. 2025a. Rag-rl: Advancing retrieval-augmented generation via rl and curriculum learning. *arXiv preprint arXiv:2503.12759*.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. 2025b. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55.

Xiaowei Huang, Wenjie Ruan, Wei Huang, Gaojie Jin, Yi Dong, Changshun Wu, Saddek Bensalem, Ronghui Mu, Yi Qi, Xingyu Zhao, et al. 2024. A survey of safety and trustworthiness of large language models through the lens of verification and validation. *Artificial Intelligence Review*, 57(7):175.

Yulong Hui, Yao Lu, and Huanchen Zhang. 2024. Uda: A benchmark suite for retrieval augmented generation in real-world document analysis. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Michael Iannelli, Sneha Kuchipudi, and Vera Dvorak. 2024. Sla management in reconfigurable multi-agent rag: A systems approach to question answering. *arXiv preprint arXiv:2412.06832*.

Shayekh Islam, Md Asib Rahman, KSM Tozammel Hossain, Enamul Hoque, Shafiq Joty, and Md Rizwan Parvez. 2024. Open-rag: Enhanced retrieval augmented reasoning with open-source large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 14231–14244.

Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. 2024. Livecodebench: Holistic and contamination free evaluation of large language models for code. *arXiv preprint arXiv:2403.07974*.

Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C Park. 2024. Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7029–7043.

Yixin Ji, Kaixin Wu, Juntao Li, Wei Chen, Mingjie Zhong, Xu Jia, and Min Zhang. 2024. Retrieval and reasoning on kgs: Integrate knowledge graphs into large language models for complex question answering. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 7598–7610.

Mingyi Jia, Junwen Duan, Yan Song, and Jianxin Wang. 2025. Find: Fine-grained information density guided adaptive retrieval-augmented generation for disease diagnosis. *arXiv preprint arXiv:2502.14614*.

Jinhao Jiang, Jiayi Chen, Junyi Li, Ruiyang Ren, Shijie Wang, Wayne Xin Zhao, Yang Song, and Tao Zhang. 2024. Rag-star: Enhancing deliberative reasoning with retrieval augmented verification and refinement. *arXiv preprint arXiv:2412.12881*.

Pengcheng Jiang, Jiacheng Lin, Lang Cao, Runchu Tian, SeongKu Kang, Zifeng Wang, Jimeng Sun, and Jiawei Han. 2025. Deepretrieval: Hacking real search engines and retrievers with large language models via reinforcement learning. *arXiv preprint arXiv:2503.00223*.

Bowen Jin, Chulin Xie, Jiawei Zhang, Kashob Kumar Roy, Yu Zhang, Zheng Li, Ruirui Li, Xianfeng Tang, Suhang Wang, Yu Meng, et al. 2024. Graph chain-of-thought: Augmenting large language models by reasoning on graphs. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 163–184.

12

Bowen Jin, Hansi Zeng, Zhenrui Yue, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. Search-r1: Training llms to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516*.

Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. 2017. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611.

Tomoyuki Kagaya, Thong Jing Yuan, Yuxuan Lou, Jayashree Karlekar, Sugiri Pranata, Akira Kinose, Koki Oguri, Felix Wick, and Yang You. 2024. Rap: Retrieval-augmented planning with contextual memory for multimodal llm agents. *arXiv preprint arXiv:2402.03610*.

Mohammed Khaliq, Paul Chang, Mingyang Ma, Bernhard Pflugfelder, and Filip Miletić. 2024. Ragar, your falsehood radar: Rag-augmented reasoning for political fact-checking using multimodal large language models. In *Proceedings of the Seventh Fact Extraction and VERification Workshop (FEVER)*, pages 280–296.

Gangwoo Kim, Sungdong Kim, Byeongguk Jeon, Joonsuk Park, and Jaewoo Kang. 2023. Tree of clarifications: Answering ambiguous questions with retrieval-augmented large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 996–1009.

Neema Kotonya and Francesca Toni. 2020. Explainable automated fact-checking for public health claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7740–7754.

Heiko Koziolek, Sten Grüner, Rhaban Hark, Virendra Ashiwal, Sofia Linsbauer, and Nafise Eskandani. 2024. Llm-based and retrieval-augmented control code generation. In *Proceedings of the 1st International Workshop on Large Language Models for Code*, pages 22–29.

Satyapriya Krishna, Kalpesh Krishna, Anhad Mohananey, Steven Schwarcz, Adam Stambler, Shyam Upadhyay, and Manaal Faruqui. 2024. Fact, fetch, and reason: A unified evaluation of retrieval-augmented generation. *arXiv preprint arXiv:2409.12941*.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466.

Sung-Min Lee, Eunhwan Park, Donghyeon Jeon, Inho Kang, and Seung-Hoon Na. 2024. Radcot: Retrieval-augmented distillation to specialization models for generating chain-of-thoughts in query expansion. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13514–13523.

Zhicheng Lee, Shulin Cao, Jinxin Liu, Jiajie Zhang, Weichuan Liu, Xiaoyin Che, Lei Hou, and Juanzi Li. 2025. Rearag: Knowledge-guided reasoning enhances factuality of large reasoning models with iterative retrieval augmented generation. *arXiv preprint arXiv:2503.21729*.

Dawei Li, Shu Yang, Zhen Tan, Jae Baik, Sukwon Yun, Joseph Lee, Aaron Chacko, Bojian Hou, Duy Duong-Tran, Ying Ding, et al. 2024a. Dalk: Dynamic co-augmentation of llms and kg to answer alzheimer's disease questions with scientific literature. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 2187–2205.

Guanghua Li, Wensheng Lu, Wei Zhang, Defu Lian, Kezhong Lu, Rui Mao, Kai Shu, and Hao Liao. 2024b. Re-search for the truth: Multi-round retrieval-augmented large language models are strong fake news detectors. *arXiv preprint arXiv:2403.09747*.

Guozheng Li, Peng Wang, Wenjun Ke, Yikai Guo, Ke Ji, Ziyu Shang, Jiajun Liu, and Zijie Xu. 2024c. Recall, retrieve and reason: towards better in-context relation extraction. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, pages 6368–6376.

Huayang Li, Pat Verga, Priyanka Sen, Bowen Yang, Vijay Viswanathan, Patrick Lewis, Taro Watanabe, and Yixuan Su. 2024d. Alr2: A retrieve-then-reason framework for long-context question answering. *arXiv preprint arXiv:2410.03227*.

Jia Li, Xianjie Shi, Kechi Zhang, Lei Li, Ge Li, Zhengwei Tao, Fang Liu, Chongyang Tao, and Zhi Jin. 2025a. Coderag: Supportive code retrieval on bigraph for real-world code generation. *arXiv preprint arXiv:2504.10046*.

Minghan Li, Honglei Zhuang, Kai Hui, Zhen Qin, Jimmy Lin, Rolf Jagerman, Xuanhui Wang, and Michael Bendersky. 2024e. Can query expansion improve generalization of strong cross-encoder rankers? In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2321–2326.

Shilong Li, Yancheng He, Hangyu Guo, Xingyuan Bu, Ge Bai, Jie Liu, Jiaheng Liu, Xingwei Qu, Yangguang Li, Wanli Ouyang, et al. 2024f. Graphreader: Building graph-based agent to enhance long-context abilities of large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 12758–12786.

Xiaoxi Li, Guanting Dong, Jiajie Jin, Yuyao Zhang, Yujia Zhou, Yutao Zhu, Peitian Zhang, and Zhicheng Dou. 2025b. Search-o1: Agentic search-enhanced large reasoning models. *arXiv preprint arXiv:2501.05366*.

Bowen Jin, Hansi Zeng, Zhenrui Yue, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. Search-r1: Training llms to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516*.

Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. 2017. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611.

Tomoyuki Kagaya, Thong Jing Yuan, Yuxuan Lou, Jayashree Karlekar, Sugiri Pranata, Akira Kinose, Koki Oguri, Felix Wick, and Yang You. 2024. Rap: Retrieval-augmented planning with contextual memory for multimodal llm agents. *arXiv preprint arXiv:2402.03610*.

Mohammed Khaliq, Paul Chang, Mingyang Ma, Bernhard Pflugfelder, and Filip Miletić. 2024. Ragar, your falsehood radar: Rag-augmented reasoning for political fact-checking using multimodal large language models. In *Proceedings of the Seventh Fact Extraction and VERification Workshop (FEVER)*, pages 280–296.

Gangwoo Kim, Sungdong Kim, Byeongguk Jeon, Joonsuk Park, and Jaewoo Kang. 2023. Tree of clarifications: Answering ambiguous questions with retrieval-augmented large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 996–1009.

Neema Kotonya and Francesca Toni. 2020. Explainable automated fact-checking for public health claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7740–7754.

Heiko Koziolek, Sten Grüner, Rhaban Hark, Virendra Ashiwal, Sofia Linsbauer, and Nafise Eskandani. 2024. Llm-based and retrieval-augmented control code generation. In *Proceedings of the 1st International Workshop on Large Language Models for Code*, pages 22–29.

Satyapriya Krishna, Kalpesh Krishna, Anhad Mohananey, Steven Schwarcz, Adam Stambler, Shyam Upadhyay, and Manaal Faruqui. 2024. Fact, fetch, and reason: A unified evaluation of retrieval-augmented generation. *arXiv preprint arXiv:2409.12941*.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466.

Sung-Min Lee, Eunhwan Park, Donghyeon Jeon, Inho Kang, and Seung-Hoon Na. 2024. Radcot: Retrieval-augmented distillation to specialization models for generating chain-of-thoughts in query expansion. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13514–13523.

Zhicheng Lee, Shulin Cao, Jinxin Liu, Jiajie Zhang, Weichuan Liu, Xiaoyin Che, Lei Hou, and Juanzi Li. 2025. Rearag: Knowledge-guided reasoning enhances factuality of large reasoning models with iterative retrieval augmented generation. *arXiv preprint arXiv:2503.21729*.

Dawei Li, Shu Yang, Zhen Tan, Jae Baik, Sukwon Yun, Joseph Lee, Aaron Chacko, Bojian Hou, Duy Duong-Tran, Ying Ding, et al. 2024a. Dalk: Dynamic co-augmentation of llms and kg to answer alzheimer's disease questions with scientific literature. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 2187–2205.

Guanghua Li, Wensheng Lu, Wei Zhang, Defu Lian, Kezhong Lu, Rui Mao, Kai Shu, and Hao Liao. 2024b. Re-search for the truth: Multi-round retrieval-augmented large language models are strong fake news detectors. *arXiv preprint arXiv:2403.09747*.

Guozheng Li, Peng Wang, Wenjun Ke, Yikai Guo, Ke Ji, Ziyu Shang, Jiajun Liu, and Zijie Xu. 2024c. Recall, retrieve and reason: towards better in-context relation extraction. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, pages 6368–6376.

Huayang Li, Pat Verga, Priyanka Sen, Bowen Yang, Vijay Viswanathan, Patrick Lewis, Taro Watanabe, and Yixuan Su. 2024d. Alr2: A retrieve-then-reason framework for long-context question answering. *arXiv preprint arXiv:2410.03227*.

Jia Li, Xianjie Shi, Kechi Zhang, Lei Li, Ge Li, Zhengwei Tao, Fang Liu, Chongyang Tao, and Zhi Jin. 2025a. Coderag: Supportive code retrieval on bigraph for real-world code generation. *arXiv preprint arXiv:2504.10046*.

Minghan Li, Honglei Zhuang, Kai Hui, Zhen Qin, Jimmy Lin, Rolf Jagerman, Xuanhui Wang, and Michael Bendersky. 2024e. Can query expansion improve generalization of strong cross-encoder rankers? In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2321–2326.

Shilong Li, Yancheng He, Hangyu Guo, Xingyuan Bu, Ge Bai, Jie Liu, Jiaheng Liu, Xingwei Qu, Yangguang Li, Wanli Ouyang, et al. 2024f. Graphreader: Building graph-based agent to enhance long-context abilities of large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 12758–12786.

Xiaoxi Li, Guanting Dong, Jiajie Jin, Yuyao Zhang, Yujia Zhou, Yutao Zhu, Peitian Zhang, and Zhicheng Dou. 2025b. Search-o1: Agentic search-enhanced large reasoning models. *arXiv preprint arXiv:2501.05366*.

13

Xiaoxi Li, Jiajie Jin, Guanting Dong, Hongjin Qian, Yutao Zhu, Yongkang Wu, Ji-Rong Wen, and Zhicheng Dou. 2025c. Webthinker: Empowering large reasoning models with deep research capability. *arXiv preprint arXiv:2504.21776*.

Yangning Li, Yinghui Li, Xinyu Wang, Yong Jiang, Zhen Zhang, Xinran Zheng, Hui Wang, Hai-Tao Zheng, Fei Huang, Jingren Zhou, and Philip S. Yu. 2025d. Benchmarking multimodal retrieval augmented generation with dynamic VQA dataset and self-adaptive planning agent. In *The Thirteenth International Conference on Learning Representations*.

Yanran Li, Hui Su, Xiaoyu Shen, Wenjie Li, Ziqiang Cao, and Shuzi Niu. 2017. Dailydialog: A manually labelled multi-turn dialogue dataset. *arXiv preprint arXiv:1710.03957*.

Zhi Li, Yicheng Li, Hequan Ye, and Yin Zhang. 2024g. Towards autonomous tool utilization in language models: A unified, efficient and scalable framework. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 16422–16432.

Zhong-Zhi Li, Duzhen Zhang, Ming-Liang Zhang, Jiaxin Zhang, Zengyan Liu, Yuxuan Yao, Haotian Xu, Junhao Zheng, Pei-Jie Wang, Xiuyi Chen, et al. 2025e. From system 1 to system 2: A survey of reasoning large language models. *arXiv preprint arXiv:2502.17419*.

Zhuoqun Li, Xuanang Chen, Haiyang Yu, Hongyu Lin, Yaojie Lu, Qiaoyu Tang, Fei Huang, Xianpei Han, Le Sun, and Yongbin Li. 2024h. Structrag: Boosting knowledge intensive reasoning of llms via inference-time hybrid information structurization. *arXiv preprint arXiv:2410.08815*.

Zijing Liang, Yanjie Xu, Yifan Hong, Penghui Shang, Qi Wang, Qiang Fu, and Ke Liu. 2024. A survey of multimodel large language models. In *Proceedings of the 3rd International Conference on Computer, Artificial Intelligence and Control Engineering*, pages 405–409.

Xi Victoria Lin, Xilun Chen, Mingda Chen, Weijia Shi, Maria Lomeli, Richard James, Pedro Rodriguez, Jacob Kahn, Gergely Szilvasy, Mike Lewis, et al. 2023. Ra-dit: Retrieval-augmented dual instruction tuning. In *The Twelfth International Conference on Learning Representations*.

Wang Ling, Dani Yogatama, Chris Dyer, and Phil Blunsom. 2017. Program induction by rationale generation: Learning to solve and explain algebraic word problems. *arXiv preprint arXiv:1705.04146*.

Aiwei Liu, Leyi Pan, Yijian Lu, Jingjing Li, Xuming Hu, Xi Zhang, Lijie Wen, Irwin King, Hui Xiong, and Philip Yu. 2024. A survey of text watermarking in the era of large language models. *ACM Computing Surveys*, 57(2):1–36.

Pei Liu, Xin Liu, Ruoyu Yao, Junming Liu, Siyuan Meng, Ding Wang, and Jun Ma. 2025. Hm-rag: Hierarchical multi-agent multimodal retrieval augmented generation. *arXiv preprint arXiv:2504.12330*.

Chang Han Low, Ziyue Wang, Tianyi Zhang, Zhitao Zeng, Zhu Zhuo, Evangelos B Mazomenos, and Yueming Jin. 2025. Surgraw: Multi-agent workflow with chain-of-thought reasoning for surgical intelligence. *arXiv preprint arXiv:2503.10265*.

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The ai scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*.

Pan Lu, Swaroop Mishra, Tanglin Xia, Liang Qiu, Kai-Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter Clark, and Ashwin Kalyan. 2022. Learn to explain: Multimodal reasoning via thought chains for science question answering. *Advances in Neural Information Processing Systems*, 35:2507–2521.

Junyu Luo, Weizhi Zhang, Ye Yuan, Yusheng Zhao, Junwei Yang, Yiyang Gu, Bohan Wu, Binqi Chen, Ziyue Qiao, Qingqing Long, et al. 2025a. Large language model agent: A survey on methodology, applications and challenges. *arXiv preprint arXiv:2503.21460*.

Man Luo, Xin Xu, Zhuyun Dai, Panupong Pasupat, Mehran Kazemi, Chitta Baral, Vaiva Imbrasaite, and Vincent Y Zhao. 2023. Dr. icl: Demonstration-retrieved in-context learning. *arXiv preprint arXiv:2305.14128*.

Ne Luo, Aryo Pradipta Gema, Xuanli He, Emile van Krieken, Pietro Lesci, and Pasquale Minervini. 2025b. Self-training large language models for tool-use without demonstrations. *arXiv preprint arXiv:2502.05867*.

Shengjie Ma, Chengjin Xu, Xuhui Jiang, Muzhi Li, Huaren Qu, Cehao Yang, Jiaxin Mao, and Jian Guo. 2024a. Think-on-graph 2.0: Deep and faithful large language model reasoning with knowledge-guided retrieval augmented generation. *arXiv preprint arXiv:2407.10805*.

Yubo Ma, Zhibin Gou, Junheng Hao, Ruochen Xu, Shuohang Wang, Liangming Pan, Yujiu Yang, Yixin Cao, and Aixin Sun. 2024b. Sciagent: Tool-augmented language models for scientific reasoning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15701–15736.

Yubo Ma, Yuhang Zang, Liangyu Chen, Meiqi Chen, Yizhu Jiao, Xinze Li, Xinyuan Lu, Ziyu Liu, Yan Ma, Xiaoyi Dong, et al. 2025. Mmlongbench-doc: Benchmarking long-context document understanding with visualizations. *Advances in Neural Information Processing Systems*, 37:95963–96010.

Kelong Mao, Zheng Liu, Hongjin Qian, Fengran Mo, Chenlong Deng, and Zhicheng Dou. 2024. Ragstudio: Towards in-domain adaptation of retrieval

Xiaoxi Li, Jiajie Jin, Guanting Dong, Hongjin Qian, Yutao Zhu, Yongkang Wu, Ji-Rong Wen, and Zhicheng Dou. 2025c. Webthinker: Empowering large reasoning models with deep research capability. *arXiv preprint arXiv:2504.21776*.

Yangning Li, Yinghui Li, Xinyu Wang, Yong Jiang, Zhen Zhang, Xinran Zheng, Hui Wang, Hai-Tao Zheng, Fei Huang, Jingren Zhou, and Philip S. Yu. 2025d. Benchmarking multimodal retrieval augmented generation with dynamic VQA dataset and self-adaptive planning agent. In *The Thirteenth International Conference on Learning Representations*.

Yanran Li, Hui Su, Xiaoyu Shen, Wenjie Li, Ziqiang Cao, and Shuzi Niu. 2017. Dailydialog: A manually labelled multi-turn dialogue dataset. *arXiv preprint arXiv:1710.03957*.

Zhi Li, Yicheng Li, Hequan Ye, and Yin Zhang. 2024g. Towards autonomous tool utilization in language models: A unified, efficient and scalable framework. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 16422–16432.

Zhong-Zhi Li, Duzhen Zhang, Ming-Liang Zhang, Jiaxin Zhang, Zengyan Liu, Yuxuan Yao, Haotian Xu, Junhao Zheng, Pei-Jie Wang, Xiuyi Chen, et al. 2025e. From system 1 to system 2: A survey of reasoning large language models. *arXiv preprint arXiv:2502.17419*.

Zhuoqun Li, Xuanang Chen, Haiyang Yu, Hongyu Lin, Yaojie Lu, Qiaoyu Tang, Fei Huang, Xianpei Han, Le Sun, and Yongbin Li. 2024h. Structrag: Boosting knowledge intensive reasoning of llms via inference-time hybrid information structurization. *arXiv preprint arXiv:2410.08815*.

Zijing Liang, Yanjie Xu, Yifan Hong, Penghui Shang, Qi Wang, Qiang Fu, and Ke Liu. 2024. A survey of multimodel large language models. In *Proceedings of the 3rd International Conference on Computer, Artificial Intelligence and Control Engineering*, pages 405–409.

Xi Victoria Lin, Xilun Chen, Mingda Chen, Weijia Shi, Maria Lomeli, Richard James, Pedro Rodriguez, Jacob Kahn, Gergely Szilvasy, Mike Lewis, et al. 2023. Ra-dit: Retrieval-augmented dual instruction tuning. In *The Twelfth International Conference on Learning Representations*.

Wang Ling, Dani Yogatama, Chris Dyer, and Phil Blunsom. 2017. Program induction by rationale generation: Learning to solve and explain algebraic word problems. *arXiv preprint arXiv:1705.04146*.

Aiwei Liu, Leyi Pan, Yijian Lu, Jingjing Li, Xuming Hu, Xi Zhang, Lijie Wen, Irwin King, Hui Xiong, and Philip Yu. 2024. A survey of text watermarking in the era of large language models. *ACM Computing Surveys*, 57(2):1–36.

Pei Liu, Xin Liu, Ruoyu Yao, Junming Liu, Siyuan Meng, Ding Wang, and Jun Ma. 2025. Hm-rag: Hierarchical multi-agent multimodal retrieval augmented generation. *arXiv preprint arXiv:2504.12330*.

Chang Han Low, Ziyue Wang, Tianyi Zhang, Zhitao Zeng, Zhu Zhuo, Evangelos B Mazomenos, and Yueming Jin. 2025. Surgraw: Multi-agent workflow with chain-of-thought reasoning for surgical intelligence. *arXiv preprint arXiv:2503.10265*.

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The ai scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*.

Pan Lu, Swaroop Mishra, Tanglin Xia, Liang Qiu, Kai-Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter Clark, and Ashwin Kalyan. 2022. Learn to explain: Multimodal reasoning via thought chains for science question answering. *Advances in Neural Information Processing Systems*, 35:2507–2521.

Junyu Luo, Weizhi Zhang, Ye Yuan, Yusheng Zhao, Junwei Yang, Yiyang Gu, Bohan Wu, Binqi Chen, Ziyue Qiao, Qingqing Long, et al. 2025a. Large language model agent: A survey on methodology, applications and challenges. *arXiv preprint arXiv:2503.21460*.

Man Luo, Xin Xu, Zhuyun Dai, Panupong Pasupat, Mehran Kazemi, Chitta Baral, Vaiva Imbrasaite, and Vincent Y Zhao. 2023. Dr. icl: Demonstration-retrieved in-context learning. *arXiv preprint arXiv:2305.14128*.

Ne Luo, Aryo Pradipta Gema, Xuanli He, Emile van Krieken, Pietro Lesci, and Pasquale Minervini. 2025b. Self-training large language models for tool-use without demonstrations. *arXiv preprint arXiv:2502.05867*.

Shengjie Ma, Chengjin Xu, Xuhui Jiang, Muzhi Li, Huaren Qu, Cehao Yang, Jiaxin Mao, and Jian Guo. 2024a. Think-on-graph 2.0: Deep and faithful large language model reasoning with knowledge-guided retrieval augmented generation. *arXiv preprint arXiv:2407.10805*.

Yubo Ma, Zhibin Gou, Junheng Hao, Ruochen Xu, Shuohang Wang, Liangming Pan, Yujiu Yang, Yixin Cao, and Aixin Sun. 2024b. Sciagent: Tool-augmented language models for scientific reasoning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15701–15736.

Yubo Ma, Yuhang Zang, Liangyu Chen, Meiqi Chen, Yizhu Jiao, Xinze Li, Xinyuan Lu, Ziyu Liu, Yan Ma, Xiaoyi Dong, et al. 2025. Mmlongbench-doc: Benchmarking long-context document understanding with visualizations. *Advances in Neural Information Processing Systems*, 37:95963–96010.

Kelong Mao, Zheng Liu, Hongjin Qian, Fengran Mo, Chenlong Deng, and Zhicheng Dou. 2024. Ragstudio: Towards in-domain adaptation of retrieval

14

augmented generation through self-alignment. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 725–735.

Maria Marina, Nikolay Ivanov, Sergey Pletenev, Mikhail Salnikov, Daria Galimzianova, Nikita Krayko, Vasily Konovalov, Alexander Panchenko, and Viktor Moskvoretskii. 2025. Llm-independent adaptive rag: Let the question speak for itself. *arXiv preprint arXiv:2505.04253*.

Costas Mavromatis and George Karypis. 2024. Gnn-rag: Graph neural retrieval for large language model reasoning. *arXiv preprint arXiv:2405.20139*.

Grégoire Mialon, Clémentine Fourrier, Thomas Wolf, Yann LeCun, and Thomas Scialom. 2023. Gaia: a benchmark for general ai assistants. In *The Twelfth International Conference on Learning Representations*.

Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. 2021. Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*.

Shashi Narayan, Shay B Cohen, and Mirella Lapata. 2018. Don't give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 1797–1807.

Xuan-Phi Nguyen, Shrey Pandit, Senthil Purushwalkam, Austin Xu, Hailin Chen, Yifei Ming, Zixuan Ke, Silvio Savarese, Caiming Xong, and Shafiq Joty. 2024. Sfr-rag: Towards contextually faithful llms. *arXiv preprint arXiv:2409.09916*.

Cheng Niu, Yang Guan, Yuanhao Wu, Juno Zhu, Juntong Song, Randy Zhong, Kaihua Zhu, Siliang Xu, Shizhe Diao, and Tong Zhang. 2024. Veract scan: Retrieval-augmented fake news detection with justifiable reasoning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 266–277.

Yasumasa Onoe, Michael J.Q. Zhang, Eunsol Choi, and Greg Durrett. 2021. Creak: A dataset for commonsense reasoning over entity knowledge. *OpenReview*.

Richard Yuanzhe Pang, Alicia Parrish, Nitish Joshi, Nikita Nangia, Jason Phang, Angelica Chen, Vishakh Padmakumar, Johnny Ma, Jana Thompson, He He, et al. 2022. Quality: Question answering with long input texts, yes! In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5336–5358.

Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, Chen Bo Calvin Zhang, Mohamed Shaaban, John Ling, Sean Shi, et al. 2025. Humanity's last exam. *arXiv preprint arXiv:2501.14249*.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A Smith, and Mike Lewis. 2023. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711.

Shuofei Qiao, Honghao Gui, Chengfei Lv, Qianghuai Jia, Huajun Chen, and Ningyu Zhang. 2024. Making language models better tool learners with execution feedback. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3550–3568.

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, et al. 2023. Toolllm: Facilitating large language models to master 16000+ real-world apis. *arXiv preprint arXiv:2307.16789*.

Leonardo Ranaldi, Marco Valentino, and Andrè Freitas. 2024. Eliciting critical reasoning in retrieval-augmented language models via contrastive explanations. *arXiv preprint arXiv:2410.22874*.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. 2024. Gpqa: A graduate-level google-proof q&a benchmark. In *First Conference on Language Modeling*.

Aniruddha Salve, Saba Attar, Mahesh Deshmukh, Sayali Shivpuje, and Arnab Mitra Utsab. 2024. A collaborative multi-agent approach to retrieval-augmented generation across diverse data. *arXiv preprint arXiv:2412.05838*.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36:68539–68551.

Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Zicheng Liu, and Emad Barsoum. 2025. Agent laboratory: Using llm agents as research assistants. *arXiv preprint arXiv:2501.04227*.

Thomas Schmied, Fabian Paischer, Vihang Patil, Markus Hofmarcher, Razvan Pascanu, and Sepp Hochreiter. 2024. Retrieval-augmented decision transformer: External memory for in-context rl. *arXiv preprint arXiv:2410.07071*.

Johannes Schneider. 2025. Generative to agentic ai: Survey, conceptualization, and challenges. *arXiv preprint arXiv:2504.18875*.

augmented generation through self-alignment. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 725–735.

Maria Marina, Nikolay Ivanov, Sergey Pletenev, Mikhail Salnikov, Daria Galimzianova, Nikita Krayko, Vasily Konovalov, Alexander Panchenko, and Viktor Moskvoretskii. 2025. Llm-independent adaptive rag: Let the question speak for itself. *arXiv preprint arXiv:2505.04253*.

Costas Mavromatis and George Karypis. 2024. Gnn-rag: Graph neural retrieval for large language model reasoning. *arXiv preprint arXiv:2405.20139*.

Grégoire Mialon, Clémentine Fourrier, Thomas Wolf, Yann LeCun, and Thomas Scialom. 2023. Gaia: a benchmark for general ai assistants. In *The Twelfth International Conference on Learning Representations*.

Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. 2021. Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*.

Shashi Narayan, Shay B Cohen, and Mirella Lapata. 2018. Don't give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 1797–1807.

Xuan-Phi Nguyen, Shrey Pandit, Senthil Purushwalkam, Austin Xu, Hailin Chen, Yifei Ming, Zixuan Ke, Silvio Savarese, Caiming Xong, and Shafiq Joty. 2024. Sfr-rag: Towards contextually faithful llms. *arXiv preprint arXiv:2409.09916*.

Cheng Niu, Yang Guan, Yuanhao Wu, Juno Zhu, Juntong Song, Randy Zhong, Kaihua Zhu, Siliang Xu, Shizhe Diao, and Tong Zhang. 2024. Veract scan: Retrieval-augmented fake news detection with justifiable reasoning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 266–277.

Yasumasa Onoe, Michael J.Q. Zhang, Eunsol Choi, and Greg Durrett. 2021. Creak: A dataset for commonsense reasoning over entity knowledge. *OpenReview*.

Richard Yuanzhe Pang, Alicia Parrish, Nitish Joshi, Nikita Nangia, Jason Phang, Angelica Chen, Vishakh Padmakumar, Johnny Ma, Jana Thompson, He He, et al. 2022. Quality: Question answering with long input texts, yes! In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5336–5358.

Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, Chen Bo Calvin Zhang, Mohamed Shaaban, John Ling, Sean Shi, et al. 2025. Humanity's last exam. *arXiv preprint arXiv:2501.14249*.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A Smith, and Mike Lewis. 2023. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711.

Shuofei Qiao, Honghao Gui, Chengfei Lv, Qianghuai Jia, Huajun Chen, and Ningyu Zhang. 2024. Making language models better tool learners with execution feedback. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3550–3568.

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, et al. 2023. Toolllm: Facilitating large language models to master 16000+ real-world apis. *arXiv preprint arXiv:2307.16789*.

Leonardo Ranaldi, Marco Valentino, and Andrè Freitas. 2024. Eliciting critical reasoning in retrieval-augmented language models via contrastive explanations. *arXiv preprint arXiv:2410.22874*.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. 2024. Gpqa: A graduate-level google-proof q&a benchmark. In *First Conference on Language Modeling*.

Aniruddha Salve, Saba Attar, Mahesh Deshmukh, Sayali Shivpuje, and Arnab Mitra Utsab. 2024. A collaborative multi-agent approach to retrieval-augmented generation across diverse data. *arXiv preprint arXiv:2412.05838*.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36:68539–68551.

Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Zicheng Liu, and Emad Barsoum. 2025. Agent laboratory: Using llm agents as research assistants. *arXiv preprint arXiv:2501.04227*.

Thomas Schmied, Fabian Paischer, Vihang Patil, Markus Hofmarcher, Razvan Pascanu, and Sepp Hochreiter. 2024. Retrieval-augmented decision transformer: External memory for in-context rl. *arXiv preprint arXiv:2410.07071*.

Johannes Schneider. 2025. Generative to agentic ai: Survey, conceptualization, and challenges. *arXiv preprint arXiv:2504.18875*.

15

Eva Sharma, Chen Li, and Lu Wang. 2019. Bigpatent: A large-scale dataset for abstractive and coherent summarization. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2204–2213.

Ola Shorinwa, Zhiting Mei, Justin Lidard, Allen Z Ren, and Anirudha Majumdar. 2025. A survey on uncertainty quantification of large language models: Taxonomy, open research challenges, and future directions. *ACM Computing Surveys*.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Cote, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. Alfworld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*.

Huatong Song, Jinhao Jiang, Yingqian Min, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, and Ji-Rong Wen. 2025. R1-searcher: Incentivizing the search capability in llms via reinforcement learning. *arXiv preprint arXiv:2503.05592*.

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, et al. 2022. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *arXiv preprint arXiv:2206.04615*.

Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Hanjie Chen, et al. 2025. Stop overthinking: A survey on efficient reasoning for large language models. *arXiv preprint arXiv:2503.16419*.

Chuanneng Sun, Songjun Huang, and Dario Pompili. 2024a. Retrieval-augmented hierarchical in-context reinforcement learning and hindsight modular reflections for task planning with llms. *arXiv preprint arXiv:2408.06520*.

Haitian Sun, Tania Bedrax-Weiss, and William Cohen. 2019. Pullnet: Open domain question answering with iterative retrieval on knowledge bases and text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2380–2390.

Hao Sun, Zile Qiao, Jiayan Guo, Xuanbo Fan, Yingyan Hou, Yong Jiang, Pengjun Xie, Fei Huang, and Yan Zhang. 2025a. Zerosearch: Incentivize the search capability of llms without searching. *arXiv preprint arXiv:2505.04588*.

Jiashuo Sun, Chengjin Xu, Lumingyuan Tang, Saizhuo Wang, Chen Lin, Yeyun Gong, Lionel Ni, Heung-Yeung Shum, and Jian Guo. 2024b. Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph. In *The Twelfth International Conference on Learning Representations*.

Qiang Sun, Tingting Bi, Sirui Li, Eun-Jung Holden, Paul Duuring, Kai Niu, and Wei Liu. 2025b. Symbioticrag: Enhancing document intelligence through human-llm symbiotic collaboration. *arXiv preprint arXiv:2505.02418*.

Zhongxiang Sun, Qipeng Wang, Weijie Yu, Xiaoxue Zang, Kai Zheng, Jun Xu, Xiao Zhang, Song Yang, and Han Li. 2025c. Rearter: Retrieval-augmented reasoning with trustworthy process rewarding. *arXiv preprint arXiv:2501.07861*.

Alon Talmor and Jonathan Berant. 2018. The web as a knowledge-base for answering complex questions. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 641–651.

Yixuan Tang and Yi Yang. 2024. Multihop-rag: Benchmarking retrieval-augmented generation for multi-hop queries. *arXiv preprint arXiv:2401.15391*.

Yicheng Tao, Haotian Liu, Shanwen Wang, and Hongteng Xu. 2025. Assisting mathematical formalization with a learning-based premise retriever. *arXiv preprint arXiv:2501.13959*.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. Fever: a large-scale dataset for fact extraction and verification. *arXiv preprint arXiv:1803.05355*.

SM Tonmoy, SM Zaman, Vinija Jain, Anku Rani, Vipula Rawte, Aman Chadha, and Amitava Das. 2024. A comprehensive survey of hallucination mitigation techniques in large language models. *arXiv preprint arXiv:2401.01313*.

Hieu Tran, Zonghai Yao, Junda Wang, Yifan Zhang, Zhichao Yang, and Hong Yu. 2024. Rare: Retrieval-augmented reasoning enhancement for large language models. *arXiv preprint arXiv:2412.02830*.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. ♫ musique: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10014–10037.

Nikolaos Tsantalis, Ameya Ketkar, and Danny Dig. 2020. Refactoringminer 2.0. *IEEE Transactions on Software Engineering*, 48(3):930–950.

Boxin Wang, Wei Ping, Lawrence Mcafee, Peng Xu, Bo Li, Mohammad Shoeybi, and Bryan Catanzaro. 2024a. Instructretro: Instruction tuning post retrieval-augmented pretraining. In *International Conference on Machine Learning*, pages 51255–51272. PMLR.

Eva Sharma, Chen Li, and Lu Wang. 2019. Bigpatent: A large-scale dataset for abstractive and coherent summarization. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2204–2213.

Ola Shorinwa, Zhiting Mei, Justin Lidard, Allen Z Ren, and Anirudha Majumdar. 2025. A survey on uncertainty quantification of large language models: Taxonomy, open research challenges, and future directions. *ACM Computing Surveys*.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Cote, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. Alfworld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*.

Huatong Song, Jinhao Jiang, Yingqian Min, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, and Ji-Rong Wen. 2025. R1-searcher: Incentivizing the search capability in llms via reinforcement learning. *arXiv preprint arXiv:2503.05592*.

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, et al. 2022. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *arXiv preprint arXiv:2206.04615*.

Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Hanjie Chen, et al. 2025. Stop overthinking: A survey on efficient reasoning for large language models. *arXiv preprint arXiv:2503.16419*.

Chuanneng Sun, Songjun Huang, and Dario Pompili. 2024a. Retrieval-augmented hierarchical in-context reinforcement learning and hindsight modular reflections for task planning with llms. *arXiv preprint arXiv:2408.06520*.

Haitian Sun, Tania Bedrax-Weiss, and William Cohen. 2019. Pullnet: Open domain question answering with iterative retrieval on knowledge bases and text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2380–2390.

Hao Sun, Zile Qiao, Jiayan Guo, Xuanbo Fan, Yingyan Hou, Yong Jiang, Pengjun Xie, Fei Huang, and Yan Zhang. 2025a. Zerosearch: Incentivize the search capability of llms without searching. *arXiv preprint arXiv:2505.04588*.

Jiashuo Sun, Chengjin Xu, Lumingyuan Tang, Saizhuo Wang, Chen Lin, Yeyun Gong, Lionel Ni, Heung-Yeung Shum, and Jian Guo. 2024b. Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph. In *The Twelfth International Conference on Learning Representations*.

Qiang Sun, Tingting Bi, Sirui Li, Eun-Jung Holden, Paul Duuring, Kai Niu, and Wei Liu. 2025b. Symbioticrag: Enhancing document intelligence through human-llm symbiotic collaboration. *arXiv preprint arXiv:2505.02418*.

Zhongxiang Sun, Qipeng Wang, Weijie Yu, Xiaoxue Zang, Kai Zheng, Jun Xu, Xiao Zhang, Song Yang, and Han Li. 2025c. Rearter: Retrieval-augmented reasoning with trustworthy process rewarding. *arXiv preprint arXiv:2501.07861*.

Alon Talmor and Jonathan Berant. 2018. The web as a knowledge-base for answering complex questions. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 641–651.

Yixuan Tang and Yi Yang. 2024. Multihop-rag: Benchmarking retrieval-augmented generation for multi-hop queries. *arXiv preprint arXiv:2401.15391*.

Yicheng Tao, Haotian Liu, Shanwen Wang, and Hongteng Xu. 2025. Assisting mathematical formalization with a learning-based premise retriever. *arXiv preprint arXiv:2501.13959*.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. Fever: a large-scale dataset for fact extraction and verification. *arXiv preprint arXiv:1803.05355*.

SM Tonmoy, SM Zaman, Vinija Jain, Anku Rani, Vipula Rawte, Aman Chadha, and Amitava Das. 2024. A comprehensive survey of hallucination mitigation techniques in large language models. *arXiv preprint arXiv:2401.01313*.

Hieu Tran, Zonghai Yao, Junda Wang, Yifan Zhang, Zhichao Yang, and Hong Yu. 2024. Rare: Retrieval-augmented reasoning enhancement for large language models. *arXiv preprint arXiv:2412.02830*.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. ♫ musique: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10014–10037.

Nikolaos Tsantalis, Ameya Ketkar, and Danny Dig. 2020. Refactoringminer 2.0. *IEEE Transactions on Software Engineering*, 48(3):930–950.

Boxin Wang, Wei Ping, Lawrence Mcafee, Peng Xu, Bo Li, Mohammad Shoeybi, and Bryan Catanzaro. 2024a. Instructretro: Instruction tuning post retrieval-augmented pretraining. In *International Conference on Machine Learning*, pages 51255–51272. PMLR.

16

Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Liu, Peter Van Katwyk, Andreea Deac, et al. 2023. Scientific discovery in the age of artificial intelligence. *Nature*, 620(7972):47–60.

Junjie Wang, Mingyang Chen, Binbin Hu, Dan Yang, Ziqi Liu, Yue Shen, Peng Wei, Zhiqiang Zhang, Jinjie Gu, Jun Zhou, et al. 2024b. Learning to plan for retrieval-augmented large language models from knowledge graphs. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 7813–7835.

Song Wang, Zihan Chen, Chengshuai Shi, Cong Shen, and Jundong Li. 2024c. Mixture of demonstrations for in-context learning. *Advances in Neural Information Processing Systems*, 37:88091–88116.

Yaoting Wang, Shengqiong Wu, Yuecheng Zhang, Shuicheng Yan, Ziwei Liu, Jiebo Luo, and Hao Fei. 2025a. Multimodal chain-of-thought reasoning: A comprehensive survey. *arXiv preprint arXiv:2503.12605*.

Yu Wang, Nedim Lipka, Ryan A Rossi, Alexa Siu, Ruiyi Zhang, and Tyler Derr. 2024d. Knowledge graph prompting for multi-document question answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19206–19214.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, et al. 2025b. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *Advances in Neural Information Processing Systems*, 37:95266–95290.

Yujing Wang, Hainan Zhang, Liang Pang, Binghui Guo, Hongwei Zheng, and Zhiming Zheng. 2025c. Maferw: Query rewriting with multi-aspect feedbacks for retrieval-augmented large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 25434–25442.

Zheng Wang, Shu Teo, Jieer Ouyang, Yongjun Xu, and Wei Shi. 2024e. M-rag: Reinforcing large language model performance through retrieval-augmented generation with multiple partitions. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1966–1978.

Zhengren Wang, Jiayang Yu, Dongsheng Ma, Zhe Chen, Yu Wang, Zhiyu Li, Feiyu Xiong, Yanfeng Wang, Linpeng Tang, Wentao Zhang, et al. 2025d. Rare: Retrieval-augmented reasoning modeling. *arXiv preprint arXiv:2503.23513*.

Zihao Wang, Shaofei Cai, Anji Liu, Yonggang Jin, Jinbing Hou, Bowei Zhang, Haowei Lin, Zhaofeng He, Zilong Zheng, Yaodong Yang, et al. 2024f. Jarvis-1: Open-world multi-task agents with memory-augmented multimodal language models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

Zihao Wang, Anji Liu, Haowei Lin, Jiaqi Li, Xiaojian Ma, and Yitao Liang. 2024g. Rat: Retrieval augmented thoughts elicit context-aware reasoning in long-horizon generation. *arXiv preprint arXiv:2403.05313*.

Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. 2024. Measuring short-form factuality in large language models. *arXiv preprint arXiv:2411.04368*.

Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. 2025a. Browsecomp: A simple yet challenging benchmark for browsing agents. *arXiv preprint arXiv:2504.12516*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.

Jiaqi Wei, Hao Zhou, Xiang Zhang, Di Zhang, Zijie Qiu, Wei Wei, Jinzhe Li, Wanli Ouyang, and Siqi Sun. 2025b. Alignrag: An adaptable framework for resolving misalignments in retrieval-aware reasoning of rag. *arXiv preprint arXiv:2504.14858*.

Zhihua Wen, Zhiliang Tian, Wei Wu, Yuxin Yang, Yanqi Shi, Zhen Huang, and Dongsheng Li. 2023. Grove: A retrieval-augmented complex story generation framework with a forest of evidence. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 3980–3998.

Nirmalie Wiratunga, Ramitha Abeyratne, Lasal Jayawardena, Kyle Martin, Stewart Massie, Ikechukwu Nkisi-Orji, Ruvan Weerasinghe, Anne Liret, and Bruno Fleisch. 2024. Cbr-rag: case-based reasoning for retrieval augmented generation in llms for legal question answering. In *International Conference on Case-Based Reasoning*, pages 445–460. Springer.

Feijie Wu, Zitao Li, Fei Wei, Yaliang Li, Bolin Ding, and Jing Gao. 2025a. Talk to right specialists: Routing and planning in multi-agent system for question answering. *arXiv preprint arXiv:2501.07813*.

Jialong Wu, Wenbiao Yin, Yong Jiang, Zhenglin Wang, Zekun Xi, Runnan Fang, Linhai Zhang, Yulan He, Deyu Zhou, Pengjun Xie, et al. 2025b. Webwalker: Benchmarking llms in web traversal. *arXiv preprint arXiv:2501.07572*.

Junde Wu, Jiayuan Zhu, and Yuyuan Liu. 2025c. Agentic reasoning: Reasoning llms with tools for the deep research. *arXiv preprint arXiv:2502.04644*.

Shirley Wu, Shiyu Zhao, Qian Huang, Kexin Huang, Michihiro Yasunaga, Kaidi Cao, Vassilis Ioannidis, Karthik Subbian, Jure Leskovec, and James Y Zou. 2024. Avatar: Optimizing llm agents for tool usage via contrastive reasoning. *Advances in Neural Information Processing Systems*, 37:25981–26010.

Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Liu, Peter Van Katwyk, Andreea Deac, et al. 2023. Scientific discovery in the age of artificial intelligence. *Nature*, 620(7972):47–60.

Junjie Wang, Mingyang Chen, Binbin Hu, Dan Yang, Ziqi Liu, Yue Shen, Peng Wei, Zhiqiang Zhang, Jinjie Gu, Jun Zhou, et al. 2024b. Learning to plan for retrieval-augmented large language models from knowledge graphs. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 7813–7835.

Song Wang, Zihan Chen, Chengshuai Shi, Cong Shen, and Jundong Li. 2024c. Mixture of demonstrations for in-context learning. *Advances in Neural Information Processing Systems*, 37:88091–88116.

Yaoting Wang, Shengqiong Wu, Yuecheng Zhang, Shuicheng Yan, Ziwei Liu, Jiebo Luo, and Hao Fei. 2025a. Multimodal chain-of-thought reasoning: A comprehensive survey. *arXiv preprint arXiv:2503.12605*.

Yu Wang, Nedim Lipka, Ryan A Rossi, Alexa Siu, Ruiyi Zhang, and Tyler Derr. 2024d. Knowledge graph prompting for multi-document question answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19206–19214.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, et al. 2025b. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *Advances in Neural Information Processing Systems*, 37:95266–95290.

Yujing Wang, Hainan Zhang, Liang Pang, Binghui Guo, Hongwei Zheng, and Zhiming Zheng. 2025c. Maferw: Query rewriting with multi-aspect feedbacks for retrieval-augmented large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 25434–25442.

Zheng Wang, Shu Teo, Jieer Ouyang, Yongjun Xu, and Wei Shi. 2024e. M-rag: Reinforcing large language model performance through retrieval-augmented generation with multiple partitions. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1966–1978.

Zhengren Wang, Jiayang Yu, Dongsheng Ma, Zhe Chen, Yu Wang, Zhiyu Li, Feiyu Xiong, Yanfeng Wang, Linpeng Tang, Wentao Zhang, et al. 2025d. Rare: Retrieval-augmented reasoning modeling. *arXiv preprint arXiv:2503.23513*.

Zihao Wang, Shaofei Cai, Anji Liu, Yonggang Jin, Jinbing Hou, Bowei Zhang, Haowei Lin, Zhaofeng He, Zilong Zheng, Yaodong Yang, et al. 2024f. Jarvis-1: Open-world multi-task agents with memory-augmented multimodal language models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

Zihao Wang, Anji Liu, Haowei Lin, Jiaqi Li, Xiaojian Ma, and Yitao Liang. 2024g. Rat: Retrieval augmented thoughts elicit context-aware reasoning in long-horizon generation. *arXiv preprint arXiv:2403.05313*.

Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. 2024. Measuring short-form factuality in large language models. *arXiv preprint arXiv:2411.04368*.

Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. 2025a. Browsecomp: A simple yet challenging benchmark for browsing agents. *arXiv preprint arXiv:2504.12516*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.

Jiaqi Wei, Hao Zhou, Xiang Zhang, Di Zhang, Zijie Qiu, Wei Wei, Jinzhe Li, Wanli Ouyang, and Siqi Sun. 2025b. Alignrag: An adaptable framework for resolving misalignments in retrieval-aware reasoning of rag. *arXiv preprint arXiv:2504.14858*.

Zhihua Wen, Zhiliang Tian, Wei Wu, Yuxin Yang, Yanqi Shi, Zhen Huang, and Dongsheng Li. 2023. Grove: A retrieval-augmented complex story generation framework with a forest of evidence. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 3980–3998.

Nirmalie Wiratunga, Ramitha Abeyratne, Lasal Jayawardena, Kyle Martin, Stewart Massie, Ikechukwu Nkisi-Orji, Ruvan Weerasinghe, Anne Liret, and Bruno Fleisch. 2024. Cbr-rag: case-based reasoning for retrieval augmented generation in llms for legal question answering. In *International Conference on Case-Based Reasoning*, pages 445–460. Springer.

Feijie Wu, Zitao Li, Fei Wei, Yaliang Li, Bolin Ding, and Jing Gao. 2025a. Talk to right specialists: Routing and planning in multi-agent system for question answering. *arXiv preprint arXiv:2501.07813*.

Jialong Wu, Wenbiao Yin, Yong Jiang, Zhenglin Wang, Zekun Xi, Runnan Fang, Linhai Zhang, Yulan He, Deyu Zhou, Pengjun Xie, et al. 2025b. Webwalker: Benchmarking llms in web traversal. *arXiv preprint arXiv:2501.07572*.

Junde Wu, Jiayuan Zhu, and Yuyuan Liu. 2025c. Agentic reasoning: Reasoning llms with tools for the deep research. *arXiv preprint arXiv:2502.04644*.

Shirley Wu, Shiyu Zhao, Qian Huang, Kexin Huang, Michihiro Yasunaga, Kaidi Cao, Vassilis Ioannidis, Karthik Subbian, Jure Leskovec, and James Y Zou. 2024. Avatar: Optimizing llm agents for tool usage via contrastive reasoning. *Advances in Neural Information Processing Systems*, 37:25981–26010.

17

Heming Xia, Yongqi Li, Chak Tou Leong, Wenjie Wang, and Wenjie Li. 2025a. Tokenskip: Controllable chain-of-thought compression in llms. *arXiv preprint arXiv:2502.12067*.

Yuan Xia, Jingbo Zhou, Zhenhui Shi, Jun Chen, and Haifeng Huang. 2025b. Improving retrieval augmented language model with self-reasoning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 39, pages 25534–25542.

Guangzhi Xiong, Qiao Jin, Xiao Wang, Yin Fang, Haolin Liu, Yifan Yang, Fangyuan Chen, Zhixing Song, Dengyu Wang, Minjia Zhang, et al. 2025. Rag-gym: Optimizing reasoning and search agents with process supervision. *arXiv preprint arXiv:2502.13957*.

Kehan Xu, Kun Zhang, Jingyuan Li, Wei Huang, and Yuanzhuo Wang. 2024. Crp-rag: A retrieval-augmented generation framework for supporting complex logical reasoning and knowledge planning. *Electronics*, 14(1):47.

Kehan Xu, Kun Zhang, Jingyuan Li, Wei Huang, and Yuanzhuo Wang. 2025a. Crp-rag: A retrieval-augmented generation framework for supporting complex logical reasoning and knowledge planning. *Electronics (2079-9292)*, 14(1).

Ran Xu, Wenqi Shi, Yuchen Zhuang, Yue Yu, Joyce C Ho, Haoyu Wang, and Carl Yang. 2025b. Collab-rag: Boosting retrieval-augmented generation for complex question answering via white-box and black-box llm collaboration. *arXiv preprint arXiv:2504.04915*.

Chen Yang, Chenyang Zhao, Quanquan Gu, and Dongruo Zhou. 2024a. Cops: Empowering llm agents with provable cross-task experience sharing. *arXiv preprint arXiv:2410.16670*.

Rui Yang. 2024. Casegpt: a case reasoning framework based on language models and retrieval-augmented generation. *arXiv preprint arXiv:2407.07913*.

Wooseong Yang, Weizhi Zhang, Yuqing Liu, Yuwei Han, Yu Wang, Junhyun Lee, and Philip S Yu. 2025. Cold-start recommendation with knowledge-guided retrieval-augmented generation. *arXiv preprint arXiv:2505.20773*.

Xiao Yang, Kai Sun, Hao Xin, Yushi Sun, Nikita Bhalla, Xiangsen Chen, Sajal Choudhary, Rongze Gui, Ziran Jiang, Ziyu Jiang, et al. 2024b. Crag-comprehensive rag benchmark. *Advances in Neural Information Processing Systems*, 37:10470–10490.

Yahe Yang and Chengyue Huang. 2025. Tree-based rag-agent recommendation system: A case study in medical test data. *arXiv preprint arXiv:2501.02727*.

Yi Yang, Wen-tau Yih, and Christopher Meek. 2015. Wikiqa: A challenge dataset for open-domain question answering. In *Proceedings of the 2015 conference on empirical methods in natural language processing*, pages 2013–2018.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. 2022. Webshop: Towards scalable real-world web interaction with grounded language agents. *Advances in Neural Information Processing Systems*, 35:20744–20757.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. 2023a. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36:11809–11822.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023b. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Michihiro Yasunaga, Hongyu Ren, Antoine Bosselut, Percy Liang, and Jure Leskovec. 2021. Qa-gnn: Reasoning with language models and knowledge graphs for question answering. In *North American Chapter of the Association for Computational Linguistics (NAACL)*.

Jaeseok Yoo, Hojae Han, Youngwon Lee, Jaejin Kim, and Seung-won Hwang. 2025. Perc: Plan-as-query example retrieval for underrepresented code generation. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 7982–7997.

Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. 2024. Making retrieval-augmented language models robust to irrelevant context. In *ICLR 2024 Workshop on Large Language Model (LLM) Agents*.

Hong Qing Yu and Frank McQuade. 2025. Rag-kg-il: A multi-agent hybrid framework for reducing hallucinations and enhancing llm reasoning through rag and incremental knowledge graph learning integration. *arXiv preprint arXiv:2503.13514*.

Wenhao Yu, Hongming Zhang, Xiaoman Pan, Peixin Cao, Kaixin Ma, Jian Li, Hongwei Wang, and Dong Yu. 2024. Chain-of-note: Enhancing robustness in retrieval-augmented language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14672–14685.

Jing Zhang, Xiaokang Zhang, Jifan Yu, Jian Tang, Jie Tang, Cuiping Li, and Hong Chen. 2022a. Subgraph retrieval enhanced model for multi-hop knowledge base question answering. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5773–5784.

Heming Xia, Yongqi Li, Chak Tou Leong, Wenjie Wang, and Wenjie Li. 2025a. Tokenskip: Controllable chain-of-thought compression in llms. *arXiv preprint arXiv:2502.12067*.

Yuan Xia, Jingbo Zhou, Zhenhui Shi, Jun Chen, and Haifeng Huang. 2025b. Improving retrieval augmented language model with self-reasoning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 39, pages 25534–25542.

Guangzhi Xiong, Qiao Jin, Xiao Wang, Yin Fang, Haolin Liu, Yifan Yang, Fangyuan Chen, Zhixing Song, Dengyu Wang, Minjia Zhang, et al. 2025. Rag-gym: Optimizing reasoning and search agents with process supervision. *arXiv preprint arXiv:2502.13957*.

Kehan Xu, Kun Zhang, Jingyuan Li, Wei Huang, and Yuanzhuo Wang. 2024. Crp-rag: A retrieval-augmented generation framework for supporting complex logical reasoning and knowledge planning. *Electronics*, 14(1):47.

Kehan Xu, Kun Zhang, Jingyuan Li, Wei Huang, and Yuanzhuo Wang. 2025a. Crp-rag: A retrieval-augmented generation framework for supporting complex logical reasoning and knowledge planning. *Electronics (2079-9292)*, 14(1).

Ran Xu, Wenqi Shi, Yuchen Zhuang, Yue Yu, Joyce C Ho, Haoyu Wang, and Carl Yang. 2025b. Collab-rag: Boosting retrieval-augmented generation for complex question answering via white-box and black-box llm collaboration. *arXiv preprint arXiv:2504.04915*.

Chen Yang, Chenyang Zhao, Quanquan Gu, and Dongruo Zhou. 2024a. Cops: Empowering llm agents with provable cross-task experience sharing. *arXiv preprint arXiv:2410.16670*.

Rui Yang. 2024. Casegpt: a case reasoning framework based on language models and retrieval-augmented generation. *arXiv preprint arXiv:2407.07913*.

Wooseong Yang, Weizhi Zhang, Yuqing Liu, Yuwei Han, Yu Wang, Junhyun Lee, and Philip S Yu. 2025. Cold-start recommendation with knowledge-guided retrieval-augmented generation. *arXiv preprint arXiv:2505.20773*.

Xiao Yang, Kai Sun, Hao Xin, Yushi Sun, Nikita Bhalla, Xiangsen Chen, Sajal Choudhary, Rongze Gui, Ziran Jiang, Ziyu Jiang, et al. 2024b. Crag-comprehensive rag benchmark. *Advances in Neural Information Processing Systems*, 37:10470–10490.

Yahe Yang and Chengyue Huang. 2025. Tree-based rag-agent recommendation system: A case study in medical test data. *arXiv preprint arXiv:2501.02727*.

Yi Yang, Wen-tau Yih, and Christopher Meek. 2015. Wikiqa: A challenge dataset for open-domain question answering. In *Proceedings of the 2015 conference on empirical methods in natural language processing*, pages 2013–2018.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. 2022. Webshop: Towards scalable real-world web interaction with grounded language agents. *Advances in Neural Information Processing Systems*, 35:20744–20757.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. 2023a. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36:11809–11822.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023b. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Michihiro Yasunaga, Hongyu Ren, Antoine Bosselut, Percy Liang, and Jure Leskovec. 2021. Qa-gnn: Reasoning with language models and knowledge graphs for question answering. In *North American Chapter of the Association for Computational Linguistics (NAACL)*.

Jaeseok Yoo, Hojae Han, Youngwon Lee, Jaejin Kim, and Seung-won Hwang. 2025. Perc: Plan-as-query example retrieval for underrepresented code generation. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 7982–7997.

Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. 2024. Making retrieval-augmented language models robust to irrelevant context. In *ICLR 2024 Workshop on Large Language Model (LLM) Agents*.

Hong Qing Yu and Frank McQuade. 2025. Rag-kg-il: A multi-agent hybrid framework for reducing hallucinations and enhancing llm reasoning through rag and incremental knowledge graph learning integration. *arXiv preprint arXiv:2503.13514*.

Wenhao Yu, Hongming Zhang, Xiaoman Pan, Peixin Cao, Kaixin Ma, Jian Li, Hongwei Wang, and Dong Yu. 2024. Chain-of-note: Enhancing robustness in retrieval-augmented language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14672–14685.

Jing Zhang, Xiaokang Zhang, Jifan Yu, Jian Tang, Jie Tang, Cuiping Li, and Hong Chen. 2022a. Subgraph retrieval enhanced model for multi-hop knowledge base question answering. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5773–5784.

18

Jinghan Zhang, Xiting Wang, Weijieying Ren, Lu Jiang, Dongjie Wang, and Kunpeng Liu. 2025a. Ratt: A thought structure for coherent and correct llm reasoning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 26733–26741.

Jintian Zhang, Yuqi Zhu, Mengshu Sun, Yujie Luo, Shuofei Qiao, Lun Du, Da Zheng, Huajun Chen, and Ningyu Zhang. 2025b. Lightthinker: Thinking step-by-step compression. *arXiv preprint arXiv:2502.15589*.

Miao Zhang, Zhenlong Fang, Tianyi Wang, Qian Zhang, Shuai Lu, Junfeng Jiao, and Tianyu Shi. 2025c. A cascading cooperative multi-agent framework for on-ramp merging control integrating large language models. *arXiv preprint arXiv:2503.08199*.

Ningning Zhang, Chi Zhang, Zhizhong Tan, Xingxing Yang, Weiping Deng, and Wenyong Wang. 2025d. Credible plan-driven rag method for multi-hop question answering. *arXiv preprint arXiv:2504.16787*.

Tianjun Zhang, Shishir G Patil, Naman Jain, Sheng Shen, Matei Zaharia, Ion Stoica, and Joseph E Gonzalez. 2024a. Raft: Adapting language model to domain specific rag. In *First Conference on Language Modeling*.

Weizhi Zhang, Yuanchen Bei, Liangwei Yang, Henry Peng Zou, Peilin Zhou, Aiwei Liu, Yinghui Li, Hao Chen, Jianling Wang, Yu Wang, et al. 2025e. Cold-start recommendation towards the era of large language models (llms): A comprehensive survey and roadmap. *arXiv preprint arXiv:2501.01945*.

Weizhi Zhang, Yangning Li, Yuanchen Bei, Junyu Luo, Guancheng Wan, Liangwei Yang, Chenxuan Xie, Yuyao Yang, Wei-Chieh Huang, Chunyu Miao, et al. 2025f. From web search towards agentic deep research: Incentivizing search with reasoning agents. *arXiv preprint arXiv:2506.18959*.

Weizhi Zhang, Xinyang Zhang, Chenwei Zhang, Liangwei Yang, Jingbo Shang, Zhepei Wei, Henry Peng Zou, Zijie Huang, Zhengyang Wang, Yifan Gao, et al. 2025g. Personaagent: When large language model agents meet personalization at test time. *arXiv preprint arXiv:2506.06254*.

Xikun Zhang, Antoine Bosselut, Michihiro Yasunaga, Hongyu Ren, Percy Liang, Christopher D Manning, and Jure Leskovec. 2022b. Greaselm: Graph reasoning enhanced language models. In *International Conference on Learning Representations*.

Xinrong Zhang, Yingfa Chen, Shengding Hu, Zihang Xu, Junhao Chen, Moo Hao, Xu Han, Zhen Thai, Shuo Wang, Zhiyuan Liu, et al. 2024b. ∞ bench: Extending long context evaluation beyond 100k tokens. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15262–15277.

Yusen Zhang, Ruoxi Sun, Yanfei Chen, Tomas Pfister, Rui Zhang, and Sercan Arik. 2024c. Chain of agents: Large language models collaborating on long-context tasks. *Advances in Neural Information Processing Systems*, 37:132208–132237.

Zhebin Zhang, Xinyu Zhang, Yuanhang Ren, Saijiang Shi, Meng Han, Yongkang Wu, Ruofei Lai, and Zhao Cao. 2023. Iag: Induction-augmented generation framework for answering reasoning questions. *arXiv preprint arXiv:2311.18397*.

Siyun Zhao, Yuqing Yang, Zilong Wang, Zhiyuan He, Luna K Qiu, and Lili Qiu. 2024a. Retrieval augmented generation (rag) and beyond: A comprehensive survey on how to make your llms use external data more wisely. *arXiv preprint arXiv:2409.14924*.

Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. 2023. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 1(2).

Xiaoyan Zhao, Lingzhi Wang, Zhanghao Wang, Hong Cheng, Rui Zhang, and Kam-Fai Wong. 2024b. Pacar: Automated fact-checking with planning and customized action reasoning using large language models. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 12564–12573.

Xinping Zhao, Dongfang Li, Yan Zhong, Boren Hu, Yibin Chen, Baotian Hu, and Min Zhang. 2024c. Seer: Self-aligned evidence extraction for retrieval-augmented generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 3027–3041.

Kunhao Zheng, Jesse Michael Han, and Stanislas Polu. 2021. Minif2f: a cross-system benchmark for formal olympiad-level mathematics. *arXiv preprint arXiv:2109.00110*.

Yuxiang Zheng, Dayuan Fu, Xiangkun Hu, Xiaojie Cai, Lyumanshan Ye, Pengrui Lu, and Pengfei Liu. 2025. Deepresearcher: Scaling deep research via reinforcement learning in real-world environments. *arXiv preprint arXiv:2504.03160*.

Jiawei Zhou and Lei Chen. 2025. Openrag: Optimizing rag end-to-end via in-context retrieval learning. *arXiv preprint arXiv:2503.08398*.

Peilin Zhou, Bruce Leon, Xiang Ying, Can Zhang, Yifan Shao, Qichen Ye, Dading Chong, Zhiling Jin, Chenxuan Xie, Meng Cao, et al. 2025a. Browsecomp-zh: Benchmarking web browsing ability of large language models in chinese. *arXiv preprint arXiv:2504.19314*.

Yifei Zhou, Song Jiang, Yuandong Tian, Jason Weston, Sergey Levine, Sainbayar Sukhbaatar, and Xian Li. 2025b. Sweet-rl: Training multi-turn llm agents on collaborative reasoning tasks. *arXiv preprint arXiv:2503.15478*.

Jinghan Zhang, Xiting Wang, Weijieying Ren, Lu Jiang, Dongjie Wang, and Kunpeng Liu. 2025a. Ratt: A thought structure for coherent and correct llm reasoning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 26733–26741.

Jintian Zhang, Yuqi Zhu, Mengshu Sun, Yujie Luo, Shuofei Qiao, Lun Du, Da Zheng, Huajun Chen, and Ningyu Zhang. 2025b. Lightthinker: Thinking step-by-step compression. *arXiv preprint arXiv:2502.15589*.

Miao Zhang, Zhenlong Fang, Tianyi Wang, Qian Zhang, Shuai Lu, Junfeng Jiao, and Tianyu Shi. 2025c. A cascading cooperative multi-agent framework for on-ramp merging control integrating large language models. *arXiv preprint arXiv:2503.08199*.

Ningning Zhang, Chi Zhang, Zhizhong Tan, Xingxing Yang, Weiping Deng, and Wenyong Wang. 2025d. Credible plan-driven rag method for multi-hop question answering. *arXiv preprint arXiv:2504.16787*.

Tianjun Zhang, Shishir G Patil, Naman Jain, Sheng Shen, Matei Zaharia, Ion Stoica, and Joseph E Gonzalez. 2024a. Raft: Adapting language model to domain specific rag. In *First Conference on Language Modeling*.

Weizhi Zhang, Yuanchen Bei, Liangwei Yang, Henry Peng Zou, Peilin Zhou, Aiwei Liu, Yinghui Li, Hao Chen, Jianling Wang, Yu Wang, et al. 2025e. Cold-start recommendation towards the era of large language models (llms): A comprehensive survey and roadmap. *arXiv preprint arXiv:2501.01945*.

Weizhi Zhang, Yangning Li, Yuanchen Bei, Junyu Luo, Guancheng Wan, Liangwei Yang, Chenxuan Xie, Yuyao Yang, Wei-Chieh Huang, Chunyu Miao, et al. 2025f. From web search towards agentic deep research: Incentivizing search with reasoning agents. *arXiv preprint arXiv:2506.18959*.

Weizhi Zhang, Xinyang Zhang, Chenwei Zhang, Liangwei Yang, Jingbo Shang, Zhepei Wei, Henry Peng Zou, Zijie Huang, Zhengyang Wang, Yifan Gao, et al. 2025g. Personaagent: When large language model agents meet personalization at test time. *arXiv preprint arXiv:2506.06254*.

Xikun Zhang, Antoine Bosselut, Michihiro Yasunaga, Hongyu Ren, Percy Liang, Christopher D Manning, and Jure Leskovec. 2022b. Greaselm: Graph reasoning enhanced language models. In *International Conference on Learning Representations*.

Xinrong Zhang, Yingfa Chen, Shengding Hu, Zihang Xu, Junhao Chen, Moo Hao, Xu Han, Zhen Thai, Shuo Wang, Zhiyuan Liu, et al. 2024b. ∞ bench: Extending long context evaluation beyond 100k tokens. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15262–15277.

Yusen Zhang, Ruoxi Sun, Yanfei Chen, Tomas Pfister, Rui Zhang, and Sercan Arik. 2024c. Chain of agents: Large language models collaborating on long-context tasks. *Advances in Neural Information Processing Systems*, 37:132208–132237.

Zhebin Zhang, Xinyu Zhang, Yuanhang Ren, Saijiang Shi, Meng Han, Yongkang Wu, Ruofei Lai, and Zhao Cao. 2023. Iag: Induction-augmented generation framework for answering reasoning questions. *arXiv preprint arXiv:2311.18397*.

Siyun Zhao, Yuqing Yang, Zilong Wang, Zhiyuan He, Luna K Qiu, and Lili Qiu. 2024a. Retrieval augmented generation (rag) and beyond: A comprehensive survey on how to make your llms use external data more wisely. *arXiv preprint arXiv:2409.14924*.

Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. 2023. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 1(2).

Xiaoyan Zhao, Lingzhi Wang, Zhanghao Wang, Hong Cheng, Rui Zhang, and Kam-Fai Wong. 2024b. Pacar: Automated fact-checking with planning and customized action reasoning using large language models. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 12564–12573.

Xinping Zhao, Dongfang Li, Yan Zhong, Boren Hu, Yibin Chen, Baotian Hu, and Min Zhang. 2024c. Seer: Self-aligned evidence extraction for retrieval-augmented generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 3027–3041.

Kunhao Zheng, Jesse Michael Han, and Stanislas Polu. 2021. Minif2f: a cross-system benchmark for formal olympiad-level mathematics. *arXiv preprint arXiv:2109.00110*.

Yuxiang Zheng, Dayuan Fu, Xiangkun Hu, Xiaojie Cai, Lyumanshan Ye, Pengrui Lu, and Pengfei Liu. 2025. Deepresearcher: Scaling deep research via reinforcement learning in real-world environments. *arXiv preprint arXiv:2504.03160*.

Jiawei Zhou and Lei Chen. 2025. Openrag: Optimizing rag end-to-end via in-context retrieval learning. *arXiv preprint arXiv:2503.08398*.

Peilin Zhou, Bruce Leon, Xiang Ying, Can Zhang, Yifan Shao, Qichen Ye, Dading Chong, Zhiling Jin, Chenxuan Xie, Meng Cao, et al. 2025a. Browsecomp-zh: Benchmarking web browsing ability of large language models in chinese. *arXiv preprint arXiv:2504.19314*.

Yifei Zhou, Song Jiang, Yuandong Tian, Jason Weston, Sergey Levine, Sainbayar Sukhbaatar, and Xian Li. 2025b. Sweet-rl: Training multi-turn llm agents on collaborative reasoning tasks. *arXiv preprint arXiv:2503.15478*.

19

Yutao Zhu, Peitian Zhang, Chenghao Zhang, Yifei Chen, Binyu Xie, Zheng Liu, Ji-Rong Wen, and Zhicheng Dou. 2024. Inters: Unlocking the power of large language models in search with instruction tuning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2782–2809.

Henry Peng Zou, Wei-Chieh Huang, Yaozu Wu, Yankai Chen, Chunyu Miao, Hoang Nguyen, Yue Zhou, Weizhi Zhang, Liancheng Fang, Langzhou He, et al. 2025. A survey on large language model based human-agent systems. *arXiv preprint arXiv:2505.00753*.

Yutao Zhu, Peitian Zhang, Chenghao Zhang, Yifei Chen, Binyu Xie, Zheng Liu, Ji-Rong Wen, and Zhicheng Dou. 2024. Inters: Unlocking the power of large language models in search with instruction tuning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2782–2809.

Henry Peng Zou, Wei-Chieh Huang, Yaozu Wu, Yankai Chen, Chunyu Miao, Hoang Nguyen, Yue Zhou, Weizhi Zhang, Liancheng Fang, Langzhou He, et al. 2025. A survey on large language model based human-agent systems. *arXiv preprint arXiv:2505.00753*.

20

A Full Benchmark

Section 6 introduces representative benchmarks for different RAG-reasoning tasks. This appendix complements that discussion with a comprehensive list of benchmarks organized by task and domain. Table 2 details each benchmark's attributes, including the publication venue, code repository, task category, domain, primary knowledge sources, knowledge type, and reasoning capabilities. By consolidating these attributes into a single table, we facilitate the selection and comparison of benchmarks, enabling researchers to identify the most suitable datasets for future studies on RAG-enhanced reasoning.

Our benchmark compilation is primarily derived from the methods surveyed in Sections 3 to 5 of this paper, with a particular focus on synergized approaches discussed in Section 5. We deliberately targeted benchmarks that require both external knowledge retrieval and internal deep reasoning, as this dual requirement reflects real-world scenarios where models must not only access relevant information but also integrate and reason over it effectively. For example, in the QA domain, we include datasets that necessitate synthesizing evidence across multiple documents to answer questions that cannot be resolved through single-sentence retrieval. HotpotQA (Yang et al., 2018) exemplifies this challenge, requiring reasoning across different Wikipedia articles. In coding tasks, benchmarks such as LiveCodeBench (Jain et al., 2024) and Refactoring Oracle (Tsantalis et al., 2020) extend beyond pure algorithmic problem-solving by demanding retrieval of external code snippets and documentation. Similarly, in mathematics, benchmarks like MATH (Hendrycks et al., 2021) and AQUA-RAT (Das et al., 2024) assess not only computational proficiency but also the retrieval of relevant theorems and formulas, testing the model's ability to integrate external mathematical knowledge with internal reasoning processes.

In addition to established benchmarks, we have incorporated newer and more challenging datasets that better mirror real-world applications. These datasets often demand extensive retrieval processes combined with expert-level or domain-specific reasoning, as seen in Humanity's Last Exam (HLE) (Phan et al., 2025) and web search evaluation tasks like BrowseComp (Wei et al., 2025a). Overall, our collection encompasses **46** benchmarks covering **13** distinct tasks across **12** domains, each explicitly annotated with features such as knowledge source, knowledge type, and reasoning capacity. This breadth ensures coverage of diverse domains and task types, forming a solid foundation for evaluating the interplay between retrieval and reasoning in RAG systems.

Within this benchmark set, single-hop QA datasets like TriviaQA (Joshi et al., 2017) focus on precise retrieval and fact recall, requiring models to locate and synthesize a single piece of evidence. In contrast, multi-hop QA benchmarks such as HotpotQA (Yang et al., 2018) and MuSiQue (Trivedi et al., 2022) challenge models to chain information from multiple documents and employ deductive reasoning to bridge disparate facts into coherent answers. Structured knowledge benchmarks, such as GraphQA (He et al., 2024c), require reasoning over relational graph representations, integrating nodes and edges to resolve complex queries beyond plain text retrieval. Complementing these open-ended tasks, multiple-choice evaluations like MMLU-Pro (Wang et al., 2025b) test domain-specific knowledge in areas such as science, history, or law, assessing the model's ability to perform various reasoning styles, including inductive and abductive inference. Multimodal QA benchmarks, like WebShop (Yao et al., 2022), test a model's capacity to align textual and visual information to determine the correct answer. Long-form QA datasets such as ∞BENCH (Zhang et al., 2024b) evaluate models' ability to maintain logical consistency and perform inductive reasoning over lengthy contexts. Collectively, these benchmarks establish a comprehensive evaluation chain for systematically assessing RAG-reasoning capabilities.

Beyond text-based QA, RAG-augmented benchmarks span diverse tasks involving long-form generation, interactive reasoning, and domain-specific challenges in mathematics and programming. Mathematics benchmarks such as MATH (Hendrycks et al., 2021) draw from competition-level problems to assess arithmetic and symbolic reasoning. Summarization tasks like XSum (Narayan et al., 2018) evaluate a model's ability to condense entire news articles into concise summaries while preserving factual correctness. Fact-checking benchmarks, such as FEVER (Thorne et al., 2018), test the capacity for evidence retrieval and claim verification. Code-focused evaluations, including LiveCodeBench (Jain et al., 2024), examine deductive and abductive reasoning in the context of algo-

一个完整的基准测试

第 6 节介绍了不同 RAG 推理任务的代表性基准测试。本附录通过一个按任务和领域组织的全面基准测试列表补充了该讨论。表 2 详细说明了每个基准测试的属性，包括发表场所、代码库、任务类别、领域、主要知识来源、知识类型和推理能力。通过将这些属性整合到一个表格中，我们简化了基准测试的选择和比较，使研究人员能够为未来 RAG 增强推理的研究识别最合适的 datasets。

我们的基准测试汇编主要来源于本文第 3 至 5 节中调查的方法，特别关注第 5 节中讨论的协同方法。我们刻意选择了需要同时进行外部知识检索和内部深度推理的基准测试，因为这种双重要求反映了现实场景，即模型不仅需要访问相关信息，还需要有效地整合和推理。例如，在问答领域，我们包括需要跨多个文档综合证据以回答无法通过单句检索解决的问题的 datasets。HotpotQA（Yang 等人，2018）体现了这一挑战，要求跨不同的维基百科文章进行推理。在编码任务中，LiveCodeBench（Jain 等人，2024）和 Refactoring Oracle（Tsantalis 等人，2020）等基准测试通过要求检索外部代码片段和文档，超越了纯粹的算法问题解决。类似地，在数学领域，MATH（Hendrycks 等人，2021）和 AQUA-RAT（Das 等人，2024）等基准测试不仅评估计算能力，还评估检索相关定理和公式，测试模型整合外部数学知识与内部推理过程的能力。

除了已有的基准测试外，我们还加入了更多具有挑战性的新数据集，这些数据集更能反映实际应用场景。这些数据集通常需要复杂的检索过程，并结合专家级或领域特定的推理能力，例如《人类最后考试》（HLE）（Phan 等人，2025）以及 BrowseComp（Wei 等人，2025a）等网络搜索评估任务。总体而言，我们的数据集包含 46 个基准测试，涵盖 12 个领域中的 13 种不同任务，每个数据集都明确标注了知识来源、知识类型和推理能力等特征。这种广泛性确保了覆盖不同领域和任务类型，为评估检索与推理在 RAG 系统中的相互作用奠定了坚实基础。

在这个基准数据集中，单跳问答数据集如 TriviaQA（Joshi 等人，2017）专注于精确检索和事实回忆，要求模型定位并综合单个证据。相比之下，多跳问答基准如 HotpotQA（Yang 等人，2018）和 MuSiQue（Trivedi 等人，2022）挑战模型从多个文档中串联信息，并运用演绎推理将不同的事实整合为连贯的答案。结构化知识基准如 GraphQA（He 等人，2024c）要求对关系图表示进行推理，整合节点和边以解决超出纯文本检索的复杂查询。补充这些开放式任务，多项选择题评估如 MMLU-Pro（Wang 等人，2025b）测试科学、历史或法律等领域的特定知识，评估模型执行归纳和溯因推理等不同推理风格的能力。多模态问答基准如 WebShop（Yao 等人，2022）测试模型整合文本和视觉信息以确定正确答案的能力。长文本问答数据集如 ∞BENCH（Zhang 等人，2024b）评估模型在长文本上下文中保持逻辑一致性和执行归纳推理的能力。总体而言，这些基准建立了一个全面的评估链，用于系统性地评估 RAG 推理能力。

超越基于文本的QA，RAG增强的基准涵盖了涉及长文本生成、交互式推理以及数学和编程领域中特定挑战的多样化任务。数学基准如MATH（Hendrycks等人，2021年）借鉴竞赛级问题来评估算术和符号推理能力。摘要任务如XSum（Narayan等人，2018年）评估模型将整篇新闻文章压缩成简洁摘要的能力，同时保持事实准确性。事实核查基准如FEVER（Thorne等人，2018年）测试证据检索和声明验证的能力。以代码为重点的评估包括Live-CodeBench（Jain等人，2024年），考察在算法背景下进行的演绎和溯因推理。

Dataset	*Venue*	Resource	Task	Domain	Knowledge Source	Knowledge Type	Reasoning Capability	Size	Input	Output
Code										
LiveCodeBench (Jain et al., 2024)	*Arxiv'24*	Link	Code	General	Internet	Logical	Deductive, Abductive	1,055	Question/Text, Code, Instruction	Code Instance, Test Output
Refactoring Oracle (Tsantalis et al., 2020)	*IEEE'22*	Link	Code	Software	Internet, Human	Logical	Deductive	7,226	Code, Instruction	Code Instance
ColBench (Zhou et al., 2025b)	*Arxiv'25*	Link	Code	Software	LLM, Human	Logical	Abductive, Inductive	10,000+	Question/Text, Links/Sources, Code	Code Instance
Math										
MATH (Hendrycks et al., 2021)	*NeurIPS'21*	Link	Domain-specific QA	Math	Exam/Competition	Logical, Arithmetic	Deductive	12,500	Question/Text, Equations	Number, Natural Language
MiniF2F (Zheng et al., 2021)	*ICLR'22*	Link	Domain-specific QA	Math	Exam/Competition, Books	Logical, Arithmetic	Deductive	488	Question/Text, Equations	Number, Natural Language
AQuA (Ling et al., 2017)	*Arxiv'17*	Link	Domain-specific QA	Math	Previous Source, Exam/Competition, Internet	Arithmetic, Logical	Deductive	100,000	Question/Text, Options, Equations	Natural Language, Options/Labels
Fact Checking										
CRAG (Yang et al., 2024b)	*NeurIPS'24*	Link	Fact Checking	General	Internet	Commonsense	Deductive, Abductive	4,409	Question/Text	Natural Language
CREAK (Onoe et al., 2021)	*NeurIPS'21*	Link	Fact Checking	General	Human	Commonsense	Deductive, Abductive, Analogical	13,000	Question/Text	Options/Labels, Natural Language
Fever (Thorne et al., 2018)	*ACL'18*	Link	Fact Checking	General	Internet	Logical	Deductive, Abductive	185,445	Question/Text, Links/Sources	Natural Language, Options/Labels
PubHealth (Kotonya and Toni, 2020)	*EMNLP'20*	Link	Fact Checking	Health	Internet	Commonsense, Logical	Abductive, Deductive	11,800	Question/Text	Natural Language, Options
Graph QA										
GraphQA (He et al., 2024c)	*NeurIPS'24*	Link	Graph QA	General	Previous Source	Commonsense, Multimodal	Deductive, Abductive	107,503	Question/Text	Natural Language
GRBENCH (Jin et al., 2024)	*ACL'24*	Link	Graph QA	General	LLM, Human	Logical	Deductive, Inductive	1,740	Question/Text	Natural Language
Long-form QA										
∞ BENCH (Zhang et al., 2024b)	*Arxiv'24*	Link	Long-form QA	General	Internet, Human	Multimodal, Logical	Inductive, Abductive	3,946	Question/Text, Code, Equations	Natural Language, Number, Code Instance
Multimodal QA										
CrisisMMD (Alam et al., 2018)	*Arxiv'18*	Link	Multimodal QA	Crisis Response	Media, Internet	Commonsense, Multimodal	Abductive	16,097	Question/Text, Figure/Image	Options, Natural Language
ALFWORLD (Shridhar et al.)	*ICLR'21*	Link	Multimodal QA	Game	Previous Source	Multimodal	Deductive, Abductive	3,827	Question/Text, Figure/Image	Natural Language
MMLongBench-DOC (Ma et al., 2025)	*NeurIPS'24*	Link	Multimodal QA	Narrative	Previous Source, Internet	Multimodal	Deductive, Abductive	1,082	Figure/Image, Question/Text, Documents	Natural Language, Number
LongDocURL (Deng et al., 2024)	*Arxiv'24*	Link	Multimodal QA	Narrative	Internet, Previous Source, LLM	Multimodal	Deductive, Abductive	2,325	Figure/Image, Question/Text, Documents	Natural Language, Number
UDA (Hui et al., 2024)	*NIPS'24*	Link	Multimodal QA	Narrative	Internet, Paper/Report	Multimodal	Deductive	29,590	Documents, Question/Text	Natural Language, Number
SCIENCEQA (Lu et al., 2022)	*NeurIPS'22*	Link	Multimodal QA	Science	Human	Logical, Multimodal	Deductive	21,000	Question/Text, Options, Figure/Image	Options, Natural Language, Number
WebShop (Yao et al., 2022)	*NeurIPS'22*	Link	Multimodal QA	E-commerce	Internet	Multimodal	Inductive, Abductive	12,087	Instruction, Question/Text	Natural Language, Image/Figure
SurgCoTBench (Low et al., 2025)	*Arxiv'25*	—	Multimodal QA	Health	Human	Multimodal, Logical	Abductive, Deductive	14,176	Question/Text, Figure/Image, Options	Options, Natural Language, Number

Table 2: Full representative knowledge and reasoning intensive benchmarks across diverse task categories (Part 1).

数据集	场地	资源	Task	领域	知识来源	知识类型推理能力		Size	输入	输出
Code										
LiveCodeBench (Jain 等，2024)	*Arxiv'24*	Link	Code	通用	互联网	逻辑	演绎，溯因	1,055	问题/文本，代码，指令	
重构神谕 (Tsantalis 等人，2020)	*IEEE'22*	Link	Code	软件	互联网，人类	逻辑	演绎	7,226	代码，指令代码实	
ColBench (Zhou 等人，2025b)	*Arxiv'25*	Link	Code	软件	大语言模型，人类	逻辑	溯因，归纳	10,000+	问题/文本，链接/来源，代码	代码实例
Math										
数学 (Hendrycks 等，2021)	*NeurIPS'21*	Link	特定领域 QA	Math	考试/竞赛 逻辑，算		演绎法	12,500	问题/文本，方程式	数量，自然语言
MiniF2F (郑等人，2021)	*ICLR'22*	Link	特定领域 QA	Math	考试/竞赛，书籍	逻辑，算术	演绎	488	问题/文本，方程式	数字，自然语言
AQuA (Ling 等人，2017)	*Arxiv'17*	Link	特定领域 QA	Math	上一个来源，考试/竞赛，互联网	算术，逻辑	演绎	100,000	问题/文本，选项，方程式	自然语言，选项/标签
事实核查										
CRAG（Yang等人，2024b）	*NeurIPS'24*	Link	事实核查	通用	互联网	常识	演绎、溯因	4,409	问题/文本	自然语言
CREAK (Onoe 等人，2021)	*NeurIPS'21*	Link	事实核查	通用	人类	常识	演绎的、归纳的、类比的	13,000	问题/文本	选项/标签，自然语言
发烧（Thorne 等人，2018）	*ACL'18*	Link	事实核查	综合	互联网	逻辑	演绎、溯因	185,445	问题/文本，链接/来源	自然语言，选项/标签
公共卫生（科顿亚和托尼，2020）	*EMNLP'20*	Link	事实核查	健康	互联网	常识，逻辑	溯因，演绎	11,800	问题/文本	自然语言，选项
图问答										
GraphQA (何等人，2024c)	*NeurIPS'24*	Link	图问答	通用	先前来源	常识，多模态	演绎，溯因	107,503	问题/文本	自然语言
GRBENCH (金等人，2024)	*ACL'24*	Link	图问答	通用	大语言模型，人类	逻辑	演绎法，归纳法	1,740	问题/文本	自然语言
长文本问答										
∞张 等，2024b)	*Arxiv'24*	Link	长文本问答	通用	互联网，人类	多模态，逻辑归纳，		3,946	问题/文本，代码，方程式	自然语言，数字，代码实例
多模态问答										
CrisisMMD (Alam 等., 2018)	*Arxiv'18*	Link	多模态问答 危		媒体、互联网	常识，多模态	溯因	16,097	问题/文本，图/图	选项，自然语言
ALFWORLD (Shridhar 等)	*ICLR'21*	Link	多模态问答	Game	先前来源	多模态	演绎式、溯因式	3,827	问题/文本，图/图像	自然语言
MMLongBench-DOC (Ma 等人，2025)	*NeurIPS'24*	Link	多模态问答	叙事	先前来源，互联网	多模态	演绎，溯因	1,082	图形/图像,问题/文本,文档	自然语言，数字
长文档URL (邓等, 2024)	*Arxiv'24*	Link	多模态问答	叙事	互联网，先前来源，大语言模型	多模态	演绎，溯因	2,325	图/像，问题/文档数量	
UDA（Hui 等人，2024）	*NIPS'24*	Link	多模态问答	叙事	互联网，论文/报告	多模态	演绎式	29,590	文档，问题/文本	自然语言，数量
SCIENCEQA (Lu 等人，2022)	*NeurIPS'22*	Link	多模态问答	科学	人类	逻辑，多模态	演绎	21,000	问题/文本，选项，图形/图像	选项，自然语言，数量
WebShop (Yao 等人，2022)	*NeurIPS'22*	Link	多模态问答	电子商务	互联网	多模态	归纳，溯因	12,087	指令，问题/文本	自然语言，图像/图形
SurgCoTBench (低等，2025)	*Arxiv'25*	—	多模态问答	健康	人类	多模态、逻辑溯因、		14,176	问题/文本、图形语言、数字	自然

表2：跨不同任务类别的完整代表性知识和推理密集型基准（第一部分）。

22

Dataset	*Venue*	Resource	Task	Domain	Knowledge Source	Knowledge Type	Reasoning Capability	Size	Input	Output
Multi-choice QA										
Bamboogle (Press et al., 2023)	*EMNLP'23*	Link	Multi-choice QA	General	Internet	Logical	Deductive, Abductive	125	Question/Text	Natural Language
BIG-Bench (Srivastava et al., 2022)	*Arxiv'22*	Link	Multi-choice QA	General	Internet	Commonsense, Logical	Deductive, Abductive, Inductive, Analogical	204	Question/Text, Options	Natural Language, Number, Options/Labels
ADQA (Li et al., 2024a)	*EMNLP'24*	Link	Multi-choice QA	Health	Previous Source	Commonsense, Logical	Deductive, Abductive	446	Question/Text, Options	Options
QuALITY (Pang et al., 2022)	*NAACL'22*	Link	Multi-choice QA	Narrative	Books	Commonsense, Logical	Deductive, Abductive	6,737	Question/Text, Options	Options
MMLU-Pro (Wang et al., 2025b)	*NeurIPS'24*	Link	Multi-choice QA	Science	Previous Source, Internet	Arithmetic, Commonsense, Logical	Deductive, Inductive	12,032	Question/Text, Options	Natural Language, Number, Options
Multi-hop QA										
FRAMES (Krishna et al., 2024)	*Arxiv'24*	Link	Multi-hop QA	General	Internet	Commonsense, Logical, Arithmetic	Deductive	824	Question/Text	Natural Language
HotpotQA (Yang et al., 2018)	*EMNLP'18*	Link	Multi-hop QA	General	Internet	Commonsense	Deductive	113,000	Question/Text	Natural Language
GPQA (Rein et al., 2024)	*Arxiv'24*	Link	Multi-hop QA	Science	Human	Logical	Deductive, Abductive	448	Question/Text, Options	Natural Language, Number, Options
HLE (Phan et al., 2025)	*Arxiv'25*	Link	Multi-hop QA	Science	Human	Logical, Arithmetic, Multimodal	Deductive, Abductive	2,500	Question/Text, Options, Figure/Image	Natural Language, Number, Options
CWQ (Talmor and Berant, 2018)	*NAACL'18*	Link	Multi-hop QA	General	Internet	Commonsense	Deductive	34,689	Question/Text	Natural Language
IIRC (Ferguson et al., 2020)	*EMNLP'20*	Link	Multi-hop QA	General	Internet	Commonsense, Logical	Deductive	13,000+	Question/Text, Links/Sources	Number, Natural Language
MINTQA (He et al., 2024b)	*Arxiv'24*	Link	Multi-hop QA	General	Internet	Commonsense, Logical	Deductive	10,479	Question/Text	Natural Language
MuSiQue (Trivedi et al., 2022)	*ACL'22*	Link	Multi-hop QA	General	Previous Source, Internet	Commonsense, Logical	Deductive	25,000	Question/Text	Natural Language
TopiOCQA (Adlakha et al., 2022)	*TACL'22*	Link	Multi-hop QA	General	Internet	Commonsense, Logical	Deductive	54,494	Question/Text	Natural Language
2WikiMultiHopQA (Ho et al., 2020)	*COLING'20*	Link	Multi-hop QA	General	Internet	Commonsense, Logical	Deductive	192,606	Question/Text	Natural Language
Multi-step QA										
StrategyQA (Geva et al., 2021)	*TACL'21*	Link	Multi-step QA	General	Internet	Commonsense, Logical	Deductive	2,780	Question/Text	Natural Language
Single-hop QA										
SimpleQA (Wei et al., 2024)	*Arxiv'24*	Link	Single-hop QA	General	LLM, Human	Commonsense	Deductive	4,326	Question/Text	Natural Language
TriviaQA (Joshi et al., 2017)	*ACL'17*	Link	Single-hop QA	General	Internet	Commonsense, Logical	Deductive	650,000+	Question/Text	Natural Language
NQ (Kwiatkowski et al., 2019)	*ACL'19*	Link	Single-hop QA	General	Internet	Commonsense, Logical	Deductive	307,373	Question/Text	Natural Language
Text Summarization										
XSum (Narayan et al., 2018)	*EMNLP'18*	Link	Text Summarization	Narrative	Internet, Media	Logical, Commonsense	Abductive	226,711	Question/Text	Natural Language
BIGPATENT (Sharma et al., 2019)	*ACL'19*	Link	Text Summarization	Patent	Internet	Commonsense, Logical	Abductive	1.3 M	Question/Text	Natural Language
Web Browsing										
BrowseComp (Wei et al., 2025a)	*Arxiv'25*	Link	Web Browsing	General	Human, Internet	Commonsense, Logical	Deductive	1,266	Question/Text	Natural Language
BrowseComp-ZH (Zhou et al., 2025a)	*Arxiv'25*	Link	Web Browsing	General	Human, Internet	Commonsense, Logical	Deductive	289	Question/Text	Natural Language
GAIA (Mialon et al., 2023)	*ICLR'23*	Link	Web Browsing	General	Internet, TooL	Commonsense, Logical	Deductive	466	Question/Text, Image/File/Code	Natural Language
WebWalkerQA (Wu et al., 2025b)	*Arxiv'25*	Link	Web Browsing	General	Human, LLM	Commonsense, Logical	Deductive	680	Question/Text	Natural Language
Dialog										
DailyDialog (Li et al., 2017)	*Arxiv'17*	Link	Dialog	General	Internet	Commonsense, Logical	–	13,118	Question/Text	Natural Language

Table 3: Full epresentative knowledge and reasoning intensive benchmarks across diverse task categories (Part 2, continued).

数据集	场地	资源	Task	领域	知识来源	知识类型推理能力		Size	输入	输出
多选问答										
Bamboogle (Press 等人, 2023)	*EMNLP'23*	Link	多选题问答	通用	互联网	逻辑	演绎、溯因	125	问题/文本	自然语言
BIG-Bench (Srivastava 等，2022)	*Arxiv'22*	Link	多选问答	通用	互联网	常识, 逻辑	演绎，归纳，归纳，类比	204	问题/文本, 选项	自然语言, 数字, 选项/标签
ADQA (Li 等人, 2024a)	*EMNLP'24*	Link	多选问答	健康	先前来源	常识, 逻辑	演绎推理，溯因推理	446	问题/文本, 选项	选项
QuALITY (Pang 等人, 2022)	*NAACL'22*	Link	多选问答	叙事	书籍	常识, 逻辑	演绎法，溯因法	6,737	问题/文本, 选项	选项
MMLU-Pro (Wang 等，2025b)	*NeurIPS'24*	Link	多选问答	科学	先前来源, 互联网	算术，常识, 逻辑	演绎，归纳	12,032	问题/文本, 选项	自然语言，数字, 选项
多跳问答										
FRAMES (Krishna 等，2024)	*Arxiv'24*	Link	多跳问答	通用	互联网	常识, 逻辑，算术	演绎	824	问题/文本	自然语言
火锅问答 (Yang 等人, 2018)	*EMNLP'18*	Link	多跳问答	通用	互联网	常识	演绎	113,000	问题/文本	自然语言
GPQA (Rein 等人, 2024)	*Arxiv'24*	Link	多跳问答	科学	人类	逻辑	演绎，溯因	448	问题/文本, 选项	自然语言，数字，选项
HLE（Phan等，2025）		Link	多跳问答	科学	人类	逻辑，算术, 多模态	演绎，溯因	2,500	问题/文本, 选项, 图形/图像	自然语言, 数字，选项
CWQ（塔尔莫尔和）Berant, 2018)	*NAACL'18*	Link	多跳问答	通用	互联网	常识	演绎	34,689	问题/文本	自然语言
据我回忆（Ferguson 等人, 2020)	*EMNLP'20*	Link	多跳问答	通用	互联网	常识, 逻辑	演绎	13,000+	问题/文本, 链接/来源	数字，自然语言
MINTQA (He 等人, 2024b)	*Arxiv'24*	Link	多跳问答	通用	互联网	常识, 逻辑	演绎	10,479	问题/文本	自然语言
MuSiQue (Trivedi 等人, 2022)	*ACL'22*	Link	多跳问答	通用	上一个来源, 互联网	常识, 逻辑	演绎	25,000	问题/文本	自然语言
TopiOCQA (Adlakha 等，2022)	*TACL'22*	Link	多跳问答	通用	互联网	常识, 逻辑	演绎	54,494	问题/文本	自然语言
2WikiMultiHopQA (Ho et al., 2020)	*COLING'20*	Link	多跳问答	通用	互联网	常识, 逻辑	演绎	192,606	问题/文本	自然语言
多步问答										
策略QA（Geva 等，2021）	*TACL'21*	Link	多步问答	通用	互联网	常识, 逻辑	演绎	2,780	问题/文本	自然语言
单跳问答										
SimpleQA (Wei 等人, 2024)	*Arxiv'24*	Link	单跳问答	通用	大语言模型，人类	常识	演绎	4,326	问题/文本	自然语言
TriviaQA (Joshi 等人, 2017)	*ACL'17*	Link	单跳问答	通用	互联网	常识, 逻辑	演绎	650,000+	问题/文本	自然语言
NQ（Kwiatkowski 等人 2019)	*ACL'19*	Link	单跳问答	通用	互联网	常识, 逻辑	演绎	307,373	问题/文本	自然语言
文本摘要										
XSum (Narayan 等人, 2018)	*EMNLP'18*	Link	Text 摘要	叙事	互联网、媒体	逻辑、常识	溯因	226,711	问题/文本	自然语言
BIGPATENT (Sharma 等，2019)	*ACL'19*	Link	Text 摘要	专利	互联网	常识, 逻辑	溯因	1.3 M	问题/文本	自然语言
网络浏览										
浏览组件（Wei 等人, 2025a）	*Arxiv'25*	Link	网页浏览	通用	人类，互联网	常识, 逻辑	演绎	1,266	问题/文本	自然语言
浏览组件-中文 (周等人，2025a)	*Arxiv'25*	Link	网页浏览	通用	人类，互联网	常识, 逻辑	演绎	289	问题/文本	自然语言
GAIA (Mialon 等人, 2023)	*ICLR'23*	Link	网络浏览	通用	互联网，工具	常识, 逻辑	演绎	466	问题/文本, 图片/文件/代码	自然语言
WebWalkerQA (吴等，2025b)	*Arxiv'25*	Link	网络浏览	通用	人类，大语言模型	常识, 逻辑	演绎	680	问题/文本	自然语言
对话										
日常对话 (Li 等人, 2017)	*Arxiv'17*	Link	对话	通用	互联网	常识，逻辑	–	13,118	问题/文本	自然语言

表3：跨不同任务类别的完整代表性知识和推理密集型基准（第2部分，续）

Benchmark	Domain	Primary Retrieval Challenge	Primary Reasoning Challenge
TriviaQA, NQ	General	**Scale & Noise:** Retrieval from massive, noisy corpora.	**Ambiguity:** Handling real-world queries that are often underspecified or ambiguous.
HotpotQA, 2WikiMultiHopQA, MuSiQue, HLE	General	**Multi-document / High-dependency Synthesis:** Requires finding and connecting evidence scattered across multiple Wikipedia articles.	**Multi-hop Deduction:** Explicitly designed to test the ability to link two or more discrete facts into a coherent reasoning path.
MMLU-Pro, QUALITY	Science, Narrative	**Expert-level Retrieval:** Requires accessing deep specialized knowledge from academic or densely written narrative sources.	**Complex & Long-form Reasoning:** MMLU-Pro demands expert-level problem-solving over rote memorization. QUALITY uniquely requires comprehension of very long texts (often >5,000 tokens).
MATH, AQUA-RAT	Math	**Formal Knowledge Retrieval:** Locating precise mathematical theorems, lemmas, or formulas in formal corpora.	**Symbolic & Deductive Reasoning:** Involves performing precise, multi-step logical and algebraic operations where each step must be correct. AQUA-RAT is unique in providing natural language rationales, thus testing the model's ability to explain its formal reasoning.
LiveCodeBench	Code	**Structural & Modal Heterogeneity:** Must retrieve from diverse, heterogeneous sources such as code repositories, documentation, and community forums like Stack Overflow.	**Tool Use & Self-correction Reasoning:** Requires applying retrieved code snippets/APIs, executing code, and reasoning based on test outputs to debug and iteratively improve solutions.
BrowseComp, WebWalkerQA	General (Web)	**Dynamism, Interactivity, and Long-tail Retrieval:** Tests agentic planning and tool use in live, unstructured web environments. BrowseComp requires creative, persistent navigation to locate hard-to-find, intertwined information, while WebWalkerQA focuses on systematic traversal of a website's subpages.	**Agentic & Strategic Reasoning:** Requires planning and executing multi-step strategies (e.g., searching, clicking, extracting) in dynamic and unpredictable contexts to achieve a defined goal.

Table 4: The primary retrieval and reasoning challenges for different RAG-Reasoning benchmarks.

rithmic problem-solving. Web-based tasks, exemplified by BrowseComp (Wei et al., 2025a), emulate real-world search behavior, requiring iterative query formulation and navigation across multiple webpages.

In addition to cataloging datasets, Table 4 provides a synthesized overview of the primary retrieval and reasoning challenges associated with each benchmark discussed in this survey. This comparative analysis reveals critical gaps in current benchmark coverage that future research must address. From a **domain perspective**, most benchmarks still focus on a limited set of general or academic scenarios, with few tackling real-world, realistic industrial or vertical-domain tasks where retrieval sources might be personalized, proprietary or highly specialized. Regarding **retrieval capabilities**, existing benchmarks rarely test systems' ability to handle heterogeneous or multimodal content, nor do they systematically evaluate robustness against noisy, evolving, or conflicting information within a unified framework for trustworthiness. In terms of **reasoning capabilities**, current benchmarks primarily assess deductive reasoning, leaving underexplored more complex forms such as deep causal reasoning, counterfactual thinking, decision-oriented reasoning, or analogical reasoning in specialized domains. Moreover, there is a lack of standardized benchmarks and metrics for evaluating the entire reasoning-retrieval trajectory, including the efficiency of retrieval steps, the quality of intermediate queries, and the logical consistency of multi-step reasoning chains.

B Deep Research Implementations

In this section, we extend the discussion of the agentic paradigm introduced in Section 5.2, in which RAG systems adopt the role of active researchers who plan multistep queries, interleave retrieval with reasoning, and coordinate specialized tools or agents. These characteristics collectively define what we refer to as deep research, representing the ability of a system to autonomously break down complex questions, iteratively gather diverse evidence, and synthesize information through multiple reasoning steps. This paradigm seeks to enhance autonomy, reduce hallucinations, and improve factual accuracy in open-domain tasks.

Such deep research systems can be realized through either single-agent or multi-agent architectures. Single-agent systems rely on a single model to manage the entire process of question decomposition, retrieval, and synthesis, offering simplicity and shared context but facing limitations in handling highly specialized or multi-modal tasks. In contrast, multi-agent systems distribute these responsibilities among specialized agents, enabling modularity and potentially greater robustness. However, this collaborative design introduces

基准测试	领域	主要检索挑战	主要推理挑战
TriviaQA, NQ	通用	缩放与噪声：从海量、含噪声的语料库中检索 语料中。	歧义性：处理现实世界中那些通常不明确或模糊的查询的问题。
HotpotQA、2WikiMultiHopQA、MuSiQue、HLE	通用	多文档/高依赖合成：需要查找并连接分散在多个维基百科文章中的证据 中。	多跳推理：专门设计用于测试将两个或多个离散事实连接成一个连贯推理路径的能力 中。
MMLU-Pro，质量	科学，叙事	专家级检索：需要访问深层知识库，从学术或内容密集的叙事资料中获取专业知识。	复杂与长文本推理：MMLU-Pro 要求专家级问题解决能力，而非死记硬背。QUALITY 独特要求理解极长文本（通常 >5,000 个 token）。
MATH, AQUA-RAT	Math	正式知识检索：在正式语料库中定位精确的数学 确的多步逻辑和代数运算，每一步都必须正确。 从而测试模型解释其正式推理的能力。	
LiveCodeBench	Code	结构化与模态异构性：必须检索工具使用与自我 Overflow等社区论坛等多样化、异构来源中应用 输出进行推理以调试并迭代改进解决方案。	
BrowseComp, WebWalkerQA	通用（网络）	动态性、交互性和长尾检索：在实时、非结构化 BrowseComp需要创造性的持久导航来定位难以 重于网站子页面的系统化遍历。代理与策略推理: 步骤策略（例如，搜索、点击、提取）以实现定	

表4：不同RAG-推理基准的主要检索和推理挑战。

算法问题解决。基于网络的任务，以BrowseComp（Wei等人，2025a）为例，模拟现实世界的搜索行为，需要迭代查询制定和跨多个网页的导航。

除了对数据集进行编目外，表4还提供了关于本调查中讨论的每个基准测试所关联的主要检索和推理挑战的综合概述。这项比较分析揭示了当前基准测试覆盖面中存在的关键差距，未来研究必须解决这些问题。从领域角度来看，大多数基准测试仍然专注于有限的一般性或学术场景，很少有测试现实世界、真实的工业或垂直领域任务，在这些任务中，检索来源可能是个性化的、专有的或高度专业的。在检索能力方面，现有基准测试很少测试系统处理异构或多模态内容的能力，也未能在一个统一的信任框架内系统地评估对噪声、演变或冲突信息的鲁棒性。在推理能力方面，当前基准测试主要评估演绎推理，而更复杂的推理形式，如深度因果推理、反事实思维、面向决策的推理或类比推理，在专业领域中的应用尚未得到充分探索。此外，缺乏用于评估整个推理-检索轨迹的标准基准测试和指标，包括检索步骤的效率、中间查询的质量以及多步推理链的逻辑一致性。

B 深度研究实现

在本节中，我们扩展了第 5.2 节中引入的代理范式讨论，其中 RAG 系统采用主动研究者的角色，规划多步查询，将检索与推理交替进行，并协调专业工具或代理。这些特征共同定义了我们所说的深度研究，代表系统自主分解复杂问题、迭代收集多样化证据、通过多重推理步骤综合信息的能力。这种范式旨在提高自主性、减少幻觉，并提高开放域任务中的事实准确性。

这种深度研究系统可以通过单智能体或多智能体架构来实现。单智能体系统依赖单一模型来管理问题分解、检索和合成全过程，具有简单性和共享上下文的特点，但在处理高度专业化或多模态任务时存在局限性。相比之下，多智能体系统将这些责任分配给专门的智能体，实现模块化并可能具有更强的鲁棒性。然而，这种协作设计引入了

24

Name	Base Model	Optimizatio	Reward	Retriever	Agent Architecture	Train Data	Evaluation Data	Link
Agentic Reasoning (Wu et al., 2025c)	N/A	Prompting	N/A	Web Search	Centralized	N/A	GPQA	Link
gpt-researcher		Prompting	N/A	Web Search, Local Retrieval	Centralized	N/A	N/A	Link
deep-searcher	Deepseek, , Claude, Gemini, Qwen	Prompting	N/A	Web Search	Hierarchical	N/A	N/A	Link
Search-R1 (Jin et al., 2025)	Qwen2.5-7B-Instruct, Qwen2.5-7B-Base, Qwen-2.5-3B-Instruct, Qwen-2.5-3B-Base	GRPO, PPO	Exact Match	Web Search	Single	NQ, HotpotQA	NQ, TriviaQA, PopQA, HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle	Link
ZeroSearch (Sun et al., 2025a)	Qwen2.5-3B-Base, Qwen2.5-7B-Base, Qwen2.5-7B-Instruct, Qwen2.5-3B-Instruct, LLaMA3.2-3B-Instruct, LLaMA3.2-3B-Base	GRPO, PPO, Reinforce	Exact Match	Web Search	Single	NQ, HotpotQA	NQ, TriviaQA, PopQA, HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle	Link
Webthinker (Li et al., 2025c)	GPT-o1, GPT-o3, Deepseek-R1, QwQ-32B, Qwen2.5-32B-Instruct	DPO	Preference Pairs	Web Search	Single	SuperGPQA, WebWalkerQA, OpenThoughts, NaturalReasoning, NuminaMath	GPQA, GAIA, WebWalkerQA, Humanity's Last Exam	Link
nanoDeepResearch	OpenAI series, Claude	Prompting	N/A	Web Search	Centralized	N/A	N/A	Link
DeerFlow	Qwen,	Prompting	N/A	Web Search	Decentralized	N/A	N/A	Link
deep-research	Deepseek,	Prompting	N/A	Web Search	Single	N/A	N/A	Link
open-deep-research	OpenAI series, Deepseek, Claude, Gemini	Prompting	N/A	Web Search	Single	N/A	N/A	Link
DeepResearcher (Zheng et al., 2025)	Qwen2.5-7B-Instruct	GRPO	Format	Web Search	Decentralized	NQ, TQ, HotpotQA, 2WikiMultiHopQA	MuSiQue, Bamboogle, PopQA, NQ, TQ, HotpotQA, 2WikiMultiHopQA	Link
R1-Searcher (Song et al., 2025)	Qwen2.5-7B-Base, Llama3.1-8B-Instruct	GRPO, Reinforce++, SFT	Retrieval, Format	Web Search, Local Retrieval	Single	HotpotQA, 2WikiMultiHopQA	HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle	Link
ReSearch (Chen et al., 2025a)	Qwen2.5-7B-Instruct, Qwen2.5-32B-Instruct	GRPO	Format, Answer	Web Search	Single	MuSiQue	HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle	Link
Search-o1 (Li et al., 2025b)	QwQ-32B-Preview	Prompting	N/A	Web Search	Single	N/A	GPQA, MATH500, AMC2023, AIME2024, LiveCodeBench, Natural Questions, TriviaQA, HotpotQA, 2Wiki, MuSiQue, Bamboogle	Link
r1-reasoning-rag	Deepseek	Prompting	N/A	Local Retrieval, Web Search	Single	N/A	N/A	Link
Open Deep Search (Alzubi et al., 2025)	Llama3.1-70B, Deepseek-R1	Prompting	N/A	Web Search	Single	N/A	SimpleQA, FRAME	Link
node-DeepResearch	Gemini,	Prompting	N/A	Web Search	Single	N/A	N/A	Link
deep-research	Gemini, OpenAI series, Deepseek, Claude, Grok	Prompt	N/A	Local Retrieval, Web Search	Single	N/A	N/A	Link

Table 5: Overview of deep research implementations.

additional complexity in coordination and communication, as well as higher computational costs.

Alongside these developments in agent orchestration, the nature of retrievers used in deep research has also evolved significantly. Early RAG systems relied on sparse keyword-based retrieval, later surpassed by dense retrievers employing bi-encoder architectures for semantic matching. More recent deep research systems increasingly integrate web search-based retrievers, allowing real-time access to open-domain information. Some retrievers have also been transformed into LLM-callable tools for flexible invocation. This evolution of retrievers has played a crucial role in enabling the sophisticated information-gathering processes required for deep research.

C Comparison of Reasoning Workflows and Agent Orchestration Strategies

Table 6 summarizes the diverse reasoning workflows and agent orchestration strategies employed in Synergized RAG-Reasoning systems, highlighting their respective strengths, limitations, and suitable application scenarios. Reasoning workflows vary from linear chain-based approaches, which are efficient but vulnerable to error propagation, to more complex tree-based and graph-based methods that offer higher recall and transparency at the cost of increased computational overhead. Similarly, agent orchestration strategies range from single-agent setups to multi-agent systems that distribute specialized roles among agents, enhancing robustness and scalability. However, these advanced designs often introduce additional communication overhead and complexity in conflict resolution. This comparison illustrates the trade-offs inherent in choosing particular workflows or orchestration architectures and underscores the need for adaptive systems that can dynamically balance efficiency, accuracy, and resource constraints in real-world applications.

Name	基础模型	优化		检索器	代理架构	训练数据	评估数据	Link
具身推理（吴等人，2025c）	N/A	提示	N/A	网络搜索	集中式	N/A	GPQA	Link
gpt-researcher		提示	N/A	网络搜索，本地检索	集中式	N/A	N/A	Link
deep-searcher	Deepseek, , Claude, Gemini , Qwen	提示	N/A	网络搜索	层级式	N/A	N/A	Link
搜索-R1 (金 et al., 2025)	Qwen2.5-7B-Instruct Qwen2.5-7B-Base, Qwen-2.5-3B-Instruct, Q wen-2.5-3B-B ase	GRPO, PPO	精确匹配	网络搜索	单选	NQ，HotpotQA	NQ，TriviaQA，PopQA，HotpotQA，2WikiMultiHopQA，MuSiQue，Bamboogle	Link
ZeroSearch（Sun等人，2025a）	Qwen2.5-3B-Base, Qwen2.5-7B-Base, Qwen2.5-7B-Instruct, Qwen2.5-3B-Instruct, LLaMA3.2-3B-Instruct, LLaMA3.2-3B-Base	GRPO, PPO, Reinforce	Exact 匹配	网络搜索	单选	NQ，热点问答	NQ，趣味问答，流行问答，热点问答，2Wiki多跳问答，MuSiQue，Bamboogle	Link
Webthinker（李等人，2025c）	GPT-o1、GPT-o3、Deepseek-R1、QwQ-32B、Qwen2.5-32B-Instruct	DPO	偏好对	网络搜索	单次	SuperGPQA、WebWalkerQA、OpenThoughts、自然推理、NuminaMath	GPQA、GAIA、WebWalkerQA、Humanity's Last Exam	Link
nanoDeepResearch	OpenAI系列、Claude	Prompting	N/A	Web搜索	集中式	N/A	N/A	Link
DeerFlow	Qwen,	提示	N/A	网络搜索	去中心化	N/A	N/A	Link
深度研究	Deepseek,	Prompting	N/A	Web Search	Single	N/A	N/A	Link
open-deep-research	OpenAI系列，Deepseek，Claude，Gemini	提示	N/A	网络搜索	单	N/A	N/A	Link
DeepResearcher (Zheng 等人，2025)	Qwen2.5-7B-Instruct	GRPO	格式	网络搜索	去中心化	NQ、TQ、HotpotQA、2WikiMultiHopQA	MuSiQue、Bamboogle、PopQA、NQ、TQ、HotpotQA、2WikiMultiHopQA	Link
R1-Searcher（Song 等人，2025年）	Qwen2.5-7B-Base, Llama3.1-8B-Instruct	GRPO, 重新启用++,SFT	检索, 格式	网络搜索, 本地检索	单次	HotpotQA, 2WikiMultiHopQA	HotpotQA，2WikiMultiHopQA，MuSiQue, Bamboogle	Link
ReSearch (陈等人，2025a)	Qwen2.5-7B-Instruct, Qwen2.5-32B-Instruct	GRPO	格式，答案	网络搜索	单次	MuSiQue	HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle	Link
Search-o1 (Li 等人，2025b)	QwQ-32B-Preview	Prompting	N/A	Web Search	单	N/A	GPQA、MATH500、AMC2023、AIME2024、LiveCodeBench、自然问题、TriviaQA、HotpotQA、2Wiki、MuSiQue、Bamboogle	Link
r1-推理-rag	Deepseek	提示	N/A	本地检索，网络搜索	单个	N/A	N/A	Link
开放深度搜索（Alzubi 等人，2025）	Llama3.1-70B, Deepseek-R1	提示	N/A	网络搜索	单次	N/A	SimpleQA, FRAME	Link
node-深度研究	Gemini,	提示	N/A	网络搜索	单选	N/A	N/A	Link
深度研究	Gemini、OpenAI系列、Deepseek、Claude、Grok	提示	N/A	本地检索、网络搜索	单	N/A	N/A	Link

表5：深度研究实现的概述。

协调和通信方面的额外复杂性，以及更高的计算成本。

在这些关于智能体编排的进展之外，深度研究中使用的检索器的性质也发生了显著变化。早期的RAG系统依赖于稀疏关键词检索，后来被采用双编码器架构进行语义匹配的密集检索器所超越。更近期的深度研究系统越来越多地集成了基于网络搜索的检索器，允许实时访问开放域信息。一些检索器也被转换为可被LLM调用的工具，以实现灵活的调用。检索器的这种演变在实现深度研究所需的复杂信息收集过程中发挥了关键作用。

C 推理工作流程和智能体编排策略的比较

表6总结了协同式RAG推理系统中采用的多样化推理工作流和代理编排策略，突出了各自的优缺点、适用场景。推理工作流从基于线性链的方法（高效但易受错误传播）到更复杂的基于树和图的方法（召回率更高、透明度更强但计算开销更大）不等。类似地，代理编排策略从单代理设置到多代理系统（在代理间分配专门角色以增强鲁棒性和可扩展性）不等。然而，这些高级设计往往引入额外的通信开销和冲突解决复杂性。这种比较说明了选择特定工作流或编排架构时固有的权衡，并强调了需要能够动态平衡效率、准确性和资源限制的适应性系统，以应对实际应用中的挑战。

Category	Sub-category	Strengths	Limitations	Suitable Scenarios
Reasoning Workflow	Chain-based	One retrieval per reasoning step; low latency and token cost. Easy to cache and monitor.	An early wrong sub-query propagates; context grows fast on long chains.	Single-hop or short multi-hop QA where each intermediate fact is easy to access.
	Tree-based (ToT)	High recall: explores multiple branches in parallel, hedges against early errors. Transparent what-if traces.	Quadratic cost; tree branches require many retrieval calls.	Ambiguous or "multiple plausible paths" tasks (e.g., HotpotQA, legal reasoning) where missing one clue kills accuracy.
	Tree-based (MCTS)	Budget-aware exploration: focuses calls on promising branches; graceful anytime stopping.	Tuning-heavy and may converge to a suboptimal subtree.	Deep-search problems under tight API-call or token budgets (e.g., biomedical QA).
	Graph-based (Walk-on-Graph)	Efficient in explicit KG/document graphs; short reasoning paths on KGs.	Requires high-quality KGs; fails if graphs lack explicit edges; less flexible for open-web contexts.	Enterprise or domain-specific QA where a curated KG exists (e.g., product catalogs).
	Graph-based (Think-on-Graph)	Adaptive and verifiable; LLM updates a live evidence graph, allowing node-level citation checks and high factual accuracy.	Higher latency; many micro-tool calls; search space can explode without pruning.	Open-domain "deep research" or fact-dense synthesis tasks (e.g., BrowseComp, systematic reviews).
Agent Orchestration	Single-agent (Prompt-only)	Simple implementation via a ReAct loop; low resource overhead.	Constrained by prompt engineering and system design flexibility.	Prototyping demos and small-scale applications where simplicity outweighs performance.
	Single-agent (SFT)	Clear, well-defined RAG and reasoning patterns; higher precision than prompt-only approaches.	Requires large synthetic data; may overfit tool schemas, reducing out-of-domain generalization.	Production chatbots with stable APIs and predictable query formats (e.g., internal customer support).
	Single-agent (RL)	Adaptive RAG and reasoning yields high recall and accuracy; learns when to retrieve and reason.	Challenging to define suitable reward signals; computationally expensive to train.	Open-domain research or long-form QA where call costs are high and optimal stop conditions matter.
	Multi-agent (Decentralized)	High recall via parallel domain experts; robustness to noisy or diverse corpora.	High communication and consensus overhead; conflicting answers require resolution.	Large-scale evidence aggregation across heterogeneous sources (e.g., meta-analysis, news tracking).
	Multi-agent (Central-ized/Hierarchical)	Budget-efficient: manager avoids duplicate searches and ensures a clear provenance chain. Scales horizontally without exponential cost growth.	Manager prompts or policies can become a single-point bottleneck, limiting performance.	Complex tasks requiring coordinated subtasks under strict API-call budgets.

Table 6: Comparison of reasoning workflows and agent orchestration in Synergized RAG-Reasoning systems.

类别	子类别	优势	限制	适用场景
推理流程	基于链式	每一步推理检索一次；低延迟和token成本。易于缓存和监控。	一个早期的错误子查询会传播；长链上的上下文增长很快。	单跳或短多跳问答，其中每个中间事实都易于访问。
	基于树（ToT）	高召回率：并行探索多个分支，防止早期错误。透明的what-if追踪。	二次成本；树分支需要许多检索调用。	模糊或"多条合理路径"的任务（例如，HotpotQA、法律推理），其中缺少一个线索就会导致准确率下降。
	基于树的（MCTS）	预算感知探索：专注于对有希望的分支进行调用；优雅的随时停止。	调优密集型，可能会收敛到一个次优子树。	在严格的 API 调用或 token 预算限制下的深度搜索问题（例如生物医学问答）。
	基于图的 (Walk-on-Graph)	高效处理显式知识图谱/文档高效；在知识图谱上具有短推理路径。	需要高质量的KG；如果图中缺少明确的边则失败；在开放网络环境下灵活性较低。	企业级或特定领域的QA，其中存在经过筛选的KG（例如产品目录）。
	基于图（Think-on-Graph）	自适应且可验证；支持LLM更新一个实时证据图，支持节点级引用检查和高事实准确性。	延迟较高；需要大量微工具调用；如果不进行剪枝，搜索空间可能爆炸。	开放域"深度研究"或事实密集型综合任务（例如，BrowseComp、系统评价）。
代理编排	单代理 (仅提示)	通过ReAct循环实现简单；资源开销低。	受限于提示工程以及系统设计的灵活性。	原型演示和小规模应用，其中简单性优先于性能。
	单代理 (SFT)	清晰、定义良好的 RAG 和推理模式；比纯提示方法具有更高的精度。	需要大量合成数据；可能过度拟合工具模式，降低域外泛化能力。	使用稳定API和可预测查询格式（例如，内部客户支持）生产聊天机器人。
	单代理（RL）	自适应RAG和推理可带来高召回率和准确率；学习何时检索和推理。	难以定义合适的奖励信号训练计算成本高。	开放域研究或长文本问答，其中通话成本高且最佳停止条件很重要。
	多智能体（去中心化）	通过并行领域专家实现高召回率；对噪声或多样化的语料库具有鲁棒性。	通信和共识开销高；冲突答案需要解决。	跨异构来源的大规模证据聚合（例如，荟萃分析、新闻追踪）。
	多智能体（集 免重复搜索并确保 成本增长。		管理者的提示或策略可能成为单点瓶颈，限制性能。	需要协调子任务且严格限制 API 调用预算的复杂任务。

表 6：Synergized RAG-Reasoning 系统中推理工作流与代理编排的比较。